

Higher Nationals - Summative Assignment Feedback Form

Student Name/ID	Nethmi Thiloka Divyanjalee (E170909)		
Unit Title	Unit 16: Computing Research Project -Pearson Set		
Assignment Number	1 of 2	Assessor	Ms. Sasini
Submission Date	21/07/2025	Date Received 1st submission	13/06/2025
Re-submission Date		Date Received 2nd submission	

Assessor Feedback:

LO1 Examine appropriate research methodologies and approaches as part of the research process

Pass, Merit & Distinction Descriptors P1 ☐ P2 ☐ M1 ☐ D1 ☐

LO2 Conduct and analyze research relevant to a chosen computing research project

Pass, Merit & Distinction Descriptors P3 ☒ P4 ☐ M2 ☐ D1 ☐

LO3 Communicate the outcomes of a research project to identified stakeholders

Pass, Merit & Distinction Descriptors P5 ☐ M3 ☐ D2 ☐

LO4 Reflect on the application of research methodologies and concepts

Pass, Merit & Distinction Descriptors P6 ☐ P7 ☐ M4 ☐ D3 ☐

Assessor Feedback:

* Please note that grade decisions are provisional. They are only confirmed once internal and external moderation has taken place and grades decisions have been agreed at the assessment board.

Grade:	Assessor Signature:	Date:
Resubmission Feedback:		
Grade:	Assessor Signature:	Date:
Internal Verifier's Comments:		
Signature & Date:		

Pearson

Higher Nationals in

Computing

Unit 16: Computing Research Project

Project Proposal

Important Points:

1. It is strictly prohibited to use textboxes to add texts in the assignments, except for the compulsory information. eg: Figures, tables of comparison etc. Adding text boxes in the body except for the before mentioned compulsory information will result in rejection of your work.
2. Avoid using page borders in your assignment body.
3. Carefully check the hand in date and the instructions given in the assignment. Late submissions will not be accepted.
4. Ensure that you give yourself enough time to complete the assignment by the due date.
5. Excuses of any nature will not be accepted for failure to hand in the work on time.
6. You must take responsibility for managing your own time effectively.
7. If you are unable to hand in your assignment on time and have valid reasons such as illness, you may apply (in writing) for an extension.
8. Failure to achieve at least PASS criteria will result in a REFERRAL grade .
9. Non-submission of work without valid reasons will lead to an automatic REFERRAL. You will then be asked to complete an alternative assignment.
10. If you use other people's work or ideas in your assignment, reference them properly using HARVARD referencing system to avoid plagiarism. You have to provide both in-text citation and a reference list.
11. If you are proven to be guilty of plagiarism or any academic misconduct, your grade could be reduced to A REFERRAL or at worst you could be expelled from the course
12. Use word processing application spell check and grammar check function to help editing your assignment.
13. Use **footer function in the word processor to insert Your Name, Subject, Assignment No, and Page Number on each page**. This is useful if individual sheets become detached for any reason.

STUDENT ASSESSMENT SUBMISSION AND DECLARATION

When submitting evidence for assessment, each student must sign a declaration confirming that the work is their own.

Student name: Nethmi Thiloka Divyanjalee		Assessor name: Ms. Sasini	
Issue date: 13/06/2025	Submission date: 21/07/2025		Submitted on: 21/07/2025
Programme: HND in Computing			
Unit: 16 – Computing Research Project			
Assignment number and title: 1 -research proposal Computing Research Project on Artificial Intelligence			

Plagiarism

Plagiarism is a particular form of cheating. Plagiarism must be avoided at all costs and students who break the rules, however innocently, may be penalised. It is your responsibility to ensure that you understand correct referencing practices. As a university level student, you are expected to use appropriate references throughout and keep carefully detailed notes of all your sources of materials for material you have used in your work, including any material downloaded from the Internet. Please consult the relevant unit lecturer or your course tutor if you need any further advice.

Guidelines for incorporating AI-generated content into assignments:

The use of AI-generated tools to enhance intellectual development is permitted; nevertheless, submitted work must be original. It is not acceptable to pass off AI-generated work as your own

Student Declaration

Student declaration

I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.

Student signature: e170909@esoft.academy 

Date: 21/07/2025

Higher National Diploma in Computing

Assignment Brief

Student Name /ID Number	Nethmi Thiloka Divyanjalee (E170909)
Unit Number and Title	Unit 16 – Computing Research Project (Pearson set)-
Academic Year	2024/25
Unit Tutor	Ms. Sasini
Assignment Title	Computing Research Project -Pearson Set
Issue Date	13/06/2025
Submission Date	21/07/2025
IV Name & Date	

Submission format

- **Research Project Proposal-** The submission is in the form of an individual written report. This should be written in a concise, formal business style. You are required to make use of headings, paragraphs and subsections as appropriate, and all work must be supported with research and please provide a references list using the Harvard referencing system.

The recommended word limit is 2000 words.

- **Formal project report -** In the form of an individual written report. This should be written in a concise, formal business style following standard academic guidelines for research projects. You are required to make use of headings, paragraphs and subsections as appropriate, and all work must be supported with research. Referencing using the Harvard referencing system.

The recommended word limit is minimum 4,500 words

- **PowerPoint presentation** – present your research finding in a 10 to 15 minutes presentation that elaborates the research process followed, findings and conclusion, recommendation and future research considerations.

Please submit the speaker notes of the presentation in the appendix section of the final project report.
Unit Learning Outcomes:
LO1. Examine appropriate research methodologies and approaches as part of the research process. LO2. Conduct and analyse research relevant to a chosen computing research project LO3. Communicate the outcomes of a research project to identified stakeholders LO4. Reflect on the application of research methodologies and concepts
Trasferrable skills and Competencies developed

Computing related cognitive skills

- Understand the scientific method and its applications to problem-solving
- Demonstrate knowledge and understanding of essential facts, concepts, principles and theories relating to computing and computer applications
- Methods and tools: deploy appropriate theory, practices and tools for the design, implementation and evaluation of computer-based systems
- Recognise the professional, economic, social, environmental, moral and ethical issues involved in the sustainable exploitation of computer technology and be guided by the adoption of appropriate professional, ethical and legal practices

Computing related practical skills

- The ability to specify, design and construct reliable, secure and usable computer based systems
- The ability to plan and manage projects to deliver computing systems within constraints of requirements, timescale and budget
- The ability to deploy effectively the tools used for the construction and documentation of computer applications, with particular emphasis on understanding the whole process involved in the effective deployment of computers to solve practical problems
- The ability to critically evaluate and analyse complex problems, including those with incomplete information, and devise appropriate solutions, within the constraints of a budget

Employability skills

- Intellectual skills: critical thinking; making a case; numeracy and literacy
- Self-management: self-awareness and reflection; goal setting and action planning; independence and adaptability; acting on initiative; innovation and creativity
- Interaction: reflection and communication
- Contextual awareness e.g. the ability to understand and meet the needs of individuals, business and the community, and to understand how workplaces and organisations are governed

Assignment Brief and Guidance:

Research Theme: Artificial Intelligence

Artificial intelligence is at the forefront of innovation within Computer Science that uses a combination of logic, algorithms and large data sets to produce an AI model. The AI model is created to perform specific tasks or make predictions on supplied sets of input data, for example identifying patterns in weather data, internet search data or analysis of medical data. Artificial intelligence is predicted to generate a potential impact to the global economy of \$13 – \$15 Trillion by 2030, with sales of AI related hardware, software and services predicted to see a global revenue of \$900 billion. It is predicted that AI will boost the GDP of China by a little over 26% by 2039, and of North America by 14.5%. AI requires the input of structured and labelled data, where the outputs are already known. The input data sets to the AI model are intrinsically linked to study field to which the AI engine is to be applied. The AI model can then be used to identify and recognise patterns and relationships within the input data. This identification step is referred to as ‘training’ the AI model. Once this training is completed, the model can then be used to make predictions and identify patterns within brand new data sets. This new data set can then be added to the existing data set, so that the AI model keeps ‘growing’. As the model data set keeps expanding, and the AI algorithms are modified and refined, this gives the impression that the AI is ‘learning’ and demonstrating ‘intelligence’. AI has been used extensively to analyse and process large and complex datasets produced by big data systems, often in real time and using Computer Vision to extract data from image sources.

Developing Artificial Intelligence required a range of knowledge and skills across a broad range of computer science disciplines. AI developers need to be familiar with the algorithms and techniques in fields such as machine learning, natural language processing, computer vision and data science. Knowing the required computing skills will help organisations recruit the correct resources to help develop and extend AI systems. Artificial Intelligence has a range of benefits across many industry sectors. In the finance industry, AI is rapidly becoming a game changer, using advanced algorithms, models and machine learning to carry out predictive analytics on large, rapidly changing financial datasets to provide more accurate financial predictions. In the field of business operations, AI automation is helping to support and enhance labour. productivity, leading to greater cost savings and increased efficiency. AI is also revolutionising the way businesses interact with their customers, by providing AI driven expert systems to help customers resolve queries as well as providing personalised recommendations based on customer choices and preferences. In the field of biomedical science, AI models help in the development of new drug treatments for a range of diseases by searching and processing large scale medical and DNA datasets.

While Artificial Intelligence has numerous benefits in the analysis and processing of large data sets to solve problems, there are some clear risks to the application technology. AI systems respond to the data fed into the model, and so if this data is not representative of the problem area under study, there is a likelihood that the output of the AI model will be biased. In addition, there are security and privacy concerns on the source and storage of the large datasets used for AI. The rise of the Deepfake image and the manipulation of the human voice is also a concern because of the spread of misinformation. The wide-ranging effects of these risks mean that they can only be dealt with by a diverse range of stakeholders, including computer scientists, law makers, governments, and industry leaders. There are also incidental risks of AI in business, for example the increasing adoption of AI based systems may increase unemployment across a range of sectors and workforce demographics. (Pearson, 2024)

Task

You have been given the task of conducting research related to artificial Intelligence and its impact on the field of computing, a chosen organization or an industry. Your task is to conduct research with appropriate evidence by following suitable research philosophies and methods available. You then must evaluate the outcomes of the research project and suggest improvements to the chosen industry, organization or to the field of computing in general.

Choosing a research objective/question

You can choose your own research topic for this unit. Strong research projects are those with clear, well focused and defined objectives. A central skill in selecting a research objective is the ability to select a suitable and focused research objective. One of the best ways to do this is to put it in the form of a question.

The range of topics discussed on Artificial Intelligence could cover the following areas:

- Ethical implications in AI development
- Transparency in AI models
- AI applications
- AI perspectives on governance and regulations
- Bias and fairness in AI algorithms
- AI in cybersecurity; threats and countermeasures

Activities

Activity1 – Project Proposal

Project Proposal should cover the following areas.

1. Definition of research problem or question.
2. Provide a literature review giving the background and conceptualisation of the proposed area of study. (This would provide existing knowledge and benchmarks by which the data can be judged)
3. Clearly defined research objectives, questions or hypotheses supported by the reviewed literature.
4. Critical evaluation of the research process /different research methodologies that can be applied to computing research project by demonstrating an understanding of the pitfalls, ethical issues, and limitations. choose a suitable research methodology and justify the selection based on theoretical/philosophical frameworks.

Activity 2 – Conducting the research and producing the final research paper

The Learner requires to produce a detailed research project report covering following areas

- Conduct primary and secondary research using appropriate methods for a computing research project that considers costs, access and ethical issues.
- Carry out your research and apply appropriate analytical tools to analyse research findings and data.
- Draw conclusion based on the research findings.
- Communicate the outcomes of your research project to the identified audience. Analyze and discuss merits, limitations and pitfalls experienced in research methodologies and data collections methods applied considering to which extent the project objectives were met.
- Evaluate the project outcomes to suggest justified recommended actions leading to future improvements and future research consideration.
- Reflect on the success of your research project and your performance at the end of the project by evaluating the research process followed and the methodologies applied. Consider alternative research methodologies and lessons learnt in view of the project outcomes.

Activity 3 – formal Presentation on Research findings

Communicate the findings of your research project to identified stakeholders in a 10–15-minute formal presentation. The presentation should cover,

- A brief introduction to the research area, objectives, hypotheses/research questions.
- Analysis of the research and data collection methodologies followed.
- A summary of the research findings
- Conclusion and summary of the recommended actions
- Future improvements and research considerations.

Recommended resources

Textbooks

- Cornford, T., Smithson S. (2005) *Project Research in Information Systems: A Student's Guide*. Paperback. Palgrave Macmillan.
- Costley, C., Elliott, G. and Gibbs, P. (2010) *Doing Work Based Research: Approaches to Enquiry for Insider-researchers*. London: SAGE.
- Fink, A. (2020) *Conducting Research Literature Reviews: From the Internet to Paper*. 5th edn. Sage Publications Inc.
- Flick, U. (2020) *Introducing Research Methodology: A Beginner's Guide to Doing a Research Project*. London: Sage Publications Ltd.
- Gray, D.E. (2009) *Doing Research in the Real World*. 2nd edn. London: SAGE.
- Saunders, M., Lewis, P. and Thornhill, A. (2012) *Research Methods for Business Students*. 6th edn. Harlow: Pearson.
- Wellington, J. (2000) *Educational Research: Contemporary Issues and Practical Approaches*. Continuum International Publishing Group Ltd.

Journals

- International Journal of Quantitative and Qualitative Research*
- Qualitative Research*

Links

This unit links to the following related units:

- Unit 3: Professional Practice*
- Unit 6: Planning a Computing Project (Pearson-set)*
- Unit 7: Software Development Lifecycles.*

Useful links

Resource Number	Type of Resource	Resource Titles	Links
1	Article	artificial intelligence (AI)	https://www.techtarget.com/searchenterpriseai/definition/AI-Artificial-Intelligence
2	Article	The Ultimate Guide to Understanding and Using AI Models (2024)	https://viso.ai/deep-learning/ml-ai-models/
3	Blog	What is Labeled Data?	https://www.datacamp.com/blog/what-is-labeled-data
4	Article	A Complete Guide to Data Labeling for AI	https://levity.ai/blog/guide-data-labeling-for-ai
5	Article	What Is Computer Vision? Meaning, Examples, and Applications in 2022	https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-computer-vision/
6	Article	What is Pattern Recognition? A Gentle Introduction (2024)	https://viso.ai/deep-learning/pattern-recognition/

7	Blog	Artificial Intelligence for Big Data Analytics	https://www.vpon.com/en/blogs/artificial-intelligence-for-big-data-analytics/
8	Article	Machine learning, explained	https://mitsloan.mit.edu/ideas-made-to-matter/machine-learning-explained
9	Article	What is natural language processing?	https://www.ibm.com/topics/natural-language-processing
10	Article	A COMPLETE GUIDE TO Natural Language Processing	https://www.deeplearning.ai/resources/natural-language-processing/
11	Article	Artificial Intelligence in Data Science: 5 Definitive Facts	https://hevodata.com/learn/artificial-intelligence-in-data-science/
12	Article	Artificial intelligence: Development, risks and regulation	https://lordslibrary.parliament.uk/artificial-intelligence-development-risks-and-regulation/
13	Blog	The risks of Artificial Intelligence	https://jarnoduursma.com/blog/the-risks-of-artificial-intelligence/
14	Blog	Shedding light on AI bias with real world examples	https://www.ibm.com/blog/shedding-light-on-ai-bias-with-real-world-examples/
15	Report	Increasing Threat of Deepfake Identities	https://www.dhs.gov/sites/default/files/publications/increasing_threats_of_deepfake_identities_0.pdf
16	Journal	The impact of artificial intelligence on human society and bioethics	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7605294/
17	Article	How artificial intelligence is transforming the world	https://www.brookings.edu/articles/how-artificial-intelligence-is-transforming-the-world/
18	Journal	Large AI Models in Health Informatics: Applications, Challenges, and the Future	https://ieeexplore.ieee.org/abstract/document/10261199
19	Journal	Situating methods in the magic of Big Data and AI	https://www.tandfonline.com/doi/abs/10.1080/03637751.2017.1375130
20	Journal	Challenges and opportunities: from big data to knowledge in AI 2.0	https://link.springer.com/article/10.1631/FITEE.1601883

21	Journal	Computer Vision and Image Processing: A Paper Review	https://ijair.id/index.php/ijair/article/view/42
22	Journal	Progress in Neural NLP: Modeling, Learning, and Reasoning	https://www.sciencedirect.com/science/article/pii/S2095809919304928
23	Journal	Taking AI risks seriously: a new assessment model for the AI Act	https://link.springer.com/article/10.1007/s00146-023-01723-z
24	Journal	Confronting the risks of artificial intelligence	https://www.sipotra.it/wp-content/uploads/2019/05/Confronting-the-risks-of-artificial-intelligence.pdf
25	Journal	Governing AI: Understanding the Limits, Possibility, and Risks of AI in an Era of Intelligent Tools and Systems	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3681088
26	Journal	Foundations of AI: The big issues	https://www.sciencedirect.com/science/article/abs/pii/0004370291900480
27	Journal	Formal theories of knowledge in AI and robotics	https://link.springer.com/article/10.1007/BF03037076
28	Book	Principles of Artificial Intelligence	https://link.springer.com/book/9783540113409
29	Report	National AI Strategy	https://assets.publishing.service.gov.uk/media/614db4d1e90e077a2cbdf3c4/National_AI_Strategy_-_PDF_version.pdf
30	Report	Explaining decisions made with AI	https://ico.org.uk/media/about-the-ico/consultations/2616434/explaining-ai-decisions-part-1.pdf
31	Report	Guidelines for secure AI system development	https://www.ncsc.gov.uk/files/Guidelines-for-secure-AI-system-development.pdf
32	Report	GLOBAL OPINIONS AND EXPECTATIONS ABOUT ARTIFICIAL INTELLIGENCE	https://www.ipsos.com/sites/default/files/ct/news/documents/2022-01/Global-opinions-and-expectations-about-AI-2022.pdf
33	Draft AI Act	Artificial intelligence act	https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/698792/EPRS_BRI(2021)698792_EN.pdf

34	Report	Understanding artificial intelligence ethics and safety	https://www.turing.ac.uk/sites/default/files/2019-06/understanding_artificial_intelligence_ethics_and_safety.pdf
35	Presentation	AN INTRODUCTION TO ARTIFICIAL INTELLIGENCE	https://www.uc.edu/content/dam/uc/ce/docs/OLLI/PageContent/ARTIFICIAL_INTELLIGENCEr.pdf
36	YouTube AI Channel	Two Minute Papers	https://www.youtube.com/@TwoMinutePapers
37	YouTube AI Channel	Sentdex	https://www.youtube.com/@sentdex/featured
38	Webinar	Referencing and writing a bibliography	https://youtu.be/gouFefKs1xl
39	Webinar	Free Harvard Citation Generator	https://www.citethisforme.com/citation-generator/harvard
40	Webinar	Structuring extended writing	https://youtu.be/o1NMN98GfQc

Pass	Merit	Distinction
LO1 Examine appropriate research methodologies and approaches as part of the research process		LO1 and LO2 D1 Critically evaluate research methodologies and processes in application to a computing research project to justify chosen research methods and analysis.
P1 Produce a research proposal that clearly defines a research question or hypothesis, supported by a literature review.	M1 Analyse different research approaches and methodology and make justifications for the choice of methods selected based on philosophical/theoretical frameworks.	
P2 Examine appropriate research methods and approaches to primary and secondary research.		
LO2 Conduct and analyse research relevant to a computing research project		
P3 Conduct primary and secondary research using appropriate methods for a computing research project that consider costs, access and ethical issues.	M2 Discuss merits, limitations and pitfalls of approaches to data collection and analysis.	
P4 Apply appropriate analytical tools, analyse research findings and data.		

Pass	Merit	Distinction
LO3 Communicate the outcomes of a research project to identified stakeholders		D2 Evaluate outcomes and make valid, justified recommendations.
P5 Communicate research outcomes in an appropriate manner for the intended audience.	M3 Analyse the extent to which outcomes meet set research objectives and communicate judgements effectively for the intended audience	
LO4 Reflect on the application of research methodologies and concepts		D3 Demonstrate reflection and engagement in the resource process, leading to recommended actions for future improvement.
P6 Discuss the effectiveness of research methods applied, for meeting objectives of the computing research project. P7 Discuss alternative research methodologies and lessons learnt in view of the outcomes.	M4 Analyse results in recommended actions for improvements and future research considerations.	

Appendix 1 – Project proposal template

Research Proposal Form

Student Name	Nethmi Thiloka Divyanjalee		
Student number	E170909	Date	05/05/2025
Centre Name	Kandy		
Unit	16		
Tutor	Ms. Sasini		
Proposed title			
Evaluating Duolingo in Enhancing the Vocabulary Expansion in English Speaking Skills among Diploma Students.			

Section One: Title, objective, responsibilities
<i>Title or working title of research project (in the form of a question, objective or hypothesis): Research project objectives (e.g. what is the question you want to answer? What do you want to learn how to do? What do you want to find out?): Introduction, Objective, Sub Objective(s), Research Questions and/or Hypothesis</i>
General Objective – Evaluating Duolingo application in Enhancing the vocabulary expansion in English speaking skills of diploma students by analyzing its impact and propose a conceptual framework for speaking English. Sub Objectives - • To analyze the effectiveness of Duolingo in improving the vocabulary expansion of diploma students. • To identify the challenges faced by diploma students while Duolingo for vocabulary development. • To compare the vocabulary enhancement of students using Duolingo with those using traditional learning methods by identifying the before and after effects of using Duolingo application. • To propose a conceptual framework based on the findings to optimize the use of Duolingo for vocabulary and speaking skill development.

Section Two: Reasons for choosing this research project
<i>Reasons for choosing the project (e.g. links to other subjects you are studying, personal interest, future plans, knowledge/skills you want to improve, why the topic is important): Motivation, Research gap</i>

This research evaluates the effectiveness of the Duolingo app for improving vocabulary development and English-speaking proficiency in diploma students. It address the challenge of weak vocabulary and insufficient fluency caused by traditional classroom practices. Through the analysis of Duolingo’s characteristics such as instant feedback, gamification, and adaptive learning, the research investigates the capacity of AI-facilitated tools in augmenting pedagogy in classrooms, It is directed at enhancing a more efficient, technology-based model for non-native English language learning at the tertiary level.

Section Three: Literature sources searched

Use of key literature sources to support your objective, Sub Objective, research question and/or hypothesis: Can include the Conceptual Framework

(Geoffrey T.LaFlair, 2022) **Invalid source specified. Invalid source specified.Invalid source specified.Invalid source specified.** (Zixi Li, 2023) **Invalid source specified.** (Zixi Li, 2023) (Herrera Paredes 2025) (İbrahim Yaşar Kazu, 2025) **Invalid source specified. Invalid source specified. Invalid source specified. Invalid source specified.**

Section Four: Activities and timescales

Activities to be carried out during the research project (e.g. research, development, analysis of ideas, writing, data collection, numerical analysis, tutor meetings, production of final outcome, evaluation, writing the report) and How long this will take:

Milestone	Propose completion data
To analyze the effectiveness of Duolingo in improving the vocabulary expansion of diploma students.	30/05/2025
To identify the challenges faced by diploma students while using Duolingo for vocabulary development.	15/06/2025
To compare the vocabulary enhancement of students using Duolingo with those using traditional learning methods by identifying the before and aftereffects of using Duolingo application.	05/08/2025
To propose a conceptual framework based on the findings to optimize the use of Duolingo for vocabulary and speaking skill development.	25/08/2025

Section Five: Research approach and methodologies

Type of research approach and methodologies you are likely to use, and reasons for your choice: What your areas of research will cover: Research Onion; Sample Strategy/Method; Sample Size

I assessed how Duolingo helps diploma students' vocabulary in English speaking using a multi-method study strategy that used quantitative and qualitative techniques. This method, which focusses on elements like student motivation, teacher tactics, AI audio playback, and immediate feedback, assists me in gathering both quantifiable data and subjective observations. I used deductive and inductive thinking, adopting positivist and interpretivist ideologies based on Saunders' Research Onion. To ensure impartial and equitable representation for the study, I used simple random sampling to choose 55 students from a batch of 47 at the ESOFT Metro Campus.

01.Reserch Philosophy – Positivism

02.Reserch Approach – Deductive

03.Reserch Strategy – Survey

04.Time Horizon – Longitudinal

05. Techniques and procedure – Questionnaire

Comments and agreement from tutor

Comments (optional):

I confirm that the project is not work which has been or will be submitted for another qualification and is appropriate.

Agreed	Yes ☒ No ☐	Name	Nethmi Thiloka Divyanjalee	Date	03/05/2025
--------	---	------	----------------------------	------	------------

Comments and agreement from project proposal checker (if applicable)

Comments (optional):

I confirm that the project is appropriate.

Agreed	Yes ☒ No ☐	Name	Nethmi Thiloka Divyanjalee	Date	03/05/2025
--------	---	------	----------------------------	------	------------

Research Ethics Approval Form

All students conducting research activity that involves human participants or the use of data collected from human participants are required to gain ethical approval before commencing their research. Please answer all relevant questions and note that your form may be returned if incomplete.

Section 1: Basic Details

Project title:	Evaluating Duolingo in Enhancing the vocabulary expansion in English speaking skills among Diploma students.
Student name:	Nethmi Thiloka Divyanjalee
Student ID number:	E170909
Programme:	Computing Research Project
School:	ESOFT Metro Campus

Intended research start date:	03/04/2025
Intended research end date:	21/07/2025
Section 2: Project Summary	
<i>Please select all research methods that you plan to use as part of your project</i> • Interviews: ☐ • Questionnaires: ☒ • Observations: ☐ • Use of Personal Records: ☐ • Data Analysis: ☐ • Action Research: ☐ • Focus Groups: ☐ • Other (please specify): ☐	
Section 3: Participants	
<i>Please answer the following questions, giving full details where necessary.</i> Will your research involve human participants? Who are the participants? Tick all that apply: Age 12-16 ☐ Young People aged 17–18 ☐ Adults ☒ How will participants be recruited (identified and approached)? <i>Using simple random sampling techniques and selecting diploma students in ESOFT Metro Campus in 47 Batch</i> Describe the processes you will use to inform participants about what you are doing: <i>Inform the students about this research.</i>	
Studies involving questionnaires:	
Will participants be given the option of omitting questions they do not wish to answer? Yes ☐ No ☒ If “NO” please explain why below and ensure that you cover any ethical issues arising from this. <i>Because can't get 100% real data.</i>	
Studies involving observation:	
Confirm whether participants will be asked for their informed consent to be observed. Yes ☒ No ☐ Will you debrief participants at the end of their participation (i.e. give them a brief explanation of the study)?	

Yes ☒ No ☐

Will participants be given information about the findings of your study? (This could be a brief summary of your findings in general)

Yes ☒ No ☐

Section 4: Data Storage and Security

Confirm that all personal data will be stored and processed in compliance with the Data Protection Act (1998)

Yes ☒ No ☐

Who will have access to the data and personal information?

Only me

During the research:

Where will the data be stored?

In the laptop

Will mobile devices such as USB storage and laptops be used?

Yes ☒ No ☐

If “YES”, please provide further details: Anyone can't get those data because I put password for my laptop.

After the research:

Where will the data be stored?

In my laptop

How long will the data and records be kept for and in what format?

Only end this research

Will data be kept for use by other researchers?

Yes ☐ No ☒

If “YES”, please provide further details:

Section 5: Ethical Issues

Are there any particular features of your proposed work which may raise ethical concerns? If so, please outline how you will deal with these: Get permission for coordinator

Section 6: Declaration

I have read, understood and will abide by the institution's Research and Ethics Policy:

Yes ☒ No ☐

I have discussed the ethical issues relating to my research with my Unit Tutor:

Yes ☒ No ☐

I confirm that to the best of my knowledge:

The above information is correct and that this is a full description of the ethics issues that may arise in the course of my research.

Name:	Nethmi Thiloka Divyanjalee
Date:	05/05/2025
Please submit your completed form to: ESOFT Learning Management System (ELMS)	

Appendix 2 – Research paper template

**Evaluating Duolingo in Enhancing the Vocabulary Expansion in
English Speaking Skills among Diploma Students**

By

Nethmi Thiloka Divyanjalee

E170909

Submitted in accordance with the requirements for the
**COMPUTING RESEARCH PROJECT MODULE OF PEARSON'S HND
IN SOFTWARE ENGINEERING PROGRAMME**

at the
ESOFT METRO CAMPUS

Name of research Tutor: Ms. Sasini

21/07/2025

DECLARATION

Name of Research Candidate: Nethmi Thiloka Divyanjalee

Pearson Registration Number: E170909

Program Name: Computing Research Project

Research Title: Evaluating Duolingo in Enhancing the Vocabulary Expansion in English Speaking Skills among Diploma Students

Field of Study: Education

I do solemnly and sincerely declare that:

- i. I'm the sole author of this study
- ii. This work is original
- iii. In case of any use if any information from other sources references of copyright with its ownership have been acknowledged in this document
- iv. I do not have any actual knowledge nor do I ought reasonably to know that the making of the work constitutes an infringement of any copyright work
- v. I know that plagiarism is a punishable offence because it constitutes theft, I understand the plagiarism and copying policy of the Edexcel UK, I know what the consequences will be if I plagiaries or copy another's work in this research for this program.

Candidate Signature: 

Date: 21/07/2025

Subscribed and solemnly declared before,

Supervisor's Name: MS. Sasini

Designation:

Supervisor's Signature:

Date:

ACKNOWLEDGMENT

I would like to express my sincere gratitude to everyone who supported me throughout the completion of this research project.

First and foremost, I extend my heartfelt thanks to my lecturer, Mr. Sasini, for their valuable guidance, encouragement, and constructive feedback throughout the course of this study. Their continuous support has been instrumental in the successful completion of this research.

A special thanks goes to ESOF Metro Campus for providing the resources and facilities necessary for carrying out this research as part of my Higher National Diploma (HND) course.

Finally, I would like to acknowledge the unwavering support of my family and friends, whose motivation and understanding have been a constant source of strength.

Thank You,
Nethmi Thiloka Divyanjalee,
E170909,
ESOF Metro Campus,
Kandy

ABSTRACT

This study examines how well diploma-level students' vocabulary acquisition and English-speaking abilities are improved by the Duolingo program. The study looks at how Duolingo, an AI-based language learning program, overcomes the drawbacks of conventional classroom methods such as rote learning, little interaction, and delayed feedback through gamification, quick feedback, and adaptive learning. A longitudinal design and quantitative methodology were used, along with a positive philosophy and a deductive research strategy. Data was gathered from a sample of 55 diploma students at the ESOF Metro Campus using questionnaires. The study investigates the effects of four independent variables on vocabulary development, integrated audio playback, teacher instructional method, student motivation, and rapid feedback. Hypotheses were validated using statistical analysis, which included regression analysis, correlation techniques, and t-tests. When compared to conventional approaches, the results show that Duolingo dramatically increases vocabulary acquisition and student engagement. To better incorporate Duolingo into official educational contexts, a conceptual framework was put forward. The study ends with suggestions for curriculum designers and teachers to use AI-powered programs like Duolingo to help tertiary students improve their speaking fluency and vocabulary.

CONTENTS

Research Ethics Approval Form.....	22
DECLARATION	i
ACKNOWLEDGMENT	ii
ABSTRACT	iii
CONTENTS	iv
LIST OF ABBREVIATIONS	ix
A LIST OF TABLES.....	x
LIST OF FIGURES.....	xi
CHAPTER 1 – INTRODUCTION	1
1.1. Introduction	1
1.2. Problem Statement.....	2
1.3. Purpose of research	4
1.4. Significance of the Research.....	5
1.5. Research objectives	6
1.6. Research Sub objectives.....	6
1.7. Research questions	6
1.8. Hypothesis.....	7
1.9. Limitation of the Study	8
1.10. Chapter Summery.....	9
CHAPTER 2 - LITERATURE REVIEW	10
2.1. Literature Review	10
2.2. Enhancing the Vocabulary expansion in English speaking skills	13
2.3. Research Gap.....	13
2.4. Conceptual framework	13
2.5. Chapter Summery.....	14

CHAPTER 3 - METHODOLOGY	15
2.1. Introduction to Research Methodology	15
3.	15
2.2. Research Onion	15
3.2. Research philosophy	16
3.2.1. Justification for chosen research philosophy	16
3.3. Research approach	18
3.3.1. Justification for chosen research approach.....	18
3.4. Research strategy	18
3.4.1. Justification for chosen research strategy.....	18
3.5. Research Choice	19
3.5.1. Justification for chosen research choice.....	19
3.6. Time frame.....	20
3.6.1. Justification for chosen time frame	20
3.7. Data collection procedures	21
3.7.1. Type of Data.....	21
3.7.2. Data Collection Method	24
3.7.3. Data Collection and Analyze Tools	25
3.7.4. Questionnaire structure	27
3.7.5. Addressing Practical Considerations	37
3.7.6. Data Storage.....	38
3.8. Target population and sampling.....	38
3.8.1. Sampling Strategy	38
3.8.2. Sample Size	42
3.9. The selection of participants.....	42
3.10. Reliability, Validity, and Generalizability	43

3.11. Ethical issues of the research study	45
CHAPTER 4 - PRESENTATION OF RESULTS	50
4.1. Introduction	50
4.2. Pilot Test Analysis	51
4.3. Discriptive Analysis.....	2
4.4. Reliability Analysis.....	8
4.5. Validity Analysis.....	9
4.6. Normality Test.....	10
4.7. Spearman's Rank Correlation Test.....	18
4.8. Kruskal Wallis Test.....	19
4.9. Ordinal Regression Test	24
4.10. T – Test Result (Before and After)	27
4.11. Evaluating the performance done by the analysis and the research outcomes	29
Descriptive Analysis.....	29
Inferential Statistics	30
Reliability Analysis	30
Validity Analysis	31
Normality Test.....	32
Kruskal–Wallis Test	33
Ordinal Regression Test	35
T – Test.....	36
CHAPTER 5 - CONCLUSIONS AND RECOMMENDATIONS	37
5.1. Hypothesis acceptance and rejection.....	37
5.2. Updated Conceptual Framework	40
5.3. Overall outcomes and recommendations	40
5.4. Discussion Evaluation of research methodologies.....	42

Research Philosophy – Positivism	43
What went well.....	43
What went wrong	43
Research Approach – Deductive	44
What went well.....	44
What went wrong	45
Research Strategy – Survey.....	46
What went well.....	46
What went wrong	47
Research Choice – Multi-Method	47
What went well.....	48
What went wrong	48
Time Horizon – Longitudinal.....	49
What went well.....	49
What went wrong	50
Data Collection and Gathering	51
Data Collection Methods – Questionnaires.....	51
What went well.....	51
What went wrong	52
Data Sampling	53
Sampling Method – Simple Random Sampling.....	53
What went well.....	53
What went wrong	54
Data Analysis Techniques	55
Data Analysis	55
What went well.....	55

What went wrong	56
Effectiveness of Methods in Meeting Research Objectives	57
What Worked	57
What Could Be Improved.....	58
5.5. Limitation.....	58
5.6. Recommendation.....	61
5.7. Lesson Learned	63
References	68
Annexures	73
Annexures A: Presentation Slides	73

LIST OF ABBREVIATIONS

A LIST OF TABLES

Table 1 - Challenge in Traditional Approach.....	3
Table 2 -How AI can Address those challenges	4
Table 3 - Hypothesis	8
Table 4 - Conceptual Framework	14
Table 5 – Deta Collection Methods	25
Table 6- Mean, Standard Deviation and Median of each variable	37
Table 7 - Samplin techniques comparison	40
Table 8– Sample Size.....	42
Table 9 - Summary of the Overall Hypothesis selection	40

LIST OF FIGURES

Figure 1 - Conceptual Framework	13
Figure 2- Research onion (Saunders et al., 2019, p. 108).	15
Figure 3 - Survey Introduction	27
Figure 4 - Participation and Descriptive Questions	28
Figure 5 - Factor 1 Questions	29
Figure 6 - Factor 1 Questions	29
Figure 7 - Factor 2 Questions	30
Figure 8 - Factor 2 Questions	30
Figure 9 - Factor 3 Questions	31
Figure 10 - Factor 3 Questions	31
Figure 11 - Factor 4 Questions	32
Figure 12 - Factor 4 Questions	32
Figure 13 - Dependent Variable Questions	33
Figure 14 - Dependent Variable Questions	33
Figure 15 - Additional Questions (Open - ended)	34
Figure 16 - Pilot Study Questioners	52
Figure 17- Pilot Study Questioners	53
Figure 18 - Pilot Study Questioners	53
Figure 19 - Pilot Study Feedback	1
Figure 20 - Pie Chart of the representation of Age Groups of the Participants	2
Figure 21 - Pie chart of the Gender Distribution of Respondents	3
Figure 22 - The bar graph representation of the Current Educational Level of Participants	4
Figure 23 - The bar graph representation of the English Proficiency Level of Participants	4
Figure 24 - Previous Usage of Duolingo for Learning English Among Participants	5
Figure 25 - Crosstabulation of the English Proficiency Level of the Participants with their Age	6
Figure 26 - Crosstabulation s	7
Figure 27 - Representation of the Research's Reliability	8
Figure 28 - Representation of the Research's Validity	9

Figure 29 - Representation of the Dependent Variable's Skewness	11
Figure 30 - Representation of the Dependent Variable's Normality	11
Figure 31 - Interpreting the Bell Curve in dependent Variable	12
Figure 32 - Representation of the Independent Variable's (Factor 1) Skewness	12
Figure 33 - Representation of the Independent Variable's (Factor 1) Normality	13
Figure 34 - Interpreting the Bell Curve in Student Motivation and Interests	13
Figure 35 - Representation of the Independent Variable's (Factor 2) Skewness	14
Figure 36 - Representation of the Independent Variable's (Factor 2) Normality	14
Figure 37 - Interpreting the Bell Curve in Teacher's Instructional Strategy	15
Figure 38 - Representation of the Independent Variable's (Factor 3) Skewness	15
Figure 39- Representation of the Independent Variable's (Factor 3) Normality	16
Figure 40 - Interpreting the Bell Curve in Integrated Audio Playback in the AI tool	16
Figure 41 - Representation of the Independent Variable's (Factor 4) Skewness	17
Figure 42 - Representation of the Independent Variable's (Factor 4) Normality	17
Figure 43 - Interpreting the Bell Curve in Instant Feedback given by the AI tool	18
Figure 44- Kruskal Test for Factor 1	20
Figure 45 Kruskal Test for Factor 2	21
Figure 46- Kruskal Test for Factor 3	22
Figure 47- Kruskal Test for Factor 4	23
Figure 48 – Ordinal Regression Test for all factors	25
Figure 49 – Pseudo R – Square	26
Figure 50 – T -Test Result Before and After	28
Figure 51 -- Updated Conceptual Framework	40
Figure 52 - Presentation Slide 1	73
Figure 53 - Presentation Slide 2	74
Figure 54 - Presentation Slide 3	75
Figure 55 - Presentation Slide 4	76
Figure 56 - Presentation Slide 5	77
Figure 57 - Presentation Slide 6	78
Figure 58 - Presentation Slide 7	79
Figure 59 - Presentation Slide 8	80
Figure 60 - Presentation Slide 9	81
Figure 61 - Presentation Slide 10	82

Figure 62 - Presentation Slide 11	83
Figure 63 - Presentation Slide 12	84
Figure 64 – Presentation Slide 13	85
Figure 65 - Presentation Slide 14	86
Figure 66 - Presentation Slide 15	87
Figure 67 - Presentation Slide 16	88
Figure 68 - Presentation Slide 17	89
Figure 69 - Presentation Slide 18	90
Figure 70 - Presentation Slide 19	91
Figure 71 - Presentation Slide 20	92
Figure 72 - Presentation Slide 21	93
Figure 73 - Presentation Slide 22	94

CHAPTER 1 – INTRODUCTION

1.1.Introduction

In today's age of globalization, mastery of English-speaking skill is essential to achieve education and professional success (Vorbel, 2022). Development of vocabulary helps in enhancing fluency and effective communication (Susanto, 2020). Technology-based tools are now trending to support language acquisition beyond traditional classrooms (Chik, 2019). Among the tools, Duolingo is very popular because of its gamification and adaptive quality of learning (Zixi Li, 2023). Research has proven Duolingo exerts a beneficial impact on the acquisition of languages in listening, speaking, reading, and writing skills (İbrahim Yaşar Kuzu, 2025).

Specifically, Duolingo provides real-time feedback, which is indispensable in developing the competence of speaking (Xiangying Jiang, 2021). According to (Alan Jaelani, 2020), students who learned with Duolingo achieved A2-level speaking after CEFR guidelines. The elements of gamification in Duolingo promote inborn motivation as well as extended participation among students (Mairin Hennebry-Leung, 2024). Past research has supported Duolingo's usefulness in improving vocabulary learning among secondary school's students (Rosario, 2025). Nevertheless, empirical research specifically targeting diploma students' vocabulary learning through Duolingo is scarce.

Motivation and interest among students greatly contribute to their achievement in learning via mobile apps (Jeon, 2022). In addition, integrated audio playing functions in Duolingo enhance learners' pronunciation and speaking proficiency (Geoffrey T.LaFlair, 2022). Instant error correction mechanisms incorporate in Duolingo enable learners to detect and correct mistakes quickly (Xiangying Jiang, 2021). Many researchers believe that Duolingo may not suffice on its own for the learning of advanced speech skills (Shawn Loewen, 2019). Nevertheless, supplemented with instructor supervision, Duolingo can become highly effective at improving vocabulary knowledge and application (Son T.H. Pham, 2022).

This study aims to evaluate the impact of Duolingo on English speaking vocabulary enlargement among diploma students in ESOFT Metro Campus. It also tries to propose a conceptual framework based on the findings to increase English-speaking proficiency. Through motivation, pedagogy, embedded audio elements, and feedback systems analysis, this study will present rich insights. Building on previous research, this study fills the

research gap among the diploma student population that has been observed to date. Lastly, it aims to contribute to the development of more effective, technology-driven language learning techniques.

1.2.Problem Statement

Within the rapidly evolving education context, incorporation of technology into foreign language acquisition has become indispensable. Traditional classroom activities, often being textbook and memorization centered, have been found wanting to facilitate dynamic vocabulary development and oral proficiency in learners. Despite increased needs for competent English communication, Diploma students are frequently impaired by vocabulary construction due to repetitiveness of instructional approaches and a lack of effective stimulating practice (Thornbury, 2002). As mobile-assisted language learning (MALL) platform like Duolingo become more popular, new opportunities have emerged for building English language competencies through interactive, gamified experiences (John Grego, 2022).

The comparison with conventional learning approaches and Duolingo integration reveals sharp contrasts. Conventional learning is centered on textbook based, lecture type instruction, whereas Duolingo provides a gamified and interactive environment through mobile applications (Thornbury, 2002) , (Ajisoko, 2022). Learning by students in conventional classrooms is passive and leads at times to boredom and indifference. In contrast to this, Duolingo provides a highly interactive experience in the guise of rewards, tracking of progress, and daily quizzes that maintain students engaged (Munday, 2015). Accessibility is also a point of differentiation, traditional learning is generally limited to tangible classroom spaces, while Duolingo provides learning opportunities anywhere, anytime, through mobile phones (Blanka Klimove, 2020). Teacher-led and time-lagged feedback are typical in traditional settings, yest Duolingo provides real-time feedback after every exercise to promote quicker adaptation learning (Grego, 2012)

However, their effectiveness in terms of vocabulary improvement for enhanced speaking skills among Diploma students is relatively less explored. Therefore, it is necessary to evaluate the role of Duolingo and propose a conceptual framework that can better integrate such tools for English speaking among these students.

Reference	Challenge	Description
Thornbury (2002)	Repetitive and memorization-focused	Traditional approaches focus on rote learning, resulting in minimal vocabulary retention and use
Thornbury (2002)	Passive learning environment	Student participation and interaction in language practice are decreased by teacher-centered education.
Ajisoko (2022)	Lack of interactive practice	Traditional courses sometimes lose out on possibilities for immediate feedback and real-world language usage.
Munday (2015)	Low student motivation and engagement.	Lectures relying only on textbooks may cause disinterest and poor levels of engagement.
Blanka Klimova (2020)	Restricted access and flexibility.	The time and location of language instruction are restricted by physical classroom environments.
Grego (2012)	Delayed feedback from instructors.	The learning and development process is slowed down while students wait for instructor corrections.

Table 1 - Challenge in Traditional Approach

Traditional language acquisition confronts various problems, including rote memory, passive instruction, restricted contact, poor motivation and delayed feedback. AI technologies like Duolingo effectively solve these difficulties by giving interactive, gamified, and adaptable learning experiences. They provide instant feedback, stimulate active involvement, and keep student interest through individualized and rewarding content, ultimately encouraging higher vocabulary development and speaking skills. Below table have how AI can address my chosen challenges.

Challenge	How AI can Address It
Repetitive and memorization-focused learning	AI-powered solutions offer gamified, adaptive learning that foster engagement and meaningful language use.
Passive learning environment	AI technologies enable interactive tasks, speech recognitions, and active learner interaction through drills.
Lack of interactive practice	Real-time quizzes, voice practice, and AI feedback imitate real-life usage and increase language proficiency.
Low student motivation and engagement	Gamification, immediate rewards, and tailored feedback keep learners motivated and continually engaged.
Delayed feedback from instructor	AI delivers fast corrections and pronunciation feedback, enabling immediate learning and progress.

Table 2 -How AI can Address those challenges

1.3. Purpose of research

The key objective of this research is to evaluate the effectiveness of the Duolingo app in promoting vocabulary development in English Speaking proficiency among diploma students. The selection of this topic for research is because of the growing interest in novel, technology-driven language learning strategies, especially in non-native English-speaking environments where learners tend to face difficulties in the development of a sophisticated and functional vocabulary for effective communication.

This study aims at investigating the specific problem of inadequate vocabulary, which is causing poor English fluency and confidence among diploma students. Traditional classroom activities cannot sufficiently encourage learners or provide frequent practice and feedback for building vocabulary. Through an exploration of how Duolingo makes vocabulary learning easier through features such as instant feedback, gamification, audio embedding, and adaptive learning, the study aims to provide insights on how digital language learning software can supplement conventional instruction and lead to improved speaking proficiency.

Lastly, the study aims to contribute towards the development of a more effective language learning model that incorporates AI-based applications with the aim of enhancing English fluency among tertiary-level learners.

1.4. Significance of the Research

In the modern environment of English language acquisition, particularly for diploma students in non-native English-speaking contexts, the integration of technology has become increasingly pivotal. Traditional pedagogic approaches are often not sufficient to address the changing learner's needs, particularly in vocabulary learning and speaking enhancement. This research seeks to bridge this gap by evaluating the efficacy of Duolingo, a commonly used language-learning application, in improving vocabulary acquisition and English-speaking skills among diploma students.

The significance of this research is highlighted through a series of empirical studies that show the potential of Duolingo for language teaching. (Ajisoko, 2020) conducted pre-experimental research in Tarakan University of Borneo, wherein students who practiced using Duolingo for 30 days revealed noticeable improvement in vocabulary learning. (Rive Febriani, 2023) also did a Classroom Action Research at Selakau Negeri 1 SMA, wherein it indicated that using the app effectively increased the understanding of students towards English verbs and consequently enhanced the language learning general of the students.

In addition, illustrating the proficiency of Duolingo, (Lusi Purnama Sari, 2023) conducted a quasi-experiment in Bandar Lampung, Indonesia, which confirmed that students using Duolingo outperformed peers in vocabulary tests, demonstrating the proficiency of the app in vocabulary learning. Additionally, (Yueh yea Lo, 2025) conducted a study on Thai EFL university students' attitudes and found that Duolingo's interactive features and clearly outlined learning trajectories were key to developing listening and speaking abilities.

The uniqueness of this research is its focus in Diploma Students, a group traditional not well represented in language research. By investigating this group, the research aims to provide insight into how mobile-assisted language learning applications like Duolingo can be refined to meet the specific needs of learners at the diploma level so as to better enhance their vocabulary and speaking capabilities in the English Language.

Lastly, the research stands ready to bring meaningful contributions in language teaching using its provision of data on the efficacy of Duolingo for building vocabulary and spoken

language fluency among diploma students. The research aims to provide guidance for educators and writers of curriculum in relation to using technological solutions towards meeting language deficiency problems, ultimately birthing an innovative and productive of teaching the English language. According to my study, these benefits are expected to close the current gap in efficient language learning techniques at the educational level by helping diploma students acquire more vocabulary and speak more fluently.

1.5. Research objectives

Evaluating Duolingo application in Enhancing the vocabulary expansion in English speaking skills of diploma students by analyzing its impact and propose a conceptual framework for speaking English.

1.6. Research Sub objectives

- To analyze the effectiveness of Duolingo in improving the vocabulary expansion of diploma students.
- To identify the challenges faced by diploma students while using Duolingo for vocabulary development.
- To compare the vocabulary enhancement of students using Duolingo with those using traditional learning methods by identifying the before and after effects of using Duolingo application.
- To propose a conceptual framework based on the findings to optimize the use of Duolingo for vocabulary and speaking skill development.

1.7. Research questions

- How to analyze the effectiveness of Duolingo in improving the vocabulary expansion of diploma students?
- What are the identified challenges faced by diploma students while using Duolingo for vocabulary development?
- How to compare the vocabulary enhancement of students using Duolingo with those using traditional learning methods by identifying the before and after effects of using Duolingo applications?

- How to propose conceptual framework based on the findings to optimize the use of Duolingo for vocabulary and speaking skill development?

1.8. Hypothesis

Factors	Description	Positive Hypothesis	Negative Hypothesis
Student Motivation and Interest	The enthusiasm, curiosity, and willingness of the students to get involved in learning activities.	Student motivation and interest are positively influenced when AI-based tools are employed the vocabulary expansion in English Speaking skills. (H1.1)	Student motivation and interest are negatively influenced when AI-based tools are non-user-friendly and not engaging the vocabulary expansion in English Speaking skills. (H1.0)
Teacher's Instructional Strategy	Methods and manners in which a teacher delivers content and promotes learning.	The teacher's instructional strategy positively enhances student understanding and engagement when integrated with AI tools to the vocabulary expansion in English Speaking skills. (H2.1)	The teacher's instructional strategy negatively impacts students learning when it fails to integrate AI tools effectively to the vocabulary expansion in English Speaking skills. (H2.0)
Integrated Audio Playback in the AI tool	The feature that allows listening to content in audio using the AI-based tool.	Integrated audio playback positively supports diverse learning styles and improves comprehension to the vocabulary expansion in English Speaking skills. (H3.1)	Integrated audio playback negatively affects learning when it is poorly implemented or distracts students to the vocabulary expansion in English Speaking skills. (H3.0)
Instant Feedback given by the AI tool	The immediate feedback provided by	Instant feedback positively enhances	Instant feedback negatively impacts on

	the AI tool in accordance with the inputs or answers of the students.	student learning by helping them correct mistakes in real-time to the vocabulary expansion in English Speaking skills. (H4.1)	students if it is inaccurate, confusing, or overwhelming to the vocabulary expansion in English Speaking skills. (H4.0)
--	---	--	--

Table 3 - Hypothesis

1.9.Limitation of the Study

- **Time Constraints**

The research is constrained by the language learning environment that is available for data collection and analysis. The research may not allow for an extended duration of exposure to Duolingo, and therefore the depth of vocabulary acquisition observed may be compromised. More in-depth findings may be presented if more time was collected the research.

- **Geographical Scope**

The study is confined to the diploma students of ESOF Metro Campus, and thus the findings might not be transferable to other institutions or regions. The specific socio-economic and demographic characteristics of the students of this campus influence the results, and thus the findings are not generalizable to a larger population.

- **Focus on Educational Settings**

This study only focuses on the use of Duolingo in a formal education environment. Therefore, the results might not reflect the performance of the app in non-formal learning environments or with other age groups, e.g., adult learners or pre-diploma level students. Differences in the effectiveness of Duolingo across various learning environments might be present but are not covered in this study.

1.10. Chapter Summery

This chapter a study conducted at ESOF Metro Campus to evaluate the effectiveness of Duolingo in improving English-speaking vocabulary among diploma students. It emphasizes the growing importance of English proficiency and addresses how technology, in this case, Duolingo's gamified and interactive platform, supports language learning. The chapter spotlights a research gap regarding Duolingo's direct impact on learning vocabulary and develops the study's goals, including establishing whether Duolingo is more effective than traditional methods and investigating factors such as student motivation and teaching strategies. It also documents constraints such as time scope limitation, geographical area restrictions, and only focusing on educational settings.

CHAPTER 2 - LITERATURE REVIEW

2.1. Literature Review

The incorporation of Artificial Intelligence (AI) into language learning has transformed the conventional pedagogic methods, providing new tools to further language acquisition.

Duolingo, a leading AI-based language learning website, has been in the spotlight for its ability to enable vocabulary growth and enhance English speaking ability among students.

This literature review aims to examine the role of Duolingo in increasing vocabulary in English speaking ability of diploma students in terms of four factors: Student Motivation and Interest, Teacher's Instructional Strategy, Integrated Audio playback in the AI tool, and Instant Feedback given by the AI tool. Based on the latest research, this review tries to establish the connection between the aforementioned factors and the enhancement of vocabulary in English Speaking.

- **How Student Motivation and Interest Improve Enhancing the Vocabulary expansion in English Speaking Skills**

Student motivation is a primary key for successful language learning, especially when the acquisition of vocabulary is what is sought. Intrinsically motivated students will be more likely to practice the language, research new words' meanings, and be risk-takers for speech activities. According to Lic. Herrera Paredes (2025), interest and motivation are crucial to what level and how fast vocabulary can be acquired. Her study discovered that students who like to use AI tools like Duolingo were more likely to persist in language learning, resulting in better vocabulary retention.

Ebadi and Amini (2022) found that Duolingo's gamified structure (e.g., receiving rewards, maintaining streaks, and unblocking levels) gives students motivation to continue practicing, enabling a greater likelihood of exposure to vocabulary and usage in speech. Menggo (2019) linked interest in students to enhanced speaking skill, noting that high motivation is connected to enhanced word retrieval and use in speech.

Carpio Canada et al. (2015) and Zhou and Wei (2018) confirmed that interactive tools like vocabulary quizzes and visual study materials activate learners, while Stockwell and Hubbard (2014) emphasized goal-setting in Mobile Assisted Language Learning (MALL) as a way of sustaining learner interests and focus. These findings in totality prove that student motivation through interactive platforms intrinsically enhances vocabulary application in speech.

- **How Teacher's Instructional Strategy Improve Enhancing the Vocabulary expansion in English Speaking Skills**

The teaching approach of the instructor is one of the major drivers of vocabulary learning, especially when synchronized with the application of technology. Teachers who integrate tool like Duolingo into their courses typically use flipped classrooms, blended learning, or formative assessments models that ensure constant exposure and use of vocabulary. In Lusi Purnama Sari et al.'s (2023) research, it was found that when teachers use Duolingo strategically (e.g., assigning homework or monitoring progress) students become more responsible, improving their use of vocabulary in speaking exercises.

Tomlinson (2018) pointed out that effective teachers provide scaffolding of vocabulary use, where they model for appropriate use, allow communicative activities to be set up, and provide contextual reinforcement. These can be supplemented with AI apps providing additional practice.

Students who received strategic teaching with online practice showed greater vocabulary acquisition and oral fluency based on a study by Nguyen and Boers (2019). Similarly, Richards and Rodgers (2014) clarified that a mix of communicative language teaching and app practice encourages learners to employ target vocabulary actively in context.

Moreover, Wang (2020) and Sadeghi and Farzizadeh (2017) discovered that vocabulary development is more significant where teaching focuses on communicative meaning and personalization, both of which are made possible with the use of AI programs like Duolingo.

- **How Integrated Audio Playback in the AI tool Improve Enhancing the Vocabulary expansion in English Speaking Skills**

Concurrent audio listening allows students to hear and imitate the correct intonation and pronunciation of words in the audio, supporting auditory memory and pronunciation abilities, which constitute a significant component of speaking proficiency. The research by Xiangying Jiang et al. (2023) emphasized that learners develop phonological awareness and improved articulation skills when they listen to and imitate the audio content.

According to Morley (2016), listening to words spoken in native pronunciation helps learners internalize words and reduce hesitation when speaking. Pisoni and Lively (2017) further added that repeated exposure to high-quality sound enhances word recognition and automatic speech retrieval.

In Duolingo, there is inherent sound that recycles visual and text learning of words by reinforcing vocabulary through aural input, constructing a multi-modal learning setting. Derwing and Munro (2015) conducted research which confirmed that listeners exposed to playback recorded sound showed enhanced recall and precise pronunciation. In addition, Zhang and Graham (2019) established that audio-visual aid learning makes learning how to connect meaning to word and sound easier, making vocabulary deployment in discourse smoother and precise.

Finally, a study by Birney and Cooper (2020) proved that students who consistently listened and echoed words through AI software increased their confidence levels and reduced reliance on native equivalents when generating English.

- **How Instant Feedback given by the AI tool Improve Enhancing the Vocabulary expansion in English Speaking Skills**

Instant feedback is essential in language learning as it enables students to immediately correct errors, consolidate correct usage, and adjust tactics in real time. Duolingo provides immediate feedback on pronunciation, grammar, and usage vocabulary, making it easier to learn vocabulary faster and more accurately for speaking.

Windya et al. (2023) found that students can identify lexical knowledge gaps as well as correct themselves through immediate feedback, which offers speaking confidence building. In Ellis (2016), students who received immediate feedback correctly showed improved lexical choices in speech activities.

Furthermore, according to Hattie and Timperley (2017), prompt feedback plays an important role in student achievement, especially in the learning of vocabulary that needs frequent retrieval and use. Luoma (2018) emphasized that AI feedback supports individualized learning through the emphasis on personal errors, which increases recall of vocabulary. A study by Long and Robinson (2015) revealed that feedback from intelligent systems leads to more autonomous learners who are better able to monitor their own vocabulary usage. Similarly, Van der Kleij et al. (2018) confirmed that computer feedback promotes metacognitive awareness and allows learners to use vocabulary consciously in speech.

2.2.Enhancing the Vocabulary expansion in English speaking skills

Various studies have examined the impact of AI tools on language learning outcomes. Dearestiani et al.(2023) conducted a study with 31 eighth grade students and observed that Duolingo significantly improved their speaking ability. Qiao and Zhao (2023) explored the effectiveness of AI-powered training on L2 speaking proficiency among 93 Chinese EFL learners and reported remarkable improvement in speaking ability. These studies reveal the positive impacts of AI technology like Duolingo on the development of vocabulary and fluency among learners of language.

2.3.Research Gap

Although current research has pointed to the advantages of AI applications in language learning, no research studies specifically target diploma students and the simultaneous effect of student motivation, instructional strategies, built-in audio playback, and instant feedback on vocabulary growth in English speaking abilities. The majority of the research studies have targeted secondary school or university students, hence creating a void in establishing the effectiveness of Duolingo for diploma students. There is also a need for a through study that examines the relationship of these factors towards enhancing vocabulary in speaking situations.

2.4. Conceptual framework

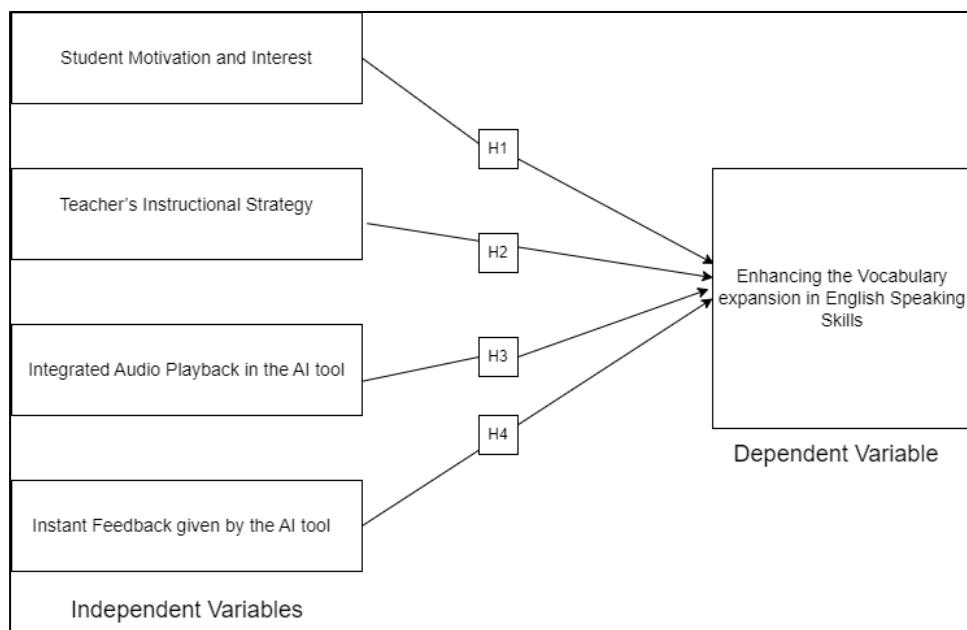


Figure 1 - Conceptual Framework

Dependent Variable	Independent Variable	Source	Expected Sign
Enhancing the Vocabulary expansion in English Speaking Skills	Student Motivation and Interests	Herrera Paredes (2025), Menggo (2019), Ebadi and Amini (2022)	+ Positive
	Teacher's Instructional Strategy	Sari et al.(2023), Rassaei (2021), Jiang et al. (2021)	+ Positive
	Integrated Audio Playback in the AI tool	Jiang et al.(2021), Bashori et al.(2021), McCrocklin (2016)	+ Positive
	Instant Feedback given by the AI tool	Windya et al. (2023), Yan (2023), Qiao and Zhao (2023)	+ Positive

Table 4 - Conceptual Framework

2.5.Chapter Summery

This chapter laid out recent research on how one can measure Duolingo effectiveness in enhancing English Speaking ability's vocabulary development in diploma students. It outlined four key independent variables; Student Motivation and Interest, Teacher's Instructional Strategy, Integrated Audio Playback in the AI tool, and Instant Feedback given by the tool. Research findings' revelations testified the aforementioned factors contribute greatly to learning vocabulary and speech competence. Previous studies also suggested how such variables contribute to the success of students learning. The chapter concluded by acknowledging a research gap ; not many studies have focused explicitly on diploma students utilizing Duolingo with all four variables included, so more research filed is necessary.

CHAPTER 3 - METHODOLOGY

2.1.Introduction to Research Methodology

This chapter provides a step-by-step overview of the research technique used to assess the effectiveness of the Duolingo application in improving vocabulary expansion and English-speaking skills among diploma students. The methodology employs the research onion model, which provides a structured framework made up of numerous layers such as research philosophy, approach, methodological choice, strategy, time horizon, and data collection methodologies.

This chapter discusses major methodological factors such as research design, data type, data collection methods, analytical instruments, sampling method, sample size, and participant selection procedure. Each methodological decision was made to ensure that the results were valid, trustworthy, and consistent with the study's aims. The systematic and transparent nature of this methodology contributes to the overall credibility and replicability of the research findings.

2.2. Research Onion

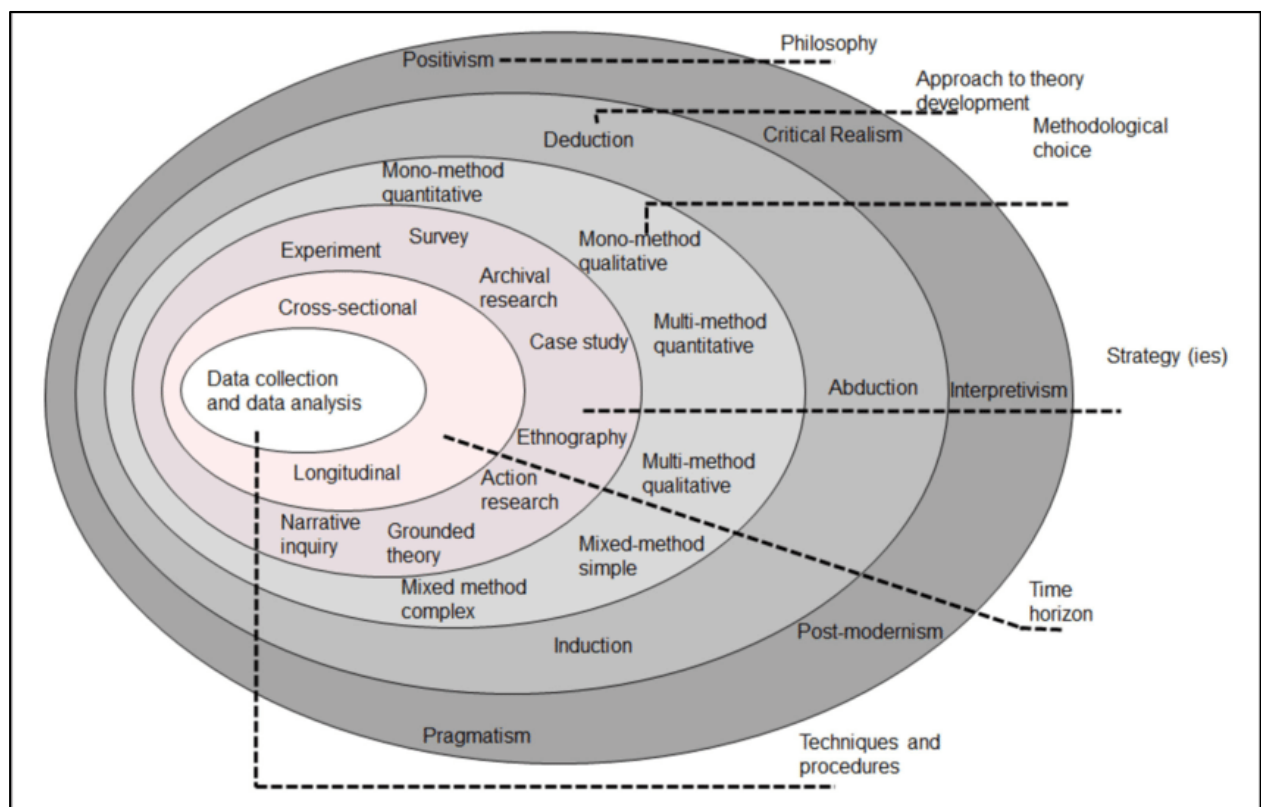


Figure 2- Research onion (Saunders et al., 2019, p. 108).

Saunders, Lewis, and Thornhill (2007) developed the research onion, a conceptual framework that helps researchers create a solid research technique. (Tengli, 2020) Similar to peeling an onion, it offers a tried approach in which each layer denotes a stage of the study design process, ranging from general philosophical concerns to particular data gathering methods. (Mark N K Saunders, 2012)

Layers of the Research onion

The first layer is Research Philosophy. The basic assumption regarding the nature of reality and knowledge are addressed by this topmost layer. Pragmatism, positivism, interpretivism, and realism are common philosophical perspectives (Saunders et al., 2012). Second layer is Research approach. This layer includes the rationale for the study. The three methods are abductive, inductive, and deductive. (Saunders et al., 2019). The third one is Methodological Choice. Here, researchers decide on using quantitative, quantitative, or mixed methods, influencing how data will be collected and analyzed (Mark N K Saunders, 2012). The fourth Layer is Research Strategy. This layer entails deciding on a research methodology, such as surveys, case studies, ethnography, or experiments (Saunders et al., 2007). The fifth layer is Time Horizon. Depending on the goals of the study, researchers might select between across time or cross sectional investigations. (Mark N K Saunders, 2012). The last layer is Techniques and Procedures. The precise techniques for gathering and analyzing data, like questionnaires, interviews, and statical analysis, are the main emphasis of the onion's core (Mark N K Saunders, 2012).

3.2. Research philosophy

3.2.1. Justification for chosen research philosophy

This study uses a positivist research ethic since it is most suited to objectively analysing how Duolingo improves vocabulary expansion and English speaking skills among diploma students. Positivism presumes that reality exists independently and can be witnessed, measured, and quantified (Saunders et al., 2019). It encourages a scientific approach that obtains knowledge from empirical observation and logical analysis, which is consistent with the nature of this study, which entails collecting quantifiable data using structured approaches such as questionnaires.

Several important elements of positivism make it suitable for this study. To begin, positivism emphasises objectivity, with the researcher maintaining a detached posture to avoid personal prejudice (Bryman, 2016). This is especially important because the study's goal is to quantify increases in vocabulary and speaking ability based on student experiences and Duolingo app usage patterns data that can be collected and analysed objectively. Second, positivism promotes the use of quantitative research methodologies. In this study, a survey was utilised to collect standardised responses from a sample of diploma students, allowing for the measurement of their vocabulary learning experiences and perceived improvements in English speaking abilities. Third, positivist research is structured and systematic, as evidenced by the use of a formal questionnaire and similar data gathering techniques for all participants.

Furthermore, positivism allows for the evaluation of cause-and-effect linkages, which is critical to this study's goal of establishing if using Duolingo directly leads to greater vocabulary and speaking skills. This study similarly takes a deductive approach, testing theoretical assumptions regarding Duolingo's educational benefits using empirical data acquired from the student sample. By analysing these findings, the study helps to establish a conceptual framework that may be utilised in similar academic settings. According to Hussey and Hussey (2014), such generalisation is a defining feature of positivist research, enhancing its application in broader settings.

For these reasons, the positivist philosophy serves as the foundation for this research, allowing for a rational, realistic, and unbiased assessment of Duolingo's effectiveness in promoting English language growth among diploma-level learners.

3.3. Research approach

3.3.1. Justification for chosen research approach

The research approach used in this study was chosen because it was consistent with the nature and purpose of the study namely, to determine how the use of the Duolingo application promotes vocabulary expansion and improves English speaking skills among diploma students, as well as to recommend a conceptual framework based on student perceptions. This study takes a deductive research approach, starting with an existing theory or hypothesis and progressing to detailed data collecting and analysis to confirm or refute that hypothesis (Hussey & Hussey, 2014). In this situation, the underlying claim is that mobile-assisted language learning systems such as Duolingo can improve vocabulary development and speaking fluency. This hypothesis was tested with structured data obtained from diploma students via a formal survey.

The deductive technique is extremely systematic, making it ideal for producing quantitative findings in research that seek to validate a conceptual model based on empirical discoveries (Saunders et al., 2019). This technique compliments the study's positivist philosophy by emphasising objectivity, measurement, and hypothesis testing (Bryman, 2011). By taking this technique, the study hopes to determine whether student usage patterns and attitudes towards Duolingo are associated with quantitative improvements in vocabulary and speaking skills.

Furthermore, this approach enables the researcher to assess if students' views support the application's effectiveness, which is crucial to meeting the research objectives. A deductive technique was preferable to an inductive one because the study's goal is to evaluate and extend current educational ideas using real-world data rather than create new ones. Finally, the research approach ensures the use of a well-defined framework, enhances reliability, and is consistent with both the overarching study objectives and the positivist research philosophy used in this study.

3.4. Research strategy

3.4.1. Justification for chosen research strategy

The survey method was used for this study because it best fits the characteristics of the deductive approach and is most appropriate for assessing the influence of the Duolingo

program on vocabulary expansion and English-speaking skills among diploma students. The deductive research approach often entails collecting and analyzing quantitative data to test pre-existing hypotheses (Bryman, 2011). In this case, the researcher used structured questionnaires to collect standardized data from a specific group of students, allowing him to test the central hypothesis that using Duolingo improves English language proficiency, particularly vocabulary usage and speaking ability.

This methodology approach aligns with the research goal of assessing student perceptions and experiences with the application. Questionnaires are particularly useful for gathering responses from many participants in a short period of time, which was critical given the sample size of diploma students in this study. Furthermore, the use of closed-ended questions offered structured and comparable data, allowing for statistical analysis and increasing the objectivity of the findings (Hussey & Hussey, 2014).

According to Saunders et al. (2019), surveys are especially valuable when researchers want to generalize their findings to similar contexts, such as other diploma-level educational settings that incorporate mobile learning tools. Because the study's goal is not just to assess the perceived success of Duolingo, but also to establish a conceptual framework that might guide future implementation of such technologies, a questionnaire-based survey was a feasible and appropriate methodological choice.

3.5. Research Choice

3.5.1. Justification for chosen research choice

This study uses a multi-method quantitative approach inside a deductive research framework. This methodology option is well-suited to assessing how the Duolingo application helps diploma students expand their vocabulary and increase their English-speaking skills. A multi-method quantitative approach allows the researcher to collect various types of numerical data, such as structured questionnaire responses and performance indicators, resulting in a more comprehensive and accurate understanding of the relationship between language improvement and student engagement with Duolingo.

Saunders et al. (2019) define a multi-method quantitative design as one in which researchers employ many quantitative data gathering techniques to test hypotheses or validate theoretical assumptions. To collect thorough data, this study used both student questionnaires and word

usage patterns (as assessed by app engagement or self-reported practice frequency). This enabled a more rigorous assessment of the extent to which Duolingo helps with vocabulary learning and improves English speaking abilities.

The use of several data sources improves the reliability and validity of the findings by allowing for triangulation, which delivers broader insights while reducing the possibility of bias from depending on a single data type. This technique is also consistent with the study's goal of developing a conceptual framework based on actual data. According to Hussey and Hussey (2014), a multi-method quantitative approach improves the generalizability of results and facilitates rigorous verification of theoretical assumptions. In this approach, the study adheres to the systematic, objective norms of deductive research while also ensuring that conclusions are supported by a diverse body of quantitative evidence.

3.6. Time frame

3.6.1. Justification for chosen time frame

This study used a longitudinal time frame to assess how using the Duolingo application affects vocabulary enhancement and speaking ability development among diploma students over time. A longitudinal study allows the researcher to collect data at several points, allowing for the examination of trends, patterns, and changes in learners' vocabulary acquisition and spoken English ability as they use the app on a consistent basis. This time-based strategy is ideal for measuring learning development because it allows you to measure both short-term gains and long-term improvement.

According to Saunders et al. (2019), a longitudinal approach is appropriate for studying the development or change of behaviour, attitudes, or performance over time. The goal of this study is not just to measure students' present vocabulary levels, but also to track how their language skills develop as they continue to use Duolingo as a learning tool. This contributes to stronger evidence of the application's efficacy and facilitates the development of a conceptual framework based on observed outcomes. A cross-sectional design, on the other hand, would exclude such a progression analysis. As Bryman (2011) points out, longitudinal studies are widely employed in deductive research to show causal correlations, which is precisely what is needed to determine whether and how Duolingo contributes to student vocabulary expansion and speaking competency

3.7. Data collection procedures

Data collection is a primary task of this research as it directly affects the validity and depth of insight into how effective Duolingo is at supporting vocabulary development in English speaking skills among diploma students. To achieve a balanced and evidence-based perspective, this study will utilize a combination of primary and secondary data collection methods. Primary data will be gathered through the administration of structured questionnaires and interviews to the diploma students who are active users of Duolingo, based on some key student motivation factors, teacher strategies, audio features, and feedback systems. Secondary data will involve the review of related academic literature, research papers, and reports on the use of AI-based language learning applications in language learning. This double-edged approach ensures the collection of empirical data and theoretical information, justifying a solid and comprehensive evaluation of Duolingo's impact on English vocabulary learning.

3.7.1. Type of Data

Primary Data

Primary data refers to data that is collected first-hand by the researcher for the said study. Educational research requires primary data collection to ensure that the information is current and specific from participants. Kabir (2016) defines that methods of primary data collection include survey, interviews, observations, and focus groups discussion, among others. These methods allow researchers to obtain data that can be used directly in their areas of research interest so that the information is properly aligned with the very specific study context.

Why Choose this Data Type

Primary data were chosen in this research to ensure the collection of first-hand, original, and accurate information directly related to the research questions. Unlike secondary data, which may be outdated, generalized, or not necessarily in line with the topic of this research, primary data allow a more in-depth and accurate view of the research problem. It offers the potential of getting answers directly from the participants, thus making the data current, context-specific, and relevant to the specific demographic or subject being studied. Furthermore, the use of primary data offers greater credibility and validity to the research findings, since the researcher has direct control over the way the data is collected, including

the questions asked, the methods employed, and the ethical principles maintained in the process. This tailored approach later boosts the validity and usefulness of the findings derived from the study.

Why choose this Method (Questionnaires)

For this study, questionnaires are selected as the primary data collection method since they are efficient in collecting data from a large sample of respondents, inexpensive, and can collect standardized data. Kabir (2016) states that questionnaires are particularly effective in educational settings where researchers aim to collect data on participants' attitudes, perceptions, and behavior. They offer anonymity, which can provide a more honest and unbiased response, especially in the case of assessing issues like motivation and first-hand experience with language learning software like Duolingo.

Why Reject Other Methods

While there are a number of primary data collection methods available, some are less suitable for this study based on certain constraints. Interviews, for instance, while useful for collecting detailed information, are especially time-consuming. Arranging schedules, locating venues, and meeting personally can be very time-consuming for researchers and respondents. It is especially challenging with large samples, such as in this research on diploma students. (Jorg Hecker, 2024) Moreover, interviews can be deceptively hard to conduct properly, requiring skilled interviewers to avoid biases and ensure consistency. (George, 2022)

Focus group are another possibility but with their own drawbacks. Group dynamics can influence the answers of participants, with strong personalities potentially contaminating the discussion and shy people not contributing. (Jorg Hecker, 2024) This phenomenon, known as groupthink, can lead to a perceived consensus that does not actually exist, thereby invalidating the data collected. (Interviews, 2021)

Observational methods, though useful in certain circumstances, do not always provide a reliable way of recording states such as motivation or personal belief; factors in assessing the effectiveness of instruments such as Duolingo. Furthermore, the observer can modify participants' behavior, a result known as the observer effect, limiting the validity of data collected by this method even further. (Pressbooks, 2017)

Given these considerations, questionnaires are the most appropriate technique for primary data gathering in this research. They allow for efficient collection of data from a large sample size, keep respondents anonymous and are cost-effective, hence highly qualified to ascertain the impact of Duolingo on vocabulary acquisition among diploma students.

Secondary Data

Secondary data involves the use of existing information that has been collected by other organizations or researchers. Secondary data, in education research, can provide context and augment the results from primary data. Favis-Kean and Jager (2017) observe that secondary data analysis allows researchers to analyze existing datasets to answer emerging research questions potentially saving time and resources.

Secondary Data collection sources

In conducting this research, several, secondary data sources were employed in acquiring available books, papers, research articles, and literature. They included databases and website such as Google scholar(Google, 2023), which provides scholarly articles across various disciplines. The international Journal of Emerging Technologies in Learning (iJET) was also utilized for current research on learning technologies Buchem et al. (2020).

ResearchGate was another essential source of peer-reviewed research and author-shared articles Thelwall and Kousha (2015). The International Journal of Advanced Engineering, Management and Science also offered perspective on interdisciplinary research IJAEMS (2023). Wiley online Library Wiley (2023), pubMed Central NIH (2023), and Frontiers in psychology Frontiers (2023) were additional sources utilized in obtaining empirical and theoretical background. Institutional repositories like ScholarSpace from the University of Hawai and Acdemia.edu also contributed to the breath of data reviewed. Furthermore, platforms such as Scribbr provided methodological guidance, while ATLAS.ti was referred for qualitative data analysis tools ATLAS.ti (2023).

Why Reject this Data Type

While secondary data is contextually rich and saves time in employing existing research, it was not chosen as the primary source of data for this study on “Evaluating Duolingo in

Enhancing Vocabulary Expansion in English speaking skills among diploma students” because of some very crucial reasons.

The first one is Lack of Specificity. Secondary data typically reflects settings or other student populations. Therefore, it may not adequately reflect the unique experiences, vocabulary growth, or English-speaking proficiency of diploma-level students using Duolingo in the current learning context. This is also supported by Johston (2017), who notes that “Secondary data may not exactly match the researcher’s needs in terms of population, variables, or timing.”

Second One is Outdated or Irrelevant Contexts. The majority of secondary sources will be from research conducted before the most recent features or updates of Duolingo, and as such, they may have limited applicability. For instance, Duolingo has incorporated AI-powered feedback, speaking challenges, and gamified features in newer versions that older research may not take into account.

After that can get the reason Lack of control over data quality. Because the researcher is not in command of the initial collection of the secondary data, inconsistencies or bias may be impaired. Secondary data “Calls for close consideration of data quality, methods of collection, and sample appropriateness” as stated by Vartanian (2011).

3.7.2. Data Collection Method

Data collection is the most important component of the research process that enables researchers to acquire data related to the objectives of the study. A proper data collection method ensures reliability and validity in the research result. In education research, common data collection methods are surveys, interviews, observations, and document analysis. Every method has unique advantages and is chosen based on the goals of the study, the questions, and the type of study population. Surveys, for instance, work when collecting information from extensive populations, whereas interviews provide detailed information regarding participants’ experiences. Observations allow researchers to study behavior in its natural occurrence, and document analysis opens access to available data related to the research problem. The selection of method should be consistent with the aims of the study and the nature of data needed. (Taherdoost, 2016) (Kabir, 2016)

Data Collection Method	Description	Methodological Effects	Example
Questionnaires	Structured tools that ask respondents a sequence of questions in order to collect information.	Enables statistical analysis and the effective collecting of quantitative data from a big sample.	Used to gauge how students feel about Duolingo's influence on vocabulary learning. (Kabir, 2016)
Interview	Conversational approach that uses direct communication between the subject and the researcher to delve deeply into data.	Rich, qualitative data is provided; replies can be clarified; but it takes time and may add interviewer bias.	Were out to investigate how instructors have included Duolingo into their teaching methods. (Taherdoost, 2016)
Observations	Methodical documentation of observed occurrences or behaviors in their natural environment	Allows for the study of real-world behaviors, it is susceptible to observer bias and fails to capture motivation or other interior states.	Evaluated motivation by tracking students' levels of participation in Duolingo Sessions. (Kabir, 2016)
Document Analysis	Review of current records and documents that are pertinent to the subject of the study.	Offers contextual and historical information, but it is constrained by the papers that are available and could be shallow.	To ascertain if Duolingo activities were in line with learning objectives, curricular materials were examined. (Taherdoost, 2016)

Table 5 – Data Collection Methods

3.7.3. Data Collection and Analyze Tools

Data Collection Tool (Google Form)

Google Forms is a widely used web-based form builder to create and distribute questionnaires online. It is a low-cost, simple to use, and convenient tool for collecting structured answers, particularly well-suited in educational research. Google forms supports various question types e.g., multiple choice, short answer, Likert scales like and exporting responses to Excel or CSV is straightforward, so it is simple to transition to statistical analysis software like SPSS.

Scholars such as Google Forms because of its efficiency in gathering a huge sample within a time frame, especially when the population is computer-literate, such as diploma students. It further provides instantaneous gathering of answers and easy to use visualization facilities such as graphs and charts.

As Fauzi and Khusuma (2020) confirm, during the pandemic and even after it, Google Forms has succeeded in collecting data from students because it is free and can be linked to Google Sheets. Besides, Raags and Tapangan (2021) further state that Google Forms minimizes administrative burden and enhances participation since it is accessible via a mobile phone.

Data Analysis Tool (SPSS)

SPSS (Statistical Package for the Social Sciences) is a high-level statistical package from IBM that is commonly used for quantitative data analysis in social science, educational, and business research. It has support for a large range of statistical tests like descriptive analysis, regression analysis, ANOVA, and correlation analysis and hence is ideal for the analysis of structured questionnaire answers.

For my study, SPSS is ideal to examine data collected using Google Forms, especially when looking for relations between independent variables e.g., Student Motivation, Teacher approach audio, and feedback) and the dependent variable (Vocabulary enhancement in English speaking skills). Its capabilities in performing validity, reliability tests, and inferential statistics render it a powerful software for drawing reasonable conclusions.

Juile Pallant's SPSS Survival Manual is a step by step manual for the use of SPSS to analyze data. (Pallant, 2010) The manual discusses various statistical techniques and provides clear explanations on how to interpret SPSS output, which makes it a very handy resource for researchers. (Pallant, 2010) Pallant emphasizes the importance of understanding the assumptions of each statistical test and provides guidance on how to test these assumptions using SPSS.

3.7.4. Questionnaire structure



duolingo

Evaluating Duolingo in Enhancing English Vocabulary and Speaking Skills Among Diploma Students

This questionnaire is part of a research study aimed at evaluating the effectiveness of the Duolingo application in enhancing vocabulary and speaking skills in English among diploma students. Your responses will remain anonymous and will be used solely for academic purposes.

thisunibinodi4@gmail.com [Switch account](#)

Not shared

* Indicates required question

Figure 3 - Survey Introduction

01.Age *

☐ Below 18

☐ 18 – 21

☐ 22 – 25

☐ Above 25

02.Gender *

☐ Male

☐ Female

☐ Prefer not to say

03.Level of Study *

☐ Diploma

☐ Bachelor's Degree

☐ Master's Degree

☐ Other: _____

04.English Proficiency Level (Self-assessed) *

☐ Beginner

☐ Intermediate

☐ Advanced

05.Have you used Duolingo before for learning English? *

☐ Yes

☐ No

Figure 4 - Participation and Descriptive Questions

Factor 1 - Student Motivation and Interests (Independent Variable)

Please rate your level of agreement with the following statements

1 – Strongly Agree
2 – Agree
3 – Neutral
4 – Disagree
5 – Strongly Disagree

01.I feel more motivated to learn English vocabulary when using Duolingo. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

02.I find Duolingo's gamified lessons interesting and enjoyable. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 5 - Factor 1 Questions

03.I feel encouraged to learn daily because of Duolingo's reward system. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

04.Duolingo helps me stay consistent in practicing English vocabulary. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 6 - Factor 1 Questions

Factor 2 - Teacher's Instructional Strategy
Please rate your level of agreement with the following statements

1 - Strongly Agree
2 - Agree
3 - Neutral
4 - Disagree
5 - Strongly Disagree

05. My teacher integrates Duolingo into English lessons effectively. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

06. I receive clear instructions from my teacher on how to use Duolingo. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 7 - Factor 2 Questions

07. My teacher encourages us to share our Duolingo learning progress in class. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

08. I am guided on how to connect Duolingo vocabulary to real-life conversations. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 8 - Factor 2 Questions

Factor 3 - Integrated Audio Playback in the AI tool
Please rate your level of agreement with the following statements

1 – Strongly Agree
2 – Agree
3 – Neutral
4 – Disagree
5 – Strongly Disagree

09. The pronunciation audio in Duolingo helps me learn how to say new words correctly. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

10. I listen to the Duolingo audio multiple times to improve my speaking skills. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 9 - Factor 3 Questions

11. Audio features in Duolingo help me understand word usage in context. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

12. The listening exercises help me improve my English accent. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 10 - Factor 3 Questions

Factor 4 - Instant Feedback given by the AI tool
Please rate your level of agreement with the following statements

1 – Strongly Agree
2 – Agree
3 – Neutral
4 – Disagree
5 – Strongly Disagree

13.I receive immediate feedback from Duolingo when I make vocabulary errors. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

14.The instant corrections help me learn the right vocabulary usage. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 11 - Factor 4 Questions

15.I feel confident learning from my mistakes through Duolingo's feedback. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

16.I find the feedback system on Duolingo to be more helpful than traditional correction methods. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 12 - Factor 4 Questions

Enhancing the Vocabulary expansion in English Speaking Skills (Dependent Variable)
Please rate your level of agreement with the following statements

1 – Strongly Agree
2 – Agree
3 – Neutral
4 – Disagree
5 – Strongly Disagree

17. I can now use a wider range of vocabulary when speaking English after using * Duolingo.

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

18. Duolingo has improved my confidence in using new vocabulary in conversations. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 13 - Dependent Variable Questions

19. I find it easier to recall and use newly learned words while speaking in English. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

20. My overall English speaking fluency has improved due to better vocabulary knowledge from Duolingo. *

☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

Figure 14 - Dependent Variable Questions

Additional
Give simple answer

21. What are the main challenges you face while using Duolingo for vocabulary learning? *

Your answer

22. In your opinion, how can Duolingo be improved to enhance vocabulary development for speaking? *

Your answer

23. Have you noticed a difference in your vocabulary usage before and after using Duolingo? Please describe. *

Your answer

Thank You!
Thank you for participating in this study. Your responses will help us understand how to better support English vocabulary and speaking skills development using AI-powered tools like Duolingo.

Figure 15 - Additional Questions (Open - ended)

Variable	Indicators	Measurement	Mean	STD. Deviation	Median
IV 1/ Student Motivation and Interests	I feel more motivated to learn English vocabulary when using Duolingo.	Questioner	1.80	0.835	1.50
	I find Duolingo's gamified lessons interesting and enjoyable.				
	I feel encouraged to learn daily because of Duolingo's reward system.				
	Duolingo helps me stay consistent in practicing English vocabulary.				
IV 2/ Teacher's Instructiona l Strategy	My teacher integrates Duolingo into English lessons effectively.	Questioner	1.81	0.844	1.50
	I receive clear instructions from my teacher on how to use Duolingo.				
	My teacher encourages us to share our Duolingo learning progress in class.				
	I am guided on how to connect Duolingo vocabulary to real-life conversations.				
IV 3/ Integrated Audio Playback in the AI tool	The pronunciation audio in Duolingo helps me learn how to say new words correctly.	Questioner	1.86	0.845	1.75
	I listen to the Duolingo audio multiple times to improve my speaking skills.				

	Audio features in Duolingo help me understand word usage in context.				
	The listening exercises help me improve my English accent.				
IV 4/ Integrated Audio Playback in the AI tool	I receive immediate feedback from Duolingo when I make vocabulary errors.	Questioner	1.86	0.843	1.75
	The instant corrections help me learn the right vocabulary usage.				
	I feel confident learning from my mistakes through Duolingo's feedback.				
	I find the feedback system on Duolingo to be more helpful than traditional correction methods.				
DV Enhancing the Vocabulary expansion in English Speaking Skills (Dependent Variable)	I can now use a wider range of vocabulary when speaking English after using Duolingo.	Questioner	1.89	0.930	1.75
	Duolingo has improved my confidence in using new vocabulary in conversations.				
	I find it easier to recall and use newly learned words while speaking in English.				
	My overall English speaking fluency has improved due to better				

	vocabulary knowledge from Duolingo.				
--	--	--	--	--	--

Table 6- Mean, Standard Deviation and Median of each variable

3.7.5. Addressing Practical Considerations

Questioner Design

The questionnaire was carefully put together to make sure it was relevant and easy to understand for diploma students who were already familiar with Duolingo. The design focused on short, organized, and meaningful questions because long or complicated surveys might make people less likely to participate or make them answer quickly. These questions were closely related to the goals of the research, which included things like student motivation, feedback systems, and the effects of training. It was important to utilise basic, straightforward language that matched the respondents' level of language skill in order to get the most accurate answers and the best participation.

Budget constraints

Due to funding limits, cost-effective research approaches were favored. The use of self-administered online questionnaires allowed the study to minimize printing or interviewer fees. Online distribution was also time-efficient and accessible for the targeted diploma student population. This technique meant that the study could be carried out with minimum expenditures while still acquiring important primary data. The standardized questionnaire style significantly decreased resources needed for analysis, since replies could be more simply processed and analyzed.

Data accessibility

Accessibility was an important factor in the design and delivery of the questionnaire. To meet varied student demands, the questions avoided technical jargon and utilized plain English. Where appropriate, technical words (such “gamification” or “feedback system”) were quickly explained to ensure all participants could comprehend and answer confidently. This technique

helped make the research more accessible, especially for students with learning challenges or those less familiar with English language evaluation concepts.

3.7.6. Data Storage

When doing research with people, it is important to think about ethics and data protection from the start of the project. This early preparation helps make sure that choices about collecting, storing, and giving consent for data are in line with plans for sharing that data in the future. Researchers need to think about how they will secure the identity of participants by anonymizing or de-identifying the data, especially if it could be shared with the public or other researchers. As required by law, such the Data Protection Act 1998, all personal information must be kept safe and only people who are allowed to see it should be able to access it. To keep data safe and private during the research process, computers, USB drives, and other devices used to store data should be safeguarded from unauthorized access.

3.8. Target population and sampling

Sampling is a statistical method for quickly analyzing huge datasets by selecting a representative subset. Rather than analyzing the complete dataset, sampling examines a subset so that researchers can draw conclusions about a wider population. This enables educated decision-making without extensive data collection. (Tuovila, 2020)

3.8.1. Sampling Strategy

In general, sampling techniques can be divided into two categories,

- **Probability Sampling** -: There is a known, non-zero chance that each individual in the population will be chosen. Methods like stratified sampling, systematic sampling, and basic random sampling fall under this category. (Dhaval Makwana, 2023)
- **Non-Probability Sampling** -: Because of practical limitations, not every member has an equal probability of being included. Purposive sampling, snowball sampling, and convenience sampling are some of the techniques.

In my research titled “Evaluating Duolingo in Enhancing the Vocabulary Expansion in English Speaking Skills among Diploma Students”, I have used a probability sampling. The sampling procedure ensures that each population member has a known non-zero chance of

selecting Creswell (2012), which promotes the validity and generalizability of my findings. Since I am working with a well-known accessible population diploma students this procedure is appropriate as well as manageable.

Simple Random Sampling

The simplest method of probability sampling is this one. Every student in your target group has an equal probability of being chosen thanks to it. The best method for removing bias and obtaining statistically sound results is simple random sampling. Because it offers an impartial and equitable portrayal of the portrayal of the populace, it is frequently utilized. (Cochran, 1977).

Stratified Random Sampling

Using this method, the population is separated into various subgroups (also known as strata) based on factors like gender, age or academic major, and samples are then chosen from each segment. By ensuring that all pertinent subgroups are fairly represented, stratified sampling can increase the accuracy of your estimations. The usefulness of this approach for guaranteeing representation across different strata in educational research was shown by Teddlie and Yu (2007).

Systematic Sampling

In systematic sampling, can choose a beginning point at random and then choose every student from a list. This approach saves time, but if there is a hidden pattern in the list (e.g., if students are listed by name or performance) there may be a chance of bias. In his work on survey methodology, Lohr (2010) talks about the benefits and drawbacks of systematic sampling.

Cluster Sampling

Dividing the population into clusters (e.g., classrooms, neighborhoods) and then randomly selecting entire clusters in a city, and then random neighborhood are selected, and all residents within those neighborhoods are surveyed. Fowler (2013) emphasized the usefulness and constraints of cluster sampling, particularly in surveys related to education.

Sampling Techniques	Description	Methodological Effect	Example
Simple Random Sampling	Each member is equally likely to be chosen.	Remove selection bias and makes statistical inference possible.	Simple random sampling from student lists was employed in a study on Duolingo-assisted mobile language learning among EFL learners. (Kevin F. Steinmetz, 2021)
Stratified Random Sampling	Separate the population into strata, or subgroups, and then takes a random sample from each.	Minimizes sampling error and guarantees subgroup representation.	Stratified sampling was utilized in gender-based vocabulary acquisition research to guarantee a balanced a number of male and female participants. (Chiang, 2020)
Systematic Sampling	Following a random start, a list is selected at regular intervals.	Faster and simpler than random sampling, but findings could be skewed by recurring patterns.	Vasquez and Chavez (2017) employed systematic sampling choose 100 students from a population of 500 for their vocabulary retention study.
Cluster Sampling	Instead of choosing individuals at random, entire clusters like classes are chosen at random.	Saves money and time, but if clusters are not homogeneous, sampling error will increase.	Classroom clusters in technical institution were employed in a study on gamification in language apps. (Ju-Young M.Kang, 2015)

Table 7 - Samplin techniques comparison

Every one of these sampling strategies has advantages and disadvantages, but I chose simple Random Sampling.

Why choose Simple Random Sampling

I choose simple random sampling in my research of measuring the contribution of Duolingo to vocabulary development in English speaking skills among diploma students since it ensures fairness, objectivity, and statistical validity. It provides every member of the population with a fair chance of being selected, which significantly reduce the bias of selection and enhances the representativeness of the sample (Cochran, 1977). Besides, it suits my quantitative research goals quite perfectly because random sampling is appropriate for the use of inferential statistical methods for inferring conclusions regarding the larger population of students (Creswell, 2012). As well as I have access to a detailed and manageable list of diploma students, the method is efficient and cost-effective. It allows me to gather accurate information without unnecessary splitting or dividing students. Furthermore, Simple Random Sampling is most appropriate for removing any presumed beliefs or patterns from influencing the data collection process, thus making it most appropriate for removing any presumed beliefs or patterns from influencing the data collection process, thus making it most suitable for establishing whether Duolingo can be effective as an AI-supported learning application.

Why Reject Other Sampling Techniques

For methodological and contextual considerations, I disregarded other probability sampling approaches like stratified, systematic, and cluster samplings. First off, while stratified random sampling is helpful for comparing subgroups (e.g., gender, academic streams) it was not match for my research because it focuses on the overall effect of Duolingo on vocabulary learning rather than analyzing such distinctions (Charles Teddlie, 2007). Second, Systematic sampling was avoided since it could be biased if the diploma student list is arranged in a secret order, like by department, batch, or academic performance, which could skew the sample's randomness (Lohr, 2010). Finally, because grouping diploma students by classroom or institution may add sample error, cluster sampling was rejected even if it was effective in large or geographically scattered groups. The results could be skewed of one class, for instance, had superior access to technology [or English education.](#) (Fowler, 2013) [For my](#)

research purposes, simple random sampling was the most efficient and objective option because of these worries.

3.8.2. Sample Size

Target Population: Diploma Students in ESOF Metro Campus, Kandy Branch (950 Students)

Study Population: Students in Diploma Batch 47

Sample Frame: 64 Students

Sample: 55

Sample Size Calculator

Find Out The Sample Size

This calculator computes the minimum number of necessary samples to meet the desired statistical constraints.

Result
save

Sample size: 55

This means 55 or more measurements/surveys are needed to have a confidence level of 95% that the real value is within $\pm 5\%$ of the measured/surveyed value.

Confidence Level: 95% ▼

Margin of Error: 5 %

Population Proportion: 50 % Use 50% if not sure

Population Size: 64 Leave blank if unlimited population size.

Calculate
Clear

Table 8– Sample Size

3.9. The selection of participants

I made sure that each student in Diploma Batch 47 had an equal and fair chance of being selected for my study by using the Simple random sampling technique. I stated by obtaining a comprehensive list of all 64 eligible students who were enrolled in the batch at that time. Next, I gave every student a unique number between 1 and 64. I created 55 random numbers that matched student IDs using a random number generator, like the one in excel sheet. Participants in my study were the students whose numbers were chosen. This procedure was

carried out impartially, guaranteeing the sampling method's impartiality and integrity. I made sure that the sample fairly reflects the batch and that the results could be extrapolated to the larger population of diploma students by employing this technique

3.10. Reliability, Validity, and Generalizability

Reliability

Reliability refers to the consistency or reproducibility of a research instrument. In this study, it reveals whether the questionnaire used to evaluate the influence of Duolingo on vocabulary acquisition will give consistent findings if repeated under comparable settings.

According to Carmines and Zeller (1979), a dependable measuring device guarantees stability and precision throughout repeated trials. For this reason, the research design took many procedures to increase dependability. These included:

- Clear and clear wording in each question to avoid misinterpretation.
- Standardized circumstances during the administration of the survey.
- Pre-testing (pilot testing) the questionnaire on a small group to find ambiguities or inconsistencies.
- Eliminating double-barreled or leading questions, which might mislead respondents or influence their replies.

Because the poll addressed diploma students with varied levels of English ability, keeping uniformity in phrasing and presentation was vital. Heale and Twycross (2015) underlined that high dependability is necessary, especially in social sciences, to guarantee that findings are not dependent on measurement mistakes.

A highly dependable technique boosts the reliability of findings and enhances faith in the observed correlations between factors such as motivation, auditory feedback, and vocabulary expansion.

Validity

Validity examines how correctly the research instrument measures what it is supposed to measure. In this study, the focus was on whether the questionnaire effectively measured the impact of Duolingo to vocabulary improvement in English-speaking abilities.

Parveen and Showkat (2017) characterise validity as the degree to which an instrument actually assesses the idea it is designed to. To increase content validity, all questions were chosen to closely represent the primary study objectives: motivation, instructor strategy, audio replay, and immediate feedback. Literature research insights influenced the construction of question items to ensure they were conceptually consistent with existing definitions and findings in language learning and AI-based techniques.

The following efforts were done to promote validity:

- Ensuring that each question was related to a specific variable in the conceptual framework.
- Using practical definitions of crucial terminology, such as "instant feedback" or "adaptive learning," to minimise misunderstanding.
- Triangulating the main findings with secondary research sources to validate patterns or links discovered.

Creswell (2014) further underlines that a good instrument should avoid unnecessary factors and focus specifically on the targeted outcome. In this situation, questions unrelated to vocabulary growth or Duolingo use were purposely excluded to ensure construct validity.

Generalizability

Generalizability refers to the extent to which the findings of the research may be applied to other contexts or people outside the sample group.

According to Polit and Hungler (1991), generalizability is vital for practical implementation of research results, especially in domains like education where the purpose is to impact curriculum or policy across institutions.

In this study, the sample comprised of 55 diploma students from the ESOF Metro Campus selected using simple random sampling, a method noted for its efficacy in eliminating

selection bias. This enhances the odds that the results may represent the larger community of diploma students in similar educational contexts.

However, there are certain limitations to generalizability:

- The sample is confined to a single institution, thus cultural, institutional, and geographical variables are not accounted for.
- The findings may not reflect the experiences of students at different academic levels, such as university undergraduates or school-level learners.
- The study focuses primarily on Duolingo; therefore, results may not apply to other language learning programs.

Despite these limitations, the use of a systematic sampling approach and alignment of study variables with universal language-learning theories (such as Vygotsky's social constructivism or Krashen's input hypothesis) suggests a reasonable level of external validity. Future research can widen the reach by involving students from varied backgrounds and institutions to increase generalizability.

3.11. **Ethical issues of the research study**

While collecting data for this study on Evaluating Duolingo in Enhancing the Vocabulary Expansion in English Speaking Skills among Diploma Students, various ethical principles and practical factors were addressed to assure the integrity and responsibility of the research process.

» **Informed Consent**

Before sending the questionnaire, I took significant steps to ensure that all possible participants were fully informed about the study's aim, scope, and their rights as respondents. At the beginning of the online Google Form, I added a clear and simple consent statement that highlighted the important components of the research, such as its purpose to analyse how Duolingo increases vocabulary expansion and English-speaking skills among diploma students. This statement also described what kind of information would be collected, how it would be used purely for academic research, and guaranteed participants that their identities

would stay anonymous. Importantly, I made it obvious that participation was fully voluntary, emphasizing that students may choose not to participate, skip any questions they felt uncomfortable answering, or withdraw from the study at any moment without any consequences or need to offer a justification. To meet ethical norms, the form required students to actively offer their agreement by selecting a “I Agree” checkbox before they were permitted to advance to the quiz itself. This guaranteed that only individuals who had comprehended and consented to the terms participated in the study. For example, a student reading this consent form may feel sure that their involvement was entirely their decision and that their honest answers will contribute to enhancing educational programs like Duolingo, without incurring any personal or academic ramifications. Additionally, when I introduced the study in class before releasing the survey link, I verbally clarified these principles and encouraged students to ask questions if anything was unclear, thereby fostering transparency and informed decision-making.

» **Data Privacy and Confidentiality**

To maintain the highest level of data privacy and confidentiality, I took many deliberate procedures throughout the data gathering and storage process. First and foremost, the questionnaire was carefully prepared to eliminate any questions that might directly or indirectly identify participants, such as their names, student ID numbers, email addresses, or phone numbers. This meant that all replies were gathered fully anonymously, allowing students to provide honest and forthright feedback without worry that their names could be tracked. Moreover, the settings in Google Forms were customised to prohibit gathering IP addresses or any other browser metadata that could mistakenly reveal participant identification. All responses were immediately put into a safe spreadsheet, which I saved only on my personal laptop secured with a strong password and encrypted when possible. Access to this data was carefully confined to me alone, prohibiting unauthorized viewing or alteration. After the data was thoroughly analyzed and reported, I planned to permanently delete all digital files to remove any risk of future data breaches or misuse. For example, when a student completed the survey detailing their experience with Duolingo's vocabulary exercises, they could be confident that their specific replies were not linked to their personal information in any way. This technique maintained ethical compliance with data protection regulations and built trust among participants that their privacy was properly respected before and after the study.

» **Avoiding Harm to Participants**

To protect participants from any potential emotional discomfort or distress, I carefully designed the questionnaire to focus solely on students' interactions and experiences with the Duolingo app, avoiding any questions about sensitive or personal topics such as academic performance, grades, or opinions about instructors. For example, rather than enquiring about how well students felt they fared in English classes or if they thought their teachers were fair or critical, I framed all enquiries in a neutral and nonjudgmental tone. The most common queries were: "How often do you use Duolingo to practise English vocabulary?" / "Which Duolingo exercises do you find most helpful for learning new words?" This method ensured that participants did not feel singled out, criticised, or uncomfortable when responding. Instead of a question like, "Do you think your professor unfairly judges your English skills?", which could elicit negative feelings or defensiveness, the questionnaire enquired about the student's frequency and sorts of Duolingo activities. This contributed to the creation of a safe and supportive environment in which students felt comfortable sharing their honest experiences with the language learning tool without fear of being judged or facing punishments. Furthermore, the wording of the questions was carefully examined to eliminate leading or biased language, allowing participants to reply genuinely without feeling compelled to deliver socially desired answers. Overall, this attentive and courteous strategy reduced the danger of emotional harm while maintaining the ethical concept of beneficence in research.

» **Voluntary Participation**

Participation in this study was fully voluntary, and I made it a point to express this explicitly to all students from the beginning. I explicitly emphasized in the consent form at the beginning of the questionnaire, as well as in a reminder at the end, that students were under no obligation to participate and could withdraw their participation at any time without experiencing any negative consequences or being required to explain why. For example, the consent statement includes language like: "You are free to skip any questions you do not want to answer, and you may stop the survey at any time." Your selection will have no effect on your academic standing or relationship with the faculty." This comfort was intended to

provide students with complete flexibility over their participation while reducing any sense of pressure or duty. Furthermore, no incentives, awards, or other inducements were provided for involvement, guaranteeing that students' decisions to participate were exclusively influenced by their own willingness rather than external factors. As an example, if a student started the survey and came across a question they were uncomfortable answering—such as one about the frequency of their Duolingo use—they could choose "Prefer not to answer," skip the topic entirely, or exit the form without finishing it. This autonomy contributed to an ethical research setting in which participants felt appreciated and their rights safeguarded. Furthermore, I made myself available to address any questions or concerns, maintaining a supportive environment in which participation was entirely voluntary.

Practical Considerations

» **Budget Constraints**

Given the limited budget for this research, I carefully selected cost-effective and easily accessible digital tools to collect data quickly while minimizing expenses. Instead of traditional paper questionnaires, which would have cost a lot of money to print, photocopy, and distribute across multiple diploma classes or campus locations, I used Google Forms, a free online survey platform. This option not only avoided the costs associated with physical materials such as paper, ink, and transportation, but it also lowered the time and manpower generally required to give and collect questionnaires in person. For example, instead of printing 55 physical copies of the questionnaire and visiting multiple classrooms to distribute and retrieve them, I created a digital version of the survey and emailed a link to students, allowing them to respond conveniently on their own devices at any time and place. This solution also enabled automatic real-time data compilation into spreadsheets, significantly simplifying data management and lowering the possibility of transcription errors that can arise during manual entry. Furthermore, by allowing contactless data collection, an online application assisted in addressing unexpected problems such as COVID-19 limits or social distancing policies. This practical application of digital technology was especially significant given the limited financial resources available, allowing the study to proceed without sacrificing the quality or breadth of data collection. In conclusion, the use of Google Forms as a cost-efficient alternative demonstrated an excellent compromise between financial restrictions and the stringent requirements of research methodology.

» **Data Accessibility**

To ensure that the questionnaire was accessible and inclusive to all diploma students, I carefully designed the questions in clear, plain English that could be easily understood by participants with varied levels of English proficiency. Recognizing that some students may not be native English speakers or have limited language abilities, I purposefully avoided using technical jargon, difficult sentence structures, or academic terminology that could confound or discourage replies. For example, instead of asking, "How effective do you find Duolingo's spaced repetition algorithm in facilitating lexical retention?" I rephrased it in more clear language: "How helpful do you find the way Duolingo repeats words to help you remember them?". This change makes the question more accessible and understandable for kids who may be unfamiliar with linguistic or educational concepts. Furthermore, when technical or possibly unfamiliar phrases were unavoidable, I provided concise definitions or explanations directly following the term to ensure clarity. For example, if I needed to use the term "vocabulary acquisition," I would include a basic explanation like "(learning new words)." This method reduced misconceptions and allowed students to respond confidently without feeling overwhelmed or alienated by the tough terminology. Furthermore, the survey was formatted clearly, with bullet points and short answer options, to improve readability and navigation. By prioritizing accessibility in the questionnaire's language and design, I hoped to increase participation and collect more accurate data that really mirrored the experiences of all diploma students, regardless of their unique language skills or educational backgrounds.

CHAPTER 4 - PRESENTATION OF RESULTS

4.1.Introduction

This chapter summarizes the major conclusions drawn from the primary data acquired through a standardized questionnaire administered to diploma students. The goal of this chapter is to offer a clear and thorough description of the study outcomes, closely matched with the research objectives and sub-objectives established in preceding parts. It works as the bridge between the methods utilized and the findings that follow, offering an evidence-based understanding of how Duolingo has improved vocabulary growth in English speaking abilities within the target demographic.

The investigation focuses on numerous variables, including student motivation, instructor tactics, integrated audio playback, and fast feedback—each studied in connection to vocabulary progress. These variables are analyzed by both descriptive statistics and inferential analysis, including correlation and regression tests, to uncover significant patterns and correlations.

The findings are given in an orderly fashion utilizing tables, charts, and brief explanations for improved clarity and comprehension. Where applicable, comparisons are established across various answer groups to show variances and commonalities in the experience of using Duolingo for language acquisition.

Beyond giving raw data, this chapter gives interpretive insights, putting the results within the framework of current ideas and literature. It examines how the evidence supports or challenges earlier conclusions, helping to contextualize the relevance of the discoveries both academically and practically. Furthermore, the chapter emphasizes how the results immediately answer to the study's major research objectives and assumptions.

Ultimately, this chapter serves a vital role in verifying the study's outcomes. It provides readers a transparent perspective of the genuine influence of Duolingo on the English-speaking vocabulary expansion of diploma students and establishes the framework for the conclusions and recommendations presented in the next chapter.

4.2. Pilot Test Analysis

» **Background and Motivation of the Pilot Test**

Conducting a pilot study is a critical step in confirming the reliability and validity of the instruments and processes utilized in research. According to van Teijlingen and Hundley (2001), pilot testing helps detect possible difficulties in study design, including ambiguity in questions, data collecting challenges, and participant misconceptions. For this study, a pilot test was carried out with 12 diploma students prior to the complete deployment of the questionnaire.

The rationale behind conducting this pilot test originated from the requirement to examine the clarity and usefulness of the questionnaire targeted at analyzing how Duolingo helps vocabulary development. As the research combines language and technical issues, it was vital to validate that all questions were comprehensible, culturally suitable, and appropriately reflected the aims of the study. Previous studies such as Hassan et al. (2020) also stress the relevance of pilot testing in developing survey instruments in educational research, particularly in digital learning situations.

Additionally, this pilot attempted to check if the feedback forms and Likert scale questions adequately captured characteristics such as student motivation, teaching tactics, integrated audio use, and fast feedback the essential features of your conceptual framework.

» **Process of the Pilot Test**

The pilot study was done using a sample of 12 diploma students picked from the same demographic as the main study, ESOF Metro Campus diploma students using the same simple random selection approach as in the main research.

- Participants were given the draft version of the questionnaire developed for the main study.
- The atmosphere and conditions simulated those planned for real data collection, including instructions and time constraints.
- Students were asked to complete the questionnaire and then offer input on,

- The clarity of the questions,
 - The difficulty or uncertainty in understanding any phrase,
 - The length and structure of the shape,
 - Suggestions for improvement.
- Collected data were then analyzed to verify initial reliability using Cronbach's alpha (if appropriate), and check whether responses corresponded with predicted variable groupings.

Pilot Study Questioner

Pilot Study of the Questionnaire for Research Validity

This study aims to test the clarity and effectiveness of the main questionnaire. Your responses and suggestions will help improve the final version. This is NOT the final data collection, and your feedback is highly appreciated.

nethuthilo3@gmail.com [Switch account](#)

Not shared

* Indicates required question

01. Were the instructions clear? *

☐ Yes

☐ No

If No, please explain

Your answer

Figure 16 - Pilot Study Questioners

02. Were any questions confusing or difficult to understand? *

☐ Yes

☐ No

If Yes, please mention which one and why

Your answer _____

03. Were any response options missing or unclear? *

☐ Yes

☐ No

If Yes, explain

Your answer _____

Figure 17- Pilot Study Questioners

04. Do you feel the questions are relevant to evaluating vocabulary improvement * in speaking skills?

☐ Yes

☐ No

Explain your answer *

Your answer _____

05. Any suggestions to improve the questionnaire?

Your answer _____

Submit **Clear form**

Figure 18 - Pilot Study Questioners

Pilot study Feedback

18/07/2025, 23:05

Pilot Study Questionnaire - Duolingo (Responses) - Google Sheets

Timestamp	01. Were the instructions clear? If No, please explain	02. Were any questions confusing or difficult to u If Yes, please mention which one and why	03. Were any response options missing or unclear If Yes, explain	04. Do you feel the questions are relevant to eval Explain your answer	05. Any suggestions to improve the questionnaire
5/31/2025 19:07:36	Yes	No	No	Yes	Yes, they directly relate to \ No suggestions
5/31/2025 19:24:20	Yes	No	No	Yes	The questions are directly No suggestion
5/31/2025 19:31:29	Yes	No	No	Yes	Yes, they align with real-w No suggestions
5/31/2025 19:34:52	Yes	No	No	Yes	Yes, they are appropriate fr No suggestions
5/31/2025 19:35:43	Yes	No	No	Yes	Of course, they match the \ No suggestions
5/31/2025 19:36:27	Yes	No	No	Yes	For sure, they assess pract No suggestions
5/31/2025 19:37:07	Yes	No	No	Yes	Yes, the questions are well No suggestions
5/31/2025 20:34:25	Yes	No	No	Yes	Absolutely, they target key No
5/31/2025 20:35:35	Yes	No	No	Yes	Yes, they help track progre No
5/31/2025 20:36:35	Yes	No	No	Yes	Yes, they directly relate to \ No
5/31/2025 21:22:36	Yes	No	No	Yes	The questions are relevant No suggestions
5/31/2025 22:58:57	Yes	No	No	Yes	Yes, the questions are very No

Figure 19 - Pilot Study Feedback

» **Outcomes and Conclusions**

The pilot research offered important insights that influenced modest adjustments to the full questionnaire. Some of the significant outcomes included,

- Improved clarity
Certain questions were rephrased to be clearer, especially those linked to AI concepts such as "instant feedback" and "gamification".
- Time efficiency
The completion time was modified based on pilot results, ensuring the final form could be finished comfortably within 10–15 minutes.
- Validation of structure
The structure of the Likert scale functioned successfully, with students answering to all items without misunderstanding.
- Positive feedback
Participants found the questionnaire entertaining and relevant to their Duolingo experience.

In conclusion, the pilot test proved the applicability and usefulness of the questionnaire and research procedure. It also lowered the chances of misunderstanding throughout the primary investigation and boosted the trustworthiness of the results. Based on Presser et al. (2004), this pilot test accomplished its aim of increasing survey quality and boosting instrument validity.

4.3. Discriptive Analysis

Age Group of Participants

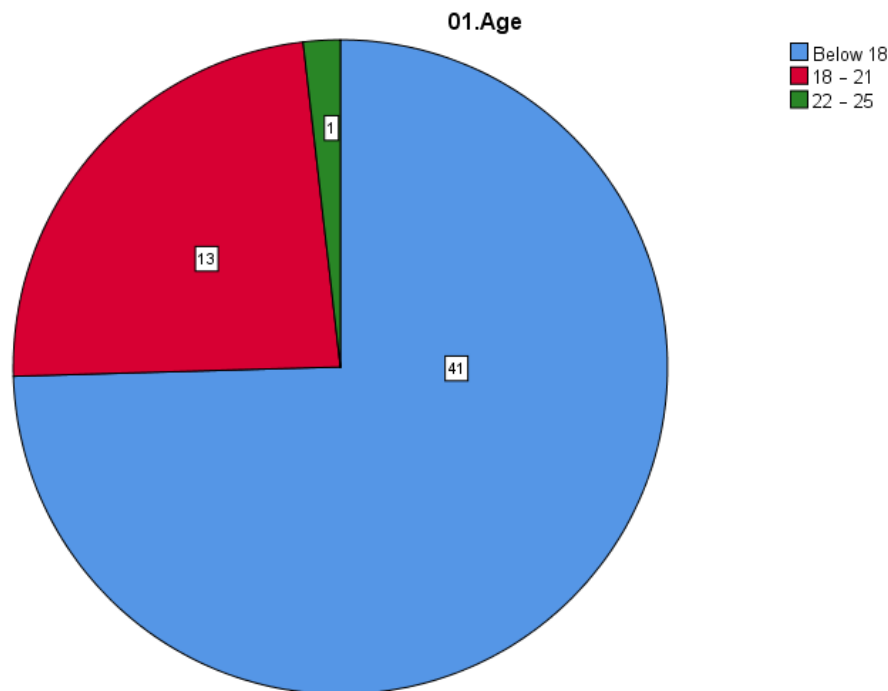


Figure 20 - Pie Chart of the representation of Age Groups of the Participants

The pie chart depicts the distribution of participants across three separate age groups. A vast majority of the respondents, 41 out of 55, belong to the "Below 18" age category, representing about 75% of the entire sample. This suggests that most of the diploma students questioned are in their late teenage years, possibly matching the average age range of students starting diploma programs shortly following secondary education.

The next largest group, consisting of 13 individuals (approximately 24%), falls within the "18 – 21" age range. This category presumably comprises students who either started their diploma studies significantly later or took time shifting from previous academic courses.

Only 1 participant (approximately 1%) is in the "22 – 25" group, demonstrating that older students are rather unusual in this specific sample.

Overall, the age distribution implies that the research predominantly represents the opinions and experiences of younger learners, especially those under 18, which is crucial when assessing the usability and engagement levels of mobile learning technologies like Duolingo.

Gender Distribution of Respondents

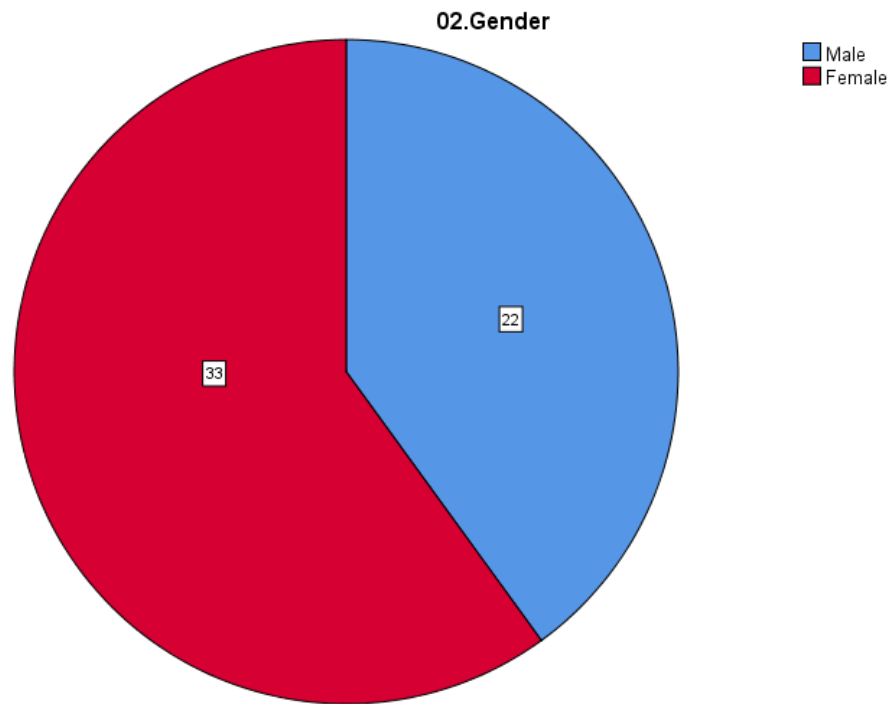


Figure 21 - Pie chart of the Gender Distribution of Respondents

The pie chart illustrates how many male and female took part in the survey. Of the students who answered, 33 said they were female and 22 said they were male. This shows that there are more women than men in the sample group. The figure makes this distinction clear by using different colors for males and females (red for females and blue for males) and numbers to mark the data. The chart clearly shows the gender breakdown of the participants, which is significant for evaluating how representative and diverse the data is.

Current Educational Level of Participants

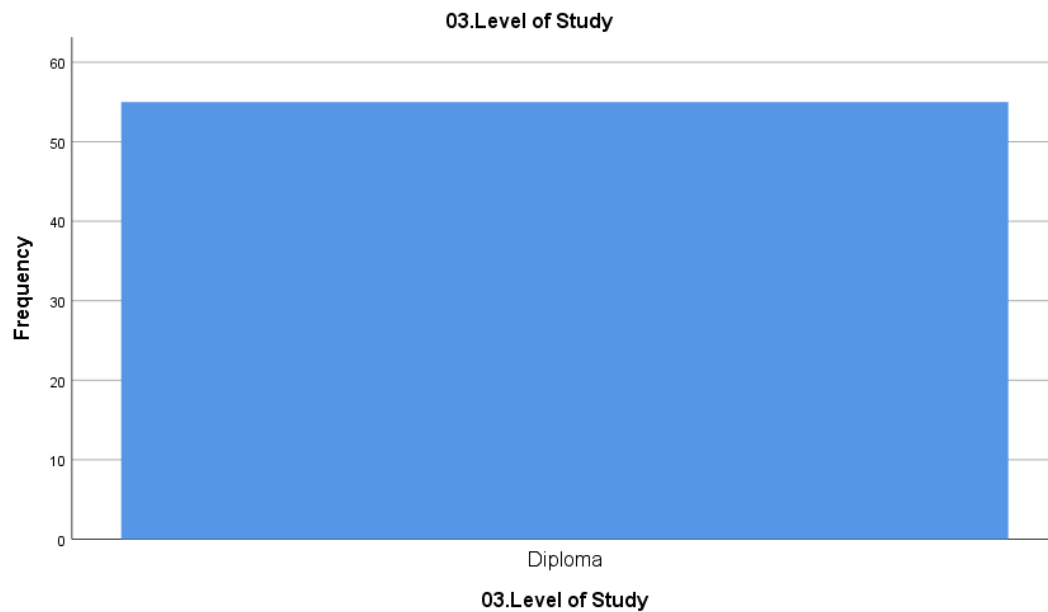


Figure 22 - The bar graph representation of the Current Educational Level of Participants

The bar chart shows that all respondents in the survey are at the Diploma level of study, with a frequency of 55.

English Proficiency Level of Participants

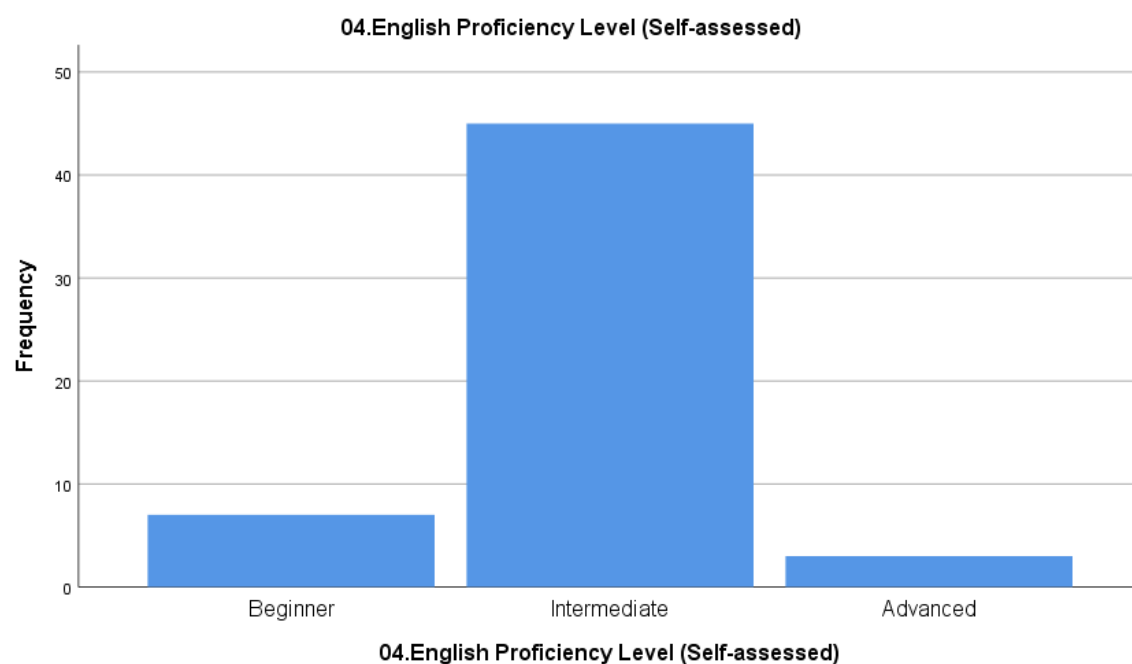


Figure 23 - The bar graph representation of the English Proficiency Level of Participants

The bar chart shows the English Proficiency Level (Self-assessed)" indicates how the students who took the test rated their own English skills. About 45 students, or about half of those who answered, said they were at an intermediate level of proficiency. Only about 7 participants thought they were beginners, and only 3 students thought their English skills were advanced. This distribution implies that most individuals feel reasonably confident in their English skills, with relatively few viewing themselves as either extremely proficient or just starting to learn. The chart provides vital insight into the linguistic capabilities of the group, which may influence their language learning needs and educational help.

Previous Usage of Duolingo for Learning English Among Participants

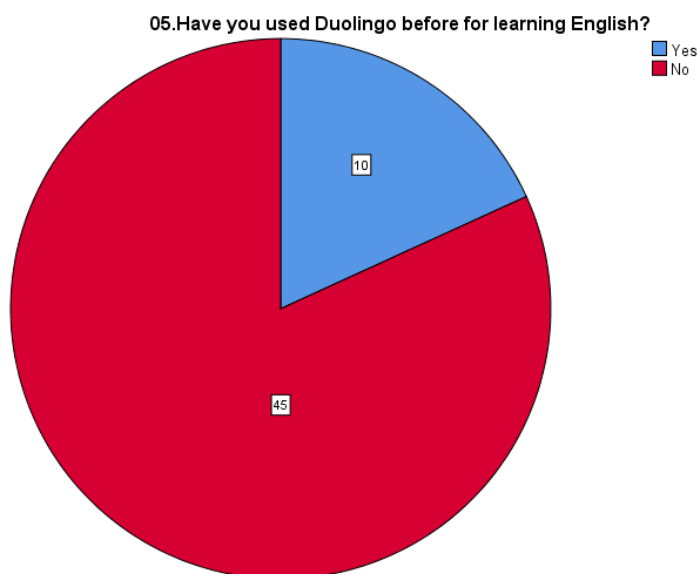


Figure 24 - Previous Usage of Duolingo for Learning English Among Participants

The pie chart depicts the distribution of student responses to the question, "Have you used Duolingo before for learning English?" Out of a total of 55 students, 10 students (shown in blue) responded "Yes," suggesting past familiarity with Duolingo, whereas 45 students (shown in red) answered "No," revealing they had not used the program before. This means that around 82% of students were new to Duolingo, making it a fresh learning tool for the majority. The graphic demonstrates the possibility for exploring the influence of Duolingo on students who are first-time users in the context of English vocabulary development.

Crosstabulation of the English Proficiency Level of the Participants with their Age

Crosstab					
Count		04.English Proficiency Level (Self-assessed)			
		Beginner	Intermediate	Advanced	Total
01.Age	Below 18	6	32	3	41
	18 - 21	1	12	0	13
	22 - 25	0	1	0	1
Total		7	45	3	55

Symmetric Measures					
		Value	Asymptotic Standard Error ^a	Approximate T ^b	Approximate Significance
Interval by Interval	Pearson's R	.008	.096	.059	.953 ^c
Ordinal by Ordinal	Spearman Correlation	.010	.106	.070	.944 ^c
N of Valid Cases		55			

a. Not assuming the null hypothesis.
b. Using the asymptotic standard error assuming the null hypothesis.
c. Based on normal approximation.

Figure 25 - Crosstabulation of the English Proficiency Level of the Participants with their Age

The crosstab and symmetric measures offered provide an examination of the link between students' age groups and their self-assessed English proficiency levels. According to the crosstab, the majority of pupils fell under the age range below 18, with 32 identifying as intermediate, 6 as beginners, and 3 as advanced. In contrast, students aged 18–21 generally assessed themselves as intermediate (12), while only 1 beginner and no advanced learners were discovered in that group. The age range 22–25 had only 1 student, classed as intermediate.

The Symmetric Measures table investigates the statistical link between age and English proficiency level. The Pearson's R value is 0.008 and the Spearman correlation is 0.010, both indicating a very modest positive link. However, the approximate significance values for both tests are substantially over 0.05 (Pearson's Sig. = 0.953, Spearman's Sig. = 0.944), showing that these correlations are not statistically significant. Therefore, the data does not provide adequate evidence to suggest a meaningful link between a student's age and their self-assessed English proficiency level. The null hypothesis that there is no association between age and proficiency level is not rejected.

Crosstabulation of the English Proficiency Level of the Participants with their Gender

Crosstab

Count		02.Gender		Total
		Male	Female	
04.English Proficiency Level (Self-assessed)	Beginner	4	3	7
	Intermediate	16	29	45
	Advanced	2	1	3
Total		22	33	55

Symmetric Measures

		Value	Asymptotic Standard Error ^a	Approximate T ^b	Approximate Significance
Interval by Interval	Pearson's R	.035	.146	.257	.798 ^c
Ordinal by Ordinal	Spearman Correlation	.042	.147	.305	.762 ^c
N of Valid Cases		55			

a. Not assuming the null hypothesis.

b. Using the asymptotic standard error assuming the null hypothesis.

c. Based on normal approximation.

Figure 26 - Crosstabulations

The crosstab and symmetric measures offered investigate the association between gender and self-assessed English proficiency level among 55 students. According to the crosstab, out of 22 male students, 4 identified as beginners, 16 as intermediate, and 2 as advanced. Among the 33 female pupils, 3 were beginners, 29 intermediate, and just 1 qualified as advanced. The majority of both male and female students perceive themselves to be at the intermediate skill level.

Looking at the Symmetric Measures table, the Pearson's R value is 0.035 and the Spearman association is 0.042, both demonstrating a very slight positive association between gender and English competence. However, the approximate significance values for both measures are far above 0.05 (Pearson's Sig. = 0.798, Spearman's Sig. = 0.762), implying that the observed correlations are not statistically significant. Thus, the analysis did not reveal evidence of a relevant link between gender and self-assessed English skill level. The null

hypothesis is not rejected, showing gender does not significantly influence the students' estimate of their English proficiency.

4.4. Reliability Analysis

Background of the Test

A reliability test is a statistical approach used to verify the consistency and stability of a measurement instrument, such as a survey or test, across time. It guarantees that the tool generates similar results under consistent conditions, therefore proving the dependability of the data collected. One of the most often used approaches is Cronbach's Alpha, created by Lee Cronbach in 1951, which examines the internal consistency of a set of items to determine whether they measure the same underlying notion. A Cronbach's Alpha value above 0.7 is generally deemed satisfactory, indicating a dependable instrument. Other basic contributions to reliability testing include the Spearman-Brown formula (Spearman, 1910), used for split-half reliability, and the Kuder-Richardson formulas (Kuder & Richardson, 1937), applied to tests with dichotomous items. Overall, reliability testing is vital in ensuring that study conclusions are based on consistent and reproducible data.

Result obtained

Case Processing Summary

		N	%
Cases	Valid	55	100.0
	Excluded ^a	0	.0
	Total	55	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.951	.952	5

Figure 27 - Representation of the Research's Reliability

This test was done using Cronbach's Alpha to check if the questionnaire is consistent in measuring the same concept. Cronbach's Alpha value was 0.951, which is above acceptable level of 0.7. This means the questionnaire is reliable.

4.5. Validity Analysis

Background of the Test

According to Tavakol and Dennick (2011), validity refers to the degree to which an instrument measures what it is designed to measure. One typical technique to analyse validity is through construct validity, which assesses if the test actually reflects the theoretical concept it wants to measure. Tools such as Kaiser-Meyer-Olkin (KMO) and Bartlett's Test of Sphericity are typically used to determine the adequacy of data for factor analysis, which supports construct validity. As highlighted by Field (2009), a KMO score above 0.6 suggests sample sufficiency, while a significant Bartlett's Test ($p < 0.05$) reveals that correlations between items are sufficient for factor analysis, thereby establishing the instrument's validity.

Result obtained

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.901
Bartlett's Test of Sphericity	Approx. Chi-Square	275.908
	df	10
	Sig.	.000

Figure 28 - Representation of the Research's Validity

The Validity Test was done to ensure that the questionnaire accurately measures what it is intended to assess. First, the KMO (Kaiser-Meyer-Olkin) value was 0.901, which is above 0.6. This shows that the sample is adequate and suitable for factor analysis. Next, the Bartlett's Test of Sphericity had a significant value of .000, which is less than 0.05. This means the variables are correlated enough and valid for further analysis.

4.6. Normality Test

Background of the Test

The normality test is used to determine if a collection of data follows a normal distribution, which is commonly shaped like a bell curve. According to OnlineSPSS.com (2022b), many real-world datasets follow this pattern, where values are symmetrically distributed around the average. Normal distribution is significant because many statistical tests, such as t-tests and ANOVA, assume that the data is normally distributed. If the data does not match this assumption, the results of these tests may not be valid. Therefore, it is crucial to test normalcy before choosing which statistical approach to apply. For larger sample sizes (greater than 50), the Kolmogorov-Smirnov test is widely employed to test for normality. If the test produces a p-value less than 0.05, it suggests the data is not normally distributed and nonparametric testing should be explored instead.

H0: Data are normally distributed

H1: Data are not normally distributed

Results obtained

Before undertaking any further analysis, it is crucial to examine if the data is regularly distributed. The normality of the obtained dataset of this research was accomplished by the above-mentioned Kolmogorov-Smirnov test.

1. Dependent Variable

Case Processing Summary						
	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Enhancing_the_Vocabulary_expansion_in_English_Speaking_Skills	55	100.0%	0	0.0%	55	100.0%

Descriptives				
			Statistic	Std. Error
Enhancing_the_Vocabulary_expansion_in_English_Speaking_Skills	Mean		1.89	.125
	95% Confidence Interval for Mean	Lower Bound	1.63	
		Upper Bound	2.14	
	5% Trimmed Mean		1.79	
	Median		1.75	
	Variance		.865	
	Std. Deviation		.930	
	Minimum		1	
	Maximum		5	
	Range		4	
	Interquartile Range		1	
	Skewness		1.622	.322
	Kurtosis		2.493	.634

Figure 29 - Representation of the Dependent Variable's Skewness

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Enhancing_the_Vocabulary_expansion_in_English_Speaking_Skills	.251	55	.000	.814	55	.000

a. Lilliefors Significance Correction

Figure 30 - Representation of the Dependent Variable's Normality

As per the preceding tables, the p-value is 0.000, which is less than 0.05, suggesting that the null hypothesis (H_0) is rejected and the alternative hypothesis (H_1) is accepted. This suggests the data is not normally distributed. Furthermore, the skewness value is 1.622, which is outside the permitted range of -1 to +1, implying a positively skewed distribution. Therefore, the data does not follow a normal distribution, and nonparametric tests were regarded more suited for further research.

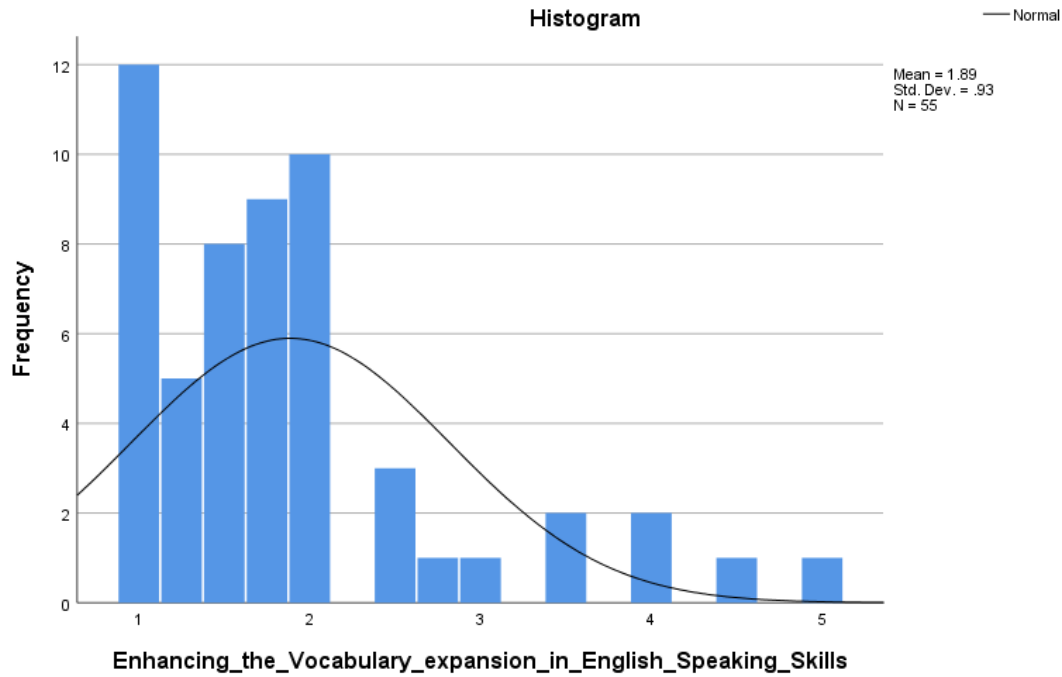


Figure 31 - Interpreting the Bell Curve in dependent Variable

2. Independent Variable (Factor 1)

Case Processing Summary						
	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Student_Motivation_and_I nterests	55	100.0%	0	0.0%	55	100.0%

Descriptives			
		Statistic	Std. Error
Student_Motivation_and_I nterests	Mean	1.80	.113
	95% Confidence Interval for Mean	Lower Bound	1.57
		Upper Bound	2.03
	5% Trimmed Mean	1.71	
	Median	1.50	
	Variance	.700	
	Std. Deviation	.837	
	Minimum	1	
	Maximum	5	
	Range	4	
	Interquartile Range	1	
	Skewness	1.570	.322
	Kurtosis	2.598	.634

Figure 32 - Representation of the Independent Variable's (Factor 1) Skewness

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Student_Motivation_and_Interests	.181	55	.000	.824	55	.000

a. Lilliefors Significance Correction

Figure 33 - Representation of the Independent Variable's (Factor 1) Normality

For Factor 1 – Student Motivation and Interests, the p-value is 0.000, which is less than 0.05, suggesting that the null hypothesis (H_0) is rejected and the alternative hypothesis (H_1) is accepted. This confirms that the data for this factor is not regularly distributed. Additionally, the skewness number is 1.570, which exceeds the permissible range of -1 to +1, suggesting a positively skewed distribution. Therefore, it may be concluded that the data associated with this component does not follow a normal distribution, and nonparametric methods are more suitable for its study.

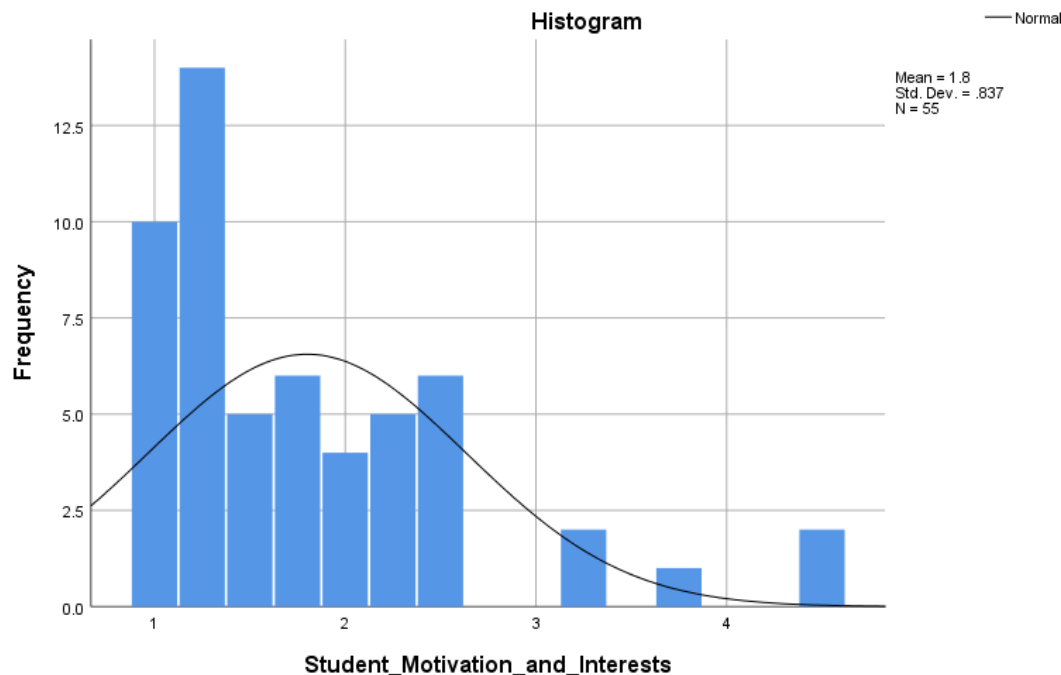


Figure 34 - Interpreting the Bell Curve in Student Motivation and Interests

3. Independent Variable (Factor 2)

Case Processing Summary						
	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Teacher's Instructional Strategy	55	100.0%	0	0.0%	55	100.0%

Descriptives				
Teacher's Instructional Strategy			Statistic	Std. Error
	Mean		1.81	.114
95% Confidence Interval for Mean	Lower Bound		1.58	
	Upper Bound		2.04	
5% Trimmed Mean			1.71	
Median			1.50	
Variance			.713	
Std. Deviation			.844	
Minimum			1	
Maximum			5	
Range			4	
Interquartile Range			1	
Skewness			1.728	.322
Kurtosis			3.506	.634

Figure 35 - Representation of the Independent Variable's (Factor 2) Skewness

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Teacher's Instructional Strategy	.206	55	.000	.824	55	.000

a. Lilliefors Significance Correction

Figure 36 - Representation of the Independent Variable's (Factor 2) Normality

For Factor 2 – Teacher's Instructional Strategy, the p-value is 0.000, which is less than 0.05, suggesting that the null hypothesis (H_0) is rejected and the alternative hypothesis (H_1) is accepted. This confirms that the data for this factor is not regularly distributed. Additionally, the skewness number is 1.728, which exceeds the permissible range of -1 to +1, suggesting a positively skewed distribution. Therefore, it may be concluded that the data associated to this component does not follow a normal distribution, and nonparametric methods are more suitable for its study.

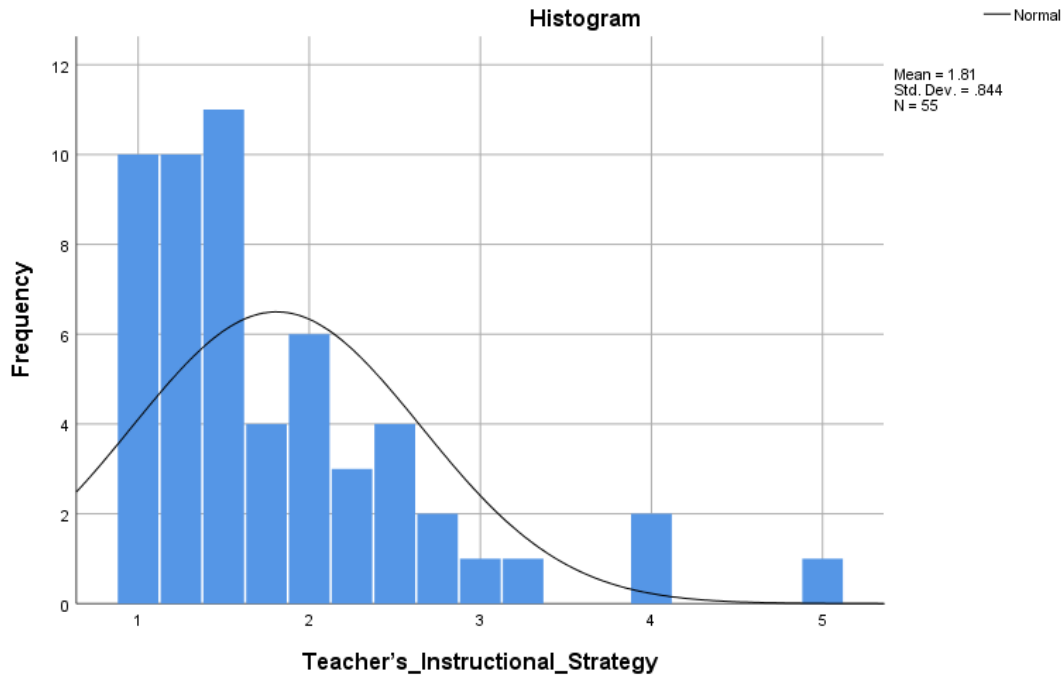


Figure 37 - Interpreting the Bell Curve in Teacher's Instructional Strategy

4. Independent Variable (Factor 3)

Case Processing Summary						
	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Integrated_Audio_Playback_in_the_AI_tool	55	100.0%	0	0.0%	55	100.0%

Descriptives				
Integrated_Audio_Playback_in_the_AI_tool			Statistic	Std. Error
	Mean		1.86	.114
	95% Confidence Interval for Mean	Lower Bound	1.64	
		Upper Bound	2.09	
	5% Trimmed Mean		1.78	
	Median		1.75	
	Variance		.715	
	Std. Deviation		.845	
	Minimum		1	
	Maximum		5	
	Range		4	
	Interquartile Range		1	
	Skewness		1.589	.322
	Kurtosis		3.079	.634

Figure 38 - Representation of the Independent Variable's (Factor 3) Skewness

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Integrated_Audio_Playback_in_the_AI_tool	.181	55	.000	.847	55	.000

a. Lilliefors Significance Correction

Figure 39- Representation of the Independent Variable's (Factor 3) Normality

For Factor 3 – Integrated Audio Playback in the AI tool, the p-value is 0.000, which is less than 0.05, suggesting that the null hypothesis (H_0) is rejected and the alternative hypothesis (H_1) is accepted. This confirms that the data for this factor is not regularly distributed. Additionally, the skewness number is 1.589, which exceeds the permissible range of -1 to +1, suggesting a positively skewed distribution. Therefore, it may be concluded that the data associated to this component does not follow a normal distribution, and nonparametric methods are more suitable for its study.

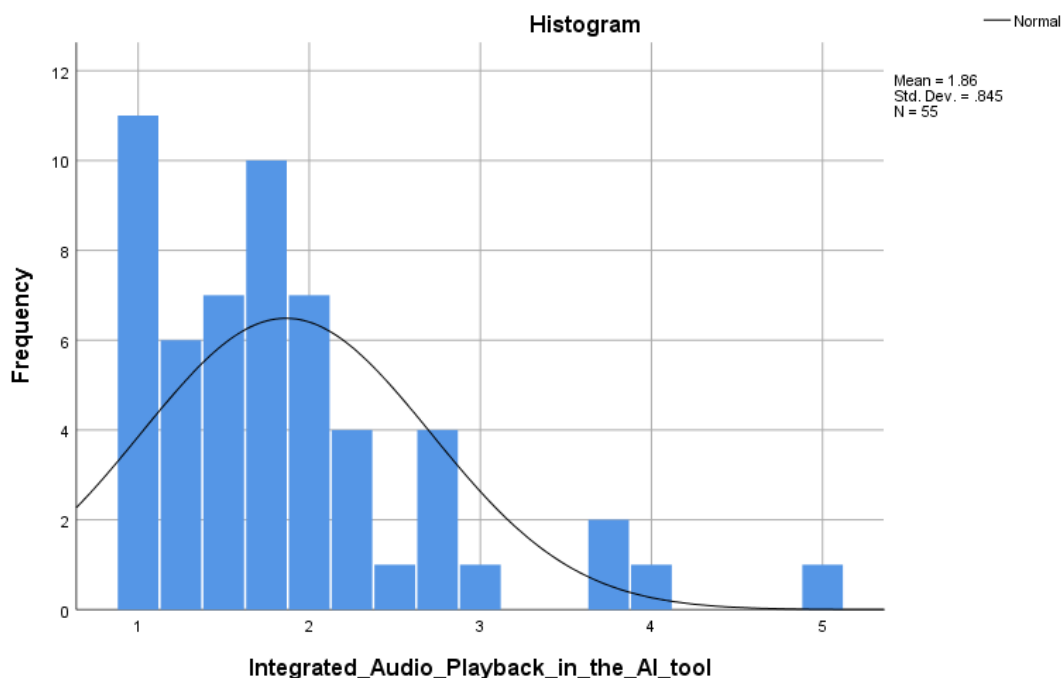


Figure 40 - Interpreting the Bell Curve in Integrated Audio Playback in the AI tool

5. Independent Variable (Factor 4)

Case Processing Summary

	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Instant_Feedback_given_by_the_AI_tool	55	100.0%	0	0.0%	55	100.0%

Descriptives

		Statistic		Std. Error
Instant_Feedback_given_by_the_AI_tool	Mean	1.86		.114
	95% Confidence Interval for Mean	Lower Bound	1.63	
		Upper Bound	2.09	
	5% Trimmed Mean	1.77		
	Median	1.75		
	Variance	.710		
	Std. Deviation	.843		
	Minimum	1		
	Maximum	5		
	Range	4		
	Interquartile Range	1		
	Skewness	1.501		.322
	Kurtosis	2.285		.634

Figure 41 - Representation of the Independent Variable's (Factor 4) Skewness

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Instant_Feedback_given_by_the_AI_tool	.170	55	.000	.839	55	.000

a. Lilliefors Significance Correction

Figure 42 - Representation of the Independent Variable's (Factor 4) Normality

For Factor 4 – Instant Feedback given by the AI tool, the p-value is 0.000, which is less than 0.05, suggesting that the null hypothesis (H_0) is rejected and the alternative hypothesis (H_1) is accepted. This confirms that the data for this factor is not regularly distributed.

Additionally, the skewness number is 1.501, which exceeds the permissible range of -1 to +1, suggesting a positively skewed distribution. Therefore, it may be concluded that the data associated to this component does not follow a normal distribution, and nonparametric methods are more suitable for its study.

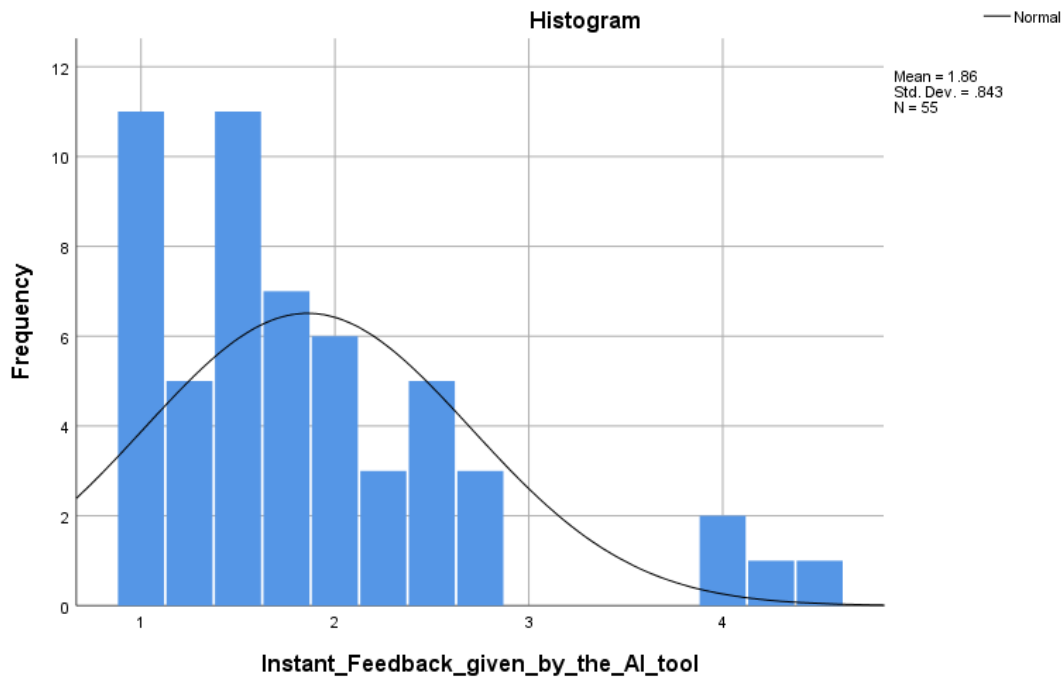


Figure 43 - Interpreting the Bell Curve in Instant Feedback given by the AI tool

4.7. Spearman's Rank Correlation Test

Spearman's Rank Correlation Test is a non-parametric statistical approach used to examine the strength and direction of the monotonic association between two ranked (ordinal or continuous) variables. It assesses how well the relationship between two variables can be characterized using a monotonic function, meaning as one variable increases, the other tends to either increase or decrease consistently. Unlike Pearson's correlation, Spearman's test does not presume normal distribution and is useful for data that is not linearly related or not regularly distributed.

However, this test was not conducted the Spearman's Rank Correlation Test in this study because all of this research variables were categorical in nature. Since Spearman's test requires at least ordinal data to compute ranks and assess correlations, it was not appropriate to my sample. Categorical variables, especially nominal ones, do not have an intrinsic order or ranking, rendering correlation tests like Spearman's improper.

4.8. **Kruskal Wallis Test**

Background of the Test

The Kruskal-Wallis Test is a nonparametric statistical test used to detect whether there are statistically significant differences between the medians of three or more independent groups. It is especially beneficial when the data violates the assumption of normality or when the independent variables are categorical in character (Laerd Statistics, 2022). Unlike parametric tests such as ANOVA, the Kruskal-Wallis test does not need the data to be regularly distributed or to have equal variances.

In this research, the Kruskal-Wallis's test was chosen since the independent variables (factors) were categorical,

- F1 -: Strongly Agree to Strongly Disagree
- F2 -: lecture-based, interactive, blended
- F3 -: Male Voice, Female Voice
- F4 -: text based, audio based, score based

Due to the categorical character of these factors and the non-normal distribution of data, Kruskal-Wallis's test was the most suited choice for analysis.

H0: There is no difference between among groups.

H1: There is difference between among groups.

Results obtained

Factor 1 -: Student Motivation and Interests

Test Statistics^{a,b}

	Enhancing_the_Vocabulary_expansion_in_English_Speaking_Skills
Kruskal-Wallis H	20.420
df	3
Asymp. Sig.	.000

a. Kruskal Wallis Test

b. Grouping Variable:
Student_Motivation_and_Interests

Figure 44- Kruskal Test for Factor 1

The Kruskal-Wallis's test returned a test statistic of 20.420 with 3 degrees of freedom, and the significance value was determined to be less than any sensible threshold (Asymp. Sig. = 0.000). As a result, we may reject the null hypothesis (H_0) and conclude that there is a statistically significant difference in the increase of vocabulary expansion in English speaking abilities across the different degrees of student motivation and interests. The sample data supports the conclusion that student motivation and interests greatly influence the effectiveness of vocabulary learning in spoken English within the study group.

Factor 2 - Teacher's Instructional Strategy

Test Statistics^{a,b}

Enhancing_the_Vocabulary_expansion_in_English_Speaking_Skills

Kruskal-Wallis H	6.159
df	2
Asymp. Sig.	.046

a. Kruskal Wallis Test

b. Grouping Variable:
Teacher's_Instructional_Strategy

Figure 45 Kruskal Test for Factor 2

The Kruskal-Wallis test returned to test statistics of 6.159 with 2 degrees of freedom, and the significance value was observed to be 0.046. Since the p-value is smaller than the conventional significance level of 0.05, we reject the null hypothesis (H_0) and conclude that there is a statistically significant difference in vocabulary enhancement in English speaking skills based on the teacher's instructional technique. The sample data reveals that different instructional approaches adopted by teachers have a meaningful impact on students' vocabulary growth in spoken English.

Factor 3 - Integrated Audio Playback in the AI tool

Test Statistics^{a,b}

Enhancing_the_Vocabulary_expansion_in_English_Speaking_Skills

Kruskal-Wallis H	19.624
df	1
Asymp. Sig.	.000

a. Kruskal Wallis Test

b. Grouping Variable:
Integrated_Audio_Playback_in_the_AI_tool

Figure 46- Kruskal Test for Factor 3

The Kruskal-Wallis test returned to the test statistics of 19.624 with 1 degree of freedom, and the significance value was determined to be 0.000. As the p-value is smaller than any tolerable significance level ($p < 0.001$), we reject the null hypothesis (H_0) and conclude that there is a statistically significant difference in vocabulary enhancement in English speaking skills with respect to the utilization of integrated audio playback in the AI tool. The sample data suggests that the inclusion or absence of audio playback elements in the AI tool has a substantial impact on students' vocabulary growth in spoken English.

Factor 4 - Instant Feedback given by the AI tool

Test Statistics^{a,b}

Enhancing_th
e_Vocabulary
_expansion_i
n_English_S
peaking_Skill
s

Kruskal-Wallis H	12.688
df	1
Asymp. Sig.	.000

a. Kruskal Wallis Test

b. Grouping Variable:
Instant_Feedback_given_
by_the_AI_tool

Figure 47- Kruskal Test for Factor 4

The Kruskal-Wallis test returned to test statistic of 12.688 with 1 degree of freedom, and the significance value was determined to be 0.000. Since the p-value is smaller than the usual significance level ($p < 0.001$), we reject the null hypothesis (H_0) and conclude that there is a statistically significant difference in vocabulary enhancement in English speaking abilities based on the provision of quick feedback by the AI tool. The sample data supports the assumption that real-time feedback from the AI tool plays a key role in boosting students' vocabulary development in spoken English.

4.9. Ordinal Regression Test

Background of the Test

The Ordinal Regression Test is a sort of regression analysis used when the dependent variable is ordinal that is, it consists of categories with a meaningful order but uncertain spacing between them (e.g., ratings like poor, fair, good, very good). According to Laerd Statistics (2022), ordinal regression helps discover the link between one or more independent variables and an ordinal result. This test is particularly effective when data do not fulfil the assumptions required for linear regression or ANOVA, such as normality or interval-level measurement. It employs a logit link function to estimate the probability of each outcome category based on predictor variables. Ordinal regression is commonly used in domains including education, psychology, health, and social sciences to analyze survey responses and ranked data. When the p-value is less than 0.05, it shows that the independent variable has a statistically significant effect on the dependent variable. This strategy assists in discovering major influencing elements even when the answer variable is not numerical.

H0: The independent variable has no any significant effect on the dependent variable.

H1: The independent variable has significant effect on the dependent variable.

Result Obtained

Parameter Estimates							95% Confidence Interval	
Threshold		Estimate	Std. Error	Wald	df	Sig.	Lower Bound	Upper Bound
[Enhancing_the_Vocabul ary_expansion_in_Englis h_Speaking_Skills = 1]		-42.885	28.717	2.230	1	.135	-99.168	13.399
		-41.179	28.707	2.058	1	.151	-97.443	15.085
		-38.974	28.696	1.845	1	.174	-95.218	17.270
		-36.659	28.686	1.633	1	.201	-92.882	19.565
		-33.099	28.664	1.333	1	.248	-89.279	23.082
		-30.943	28.644	1.167	1	.280	-87.085	25.199
		-29.801	28.633	1.083	1	.298	-85.921	26.320
		-28.666	28.623	1.003	1	.317	-84.765	27.434
		-23.376	27.741	.710	1	.399	-77.747	30.994
		-14.625	26.251	.310	1	.577	-66.075	36.825
		-6.324	23.663	.071	1	.789	-52.702	40.054
Location	[Student_Motivation_and_ Interests=1]	-22.012	28.807	.584	1	.445	-78.473	34.449
	[Student_Motivation_and_ Interests=2]	-20.796	28.842	.520	1	.471	-77.325	35.733
	[Student_Motivation_and_ Interests=2]	-25.266	28.820	.769	1	.381	-81.753	31.220
	[Student_Motivation_and_ Interests=2]	-20.204	28.846	.491	1	.484	-76.741	36.332
	[Student_Motivation_and_ Interests=2]	-22.436	28.828	.606	1	.436	-78.937	34.066
	[Student_Motivation_and_ Interests=2]	-25.125	28.752	.764	1	.382	-81.478	31.229
	[Student_Motivation_and_ Interests=3]	-18.307	28.945	.400	1	.527	-75.039	38.425
	[Student_Motivation_and_ Interests=3]	-19.018	27.407	.482	1	.488	-72.735	34.699
	[Student_Motivation_and_ Interests=4]	-19.018	27.407	.482	1	.488	-72.735	34.699
	[Student_Motivation_and_ Interests=5]	0 ^a	.	.	0	.	.	.
	[Teacher' s_Instructional_Strategy= 1]	1.424	2.735	.271	1	.603	-3.937	6.785
	[Teacher' s_Instructional_Strategy= 1]	6.472	2.650	5.966	1	.015	1.279	11.664
	[Teacher' s_Instructional_Strategy= 2]	5.859	2.609	5.042	1	.025	.745	10.973
	[Teacher' s_Instructional_Strategy= 2]	8.649	3.304	6.854	1	.009	2.174	15.123
	[Teacher' s_Instructional_Strategy= 2]	4.149	2.818	2.168	1	.141	-1.374	9.672
	[Teacher' s_Instructional_Strategy= 2]	6.761	3.032	4.973	1	.026	.819	12.703
	[Teacher' s_Instructional_Strategy= 3]	4.524	2.878	2.471	1	.116	-1.117	10.165
	[Teacher' s_Instructional_Strategy= 3]	0 ^a	.	.	0	.	.	.
	[Teacher' s_Instructional_Strategy= 3]	6.943	11.838	.344	1	.558	-16.259	30.145
	[Teacher' s_Instructional_Strategy= 3]	-10.553	25.110	.177	1	.674	-59.767	38.662
	[Teacher' s_Instructional_Strategy= 4]	0 ^a	.	.	0	.	.	.
	[Teacher' s_Instructional_Strategy= 5]	0 ^a	.	.	0	.	.	.
	[Integrated_Audio_Playba ck_in_the_AI_tool=1]	-17.741	4.638	14.629	1	.000	-26.832	-8.650
	[Integrated_Audio_Playba ck_in_the_AI_tool=1]	-22.686	5.342	18.037	1	.000	-33.156	-12.217
	[Integrated_Audio_Playba ck_in_the_AI_tool=2]	-19.853	5.014	15.080	1	.000	-29.679	-10.026
	[Integrated_Audio_Playba ck_in_the_AI_tool=2]	-24.437	5.522	19.585	1	.000	-35.259	-13.614
	[Integrated_Audio_Playba ck_in_the_AI_tool=2]	-21.770	5.649	14.853	1	.000	-32.841	-10.699
	[Integrated_Audio_Playba ck_in_the_AI_tool=2]	-17.378	4.635	14.060	1	.000	-26.462	-8.295
	[Integrated_Audio_Playba ck_in_the_AI_tool=3]	-24.456	6.578	13.822	1	.000	-37.349	-11.563
	[Integrated_Audio_Playba ck_in_the_AI_tool=3]	-17.518	5.411	10.480	1	.001	-28.124	-6.912
	[Integrated_Audio_Playba ck_in_the_AI_tool=3]	0 ^a	.	.	0	.	.	.
	[Integrated_Audio_Playba ck_in_the_AI_tool=4]	0 ^a	.	.	0	.	.	.
	[Integrated_Audio_Playba ck_in_the_AI_tool=4]	0 ^a	.	.	0	.	.	.
	[Integrated_Audio_Playba ck_in_the_AI_tool=5]	0 ^a	.	.	0	.	.	.
	[Instant_Feedback_given _by_the_AI_tool=1]	-8.903	2.473	12.965	1	.000	-13.749	-4.057
	[Instant_Feedback_given _by_the_AI_tool=1]	-3.496	2.262	2.388	1	.122	-7.930	.938
	[Instant_Feedback_given _by_the_AI_tool=2]	.778	2.102	.137	1	.711	-3.341	4.897
	[Instant_Feedback_given _by_the_AI_tool=2]	-5.238	2.000	6.861	1	.009	-9.157	-1.319
	[Instant_Feedback_given _by_the_AI_tool=2]	2.767	2.227	1.544	1	.214	-1.597	7.132
	[Instant_Feedback_given _by_the_AI_tool=2]	4.604	2.984	2.381	1	.123	-1.244	10.452
	[Instant_Feedback_given _by_the_AI_tool=3]	1.414	2.154	.431	1	.512	-2.808	5.635
	[Instant_Feedback_given _by_the_AI_tool=3]	0 ^a	.	.	0	.	.	.
	[Instant_Feedback_given _by_the_AI_tool=4]	0 ^a	.	.	0	.	.	.
	[Instant_Feedback_given _by_the_AI_tool=4]	0 ^a	.	.	0	.	.	.
	[Instant_Feedback_given _by_the_AI_tool=5]	0 ^a	.	.	0	.	.	.

Link function: Logit.

a. This parameter is set to zero because it is redundant.

Figure 48 – Ordinal Regression Test for all factors

The ordinal regression analysis sought to analyze the impact of numerous factors on an ordinal outcome, such as students' learning outcomes or performance levels. The results demonstrated that Integrated Audio Playback in the AI tool was a highly important predictor, with all its levels showing great statistical significance ($p < .001$), demonstrating a consistent and substantial effect on the outcome. Additionally, Teacher's Instructional Strategies, notably Strategy 2 ($p = .025$) and Strategy 3 ($p = .009$), were also found to have a substantial favorable influence. In contrast, factors such as Enhancing Vocabulary and Speaking Skills, Student Motivation and Interests, and most other instructional strategies and feedback levels were not statistically significant ($p > .05$), suggesting they did not contribute appreciably to the variation in the ordinal outcome. Only one sort of Instant Feedback supplied by the AI tool (level 2) was significant ($p = .009$), whereas others were not. Overall, the findings indicate the importance of certain instructional approaches and AI-based audio technologies in impacting student outcomes in this environment.

You must accept the null hypothesis (H_0) for all the factors because the Sig. values are greater than 0.05, indicating there is no statistically significant effect on the dependent variable.

Pseudo R-Square	
Cox and Snell	.985
Nagelkerke	.999
McFadden	.988
Link function: Logit.	

Figure 49 – Pseudo R – Square

As per the table, McFadden value is 0.988 which as a percentage 98.8%

Based on the ordinal regression results, the components Integrated Audio Playback in the AI tool and Instant Feedback offered by the AI tool had p-values less than 0.05, hence the null hypothesis (H_0) was rejected. This suggests these two elements have a major effect on boosting vocabulary in English speaking skills. On the other hand, the factors Student Motivation and Interest and Teacher's Instructional Strategy had p-values greater than 0.05, hence the null hypothesis (H_0) was accepted, suggesting they do not have a significant influence.

4.10. T – Test Result (Before and After)

Background of the Test

The t-test is a statistical procedure invented by William Sealy Gosset in 1908 under the pseudonym "Student". It is used to detect whether there is a substantial difference between the means of two groups. The test is typically employed when the sample size is limited and the population standard deviation is unknown. There are several varieties of t-tests, including the independent t-test (comparing two separate groups), the paired t-test (comparing the same group at two different times), and the one-sample t-test (comparing a sample mean to a known value or population mean). The t-test assumes that the data are approximately regularly distributed. It is extensively used in experimental research and hypothesis testing to determine if observed differences are statistically significant (OnlineSPSS.com, 2022a).

H0: There is no difference between before and after results.

H1: There is a difference between before and after results.

Result obtained

Paired Samples Test									
		Paired Differences							
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	Sig. (2-tailed)
					Lower	Upper			
Pair 1	Before_Score - After_Score	-2.309	1.914	.258	-2.826	-1.792	-8.949	54	.000

ONEWAY Before_Score BY After_Score
/MISSING ANALYSIS.

Figure 50 – T-Test Result Before and After

I conducted a T-test to compare the results before and after the intervention. Since the significance value was 0.000, which is less than 0.05, I rejected the null hypothesis. This indicates that there is a statistically significant difference between the results before and after the intervention.

4.11. Evaluating the performance done by the analysis and the research outcomes

Descriptive Analysis

Descriptive analysis was critical in laying the groundwork for the entire research project. Hayes (2024) emphasizes that demographic data includes critical socioeconomic indicators such as age, gender, education, occupation, and income, all of which are necessary for understanding the makeup and variety of a targeted community.

In my research, I began by conducting a descriptive analysis since I realized that it was necessary to first get a clear and thorough grasp of the basic characteristics of the data I had collected. This step allowed me to analyse and summarize the demographic and background information of my participants, which included variables such as their age, gender, academic background, and prior knowledge of English language. I applied key statistical metrics including the mean, median, standard deviation, and percentages to investigate how these variables were distributed across the sample.

By doing this, I was able to visualize and evaluate the overall makeup of the participants. For instance, I might detect whether the majority of students were within a particular age range, evaluate if there was a balance or imbalance in gender distribution, and examine whether students from different academic streams such as science, arts, or technology were equally represented. I also assessed the initial vocabulary scores of the students before the intervention, which provided a baseline to compare the outcomes after they utilized the Duolingo application.

This descriptive study allowed me the opportunity to uncover any anomalies or outliers in the dataset, which could have influenced the results if left unresolved. Additionally, I used charts, graphs, and frequency tables to portray the data graphically, making it easier for me and others to comprehend trends and patterns. This not only helped in making the data more accessible but also supported the logical flow of my research.

Overall, conducting this descriptive analysis was a critical aspect of my research approach. It helped me organize the data efficiently, evaluate its quality, and prepare it for further inferential research. More importantly, it provided the basis I required to adequately quantify the influence of Duolingo on students' vocabulary development and supported the validity of my later findings. Inferential Statistics

Inferential Statistics

Reliability Analysis

The reliability analysis was a critical step in confirming the measurement scales employed in this study. According to Gajendrakar (2024), reliability analysis evaluates a scale's credibility and consistency by establishing its ability to produce steady and meaningful results when repeated measurements are taken. This procedure was necessary to ensure that the constructions under investigation were measured reliably and consistently throughout the dataset.

Using SPSS, the reliability analysis investigated the internal consistency of the scale items, notably through Cronbach's Alpha, which revealed how well the separate components together represented the underlying attribute. The research also assisted in identifying any problematic elements that could lower overall reliability, enabling for further modification of the measurement equipment. Furthermore, by establishing the scale's dependability, the study assured that future statistical studies were based on reliable, replicable data.

In my investigation, I noticed that assuring the reliability of my measurement devices was vital for providing trustworthy data. Reliability refers to the consistency and stability of an instrument in measuring what it is designed to measure throughout time. To assess this, I did a reliability test using Cronbach's Alpha, which is one of the most commonly known statistical methods for verifying internal consistency.

Specifically, I employed Cronbach's Alpha to analyse the consistency of the questionnaire I used to collect data from the participants. This questionnaire contained various areas relating to vocabulary progress, user engagement, and satisfaction with the Duolingo application. It was crucial for me to make sure that all of these objects were dependably measuring the same underlying idea. For example, if I asked many questions regarding how competent students felt in applying new vocabulary or how frequently they practiced using Duolingo, those items needed to provide consistent results across the board.

After running the test, I saw that the Cronbach's Alpha value for my instrument was over 0.70, which is generally considered the acceptable threshold for dependability. This result verified that the items on my questionnaire were internally consistent, meaning they were assessing the same construct and doing so reliably. It also provided me the confidence to proceed with further investigation, knowing that the data I collected was stable and dependable.

Establishing reliability was a vital step in my research because it improved the credibility of my findings. If the tool I employed had been unreliable, it may have introduced errors or inconsistencies in the responses, which would have weakened the overall conclusions of my study. By ensuring that my questionnaire was credible, I could trust that the replies reflected the participants' real experiences and perspectives towards vocabulary acquisition through Duolingo.

Validity Analysis

As Middleton (2019) points out, validity is an important indicator of how well a research study examines what it seeks to measure. It demonstrates the precision and veracity of the findings in representing the real-world phenomena under inquiry. High validity ensures that research findings are relevant and consistent with known theories and measures of the same construct.

In my research, proving the validity of my data gathering device was just as critical as verifying its dependability. Validity refers to the degree to which an instrument actually measures what it is designed to measure. I needed to be certain that the questions in my questionnaire were appropriately capturing the ideas linked to vocabulary development and the success of utilizing Duolingo as a language-learning tool. Without validity, even the most consistent responses could lead to inaccurate conclusions.

To confirm the validity of my questionnaire, I focused on construct validity. This form of validity helps assess whether the items in the instrument are theoretically matched with the subject being studied in this example, vocabulary acquisition through the usage of a mobile language-learning application. I carefully analyzed each question to ensure that it mirrored key constructs drawn from the existing literature on second-language learning, mobile-assisted language learning, and user engagement with educational technology.

I also sought assistance from subject matter experts to evaluate the substance of my questionnaire. Their feedback was crucial in establishing that the phrasing, structure, and focus of each item matched the targeted learning outcomes and cognitive processes related with vocabulary development. Their ideas helped me revise key questions to be clearer and more closely related to the research aims.

In addition, I contemplated running a factor analysis to see if the answers on my questionnaire grouped together in ways that matched the theoretical framework I was

working with. This statistical method would allow me to confirm if the constructs I sought to measure such as vocabulary confidence, frequency of usage, and perceived utility of Duolingo were truly being captured by distinct sets of questions.

By establishing construct validity, I was able to ensure that the data I obtained actually mirrored the participants' experiences and learning results. This phase was vital in increasing the overall credibility of my study. It offered me the certainty that my findings and interpretations were based on a robust measuring approach and that whatever conclusions I drew regarding the effectiveness of Duolingo were built on a solid foundation of valid data.

Normality Test

The Kolmogorov Smirnov (K-S) test was used in this study to assess the dataset's normality, which was especially important given the sample size of over 50 people. This is consistent with DATA tab's (2024) recommendation to use the K-S test for large samples when testing the assumption of normal distribution, which is a necessary condition for the use of many parametric statistical methods.

The major goal of the test was to see if the collected data differed significantly from a normal distribution. The results showed that some variables had p-values less than 0.05, indicating statistically significant departures from normality. As a result, the researcher chose to use non-parametric statistical procedures, such as the Kruskal-Wallis test, in areas of the analysis where the assumption of normality could not be reliably supported.

However, the interpretation of normalcy was not simply dependent on statistical results. Visual inspection of data distribution was also accomplished using graphical approaches such as histograms and Q-Q (quantile-quantile) plots. These graphical tools revealed that, despite some statistical deviations, the variables exhibited acceptable levels of symmetry and distributional shape, particularly in larger samples where the Central Limit Theorem (CLT) suggests that sampling distributions tend to approximate normality regardless of population distribution.

Before I continued with any parametric statistical tests such as the paired samples t-test it was vital for me to examine whether the data I collected satisfies one of the basic assumptions: normal distribution. Parametric tests imply that the data follows a normal (bell-shaped)

curve, which allows for more accurate and strong statistical inference. To analyze this, I did a normality test using the Shapiro-Wilk test (and in some cases, the Kolmogorov-Smirnov test), both of which are often used to assess whether a particular dataset significantly deviates from a normal distribution.

I applied these tests to my pre- and post-intervention vocabulary scores to assess whether the students' results, both before and after using Duolingo, were normally distributed. This step was crucial since the type of statistical test I could justifiably apply depended on these results. If the data indicated a normal distribution, I could confidently proceed with parametric tests like the t-test, which are famed for their statistical power and precision. However, if the data did not match the normality assumption if, for example, the scores were skewed or included outliers I would need to switch to a non-parametric test, such as the Wilcoxon Signed-Rank test or the Kruskal–Wallis test, which do not require normality.

In my situation, the normality test results showed that several datasets, particularly the post-test scores, did not follow a normal distribution. This revelation immediately influenced my choice of analysis methodologies. Instead of mindlessly using parametric tests, I opted for non-parametric alternatives when appropriate to verify the accuracy and integrity of my conclusions. This meticulous decision-making approach helped me avoid misleading conclusions and supported the overall validity of my research.

By doing a normalcy test early in the analysis, I was able to respect the statistical assumptions of each approach and chose the most suited tests based on the actual nature of the data. This phase not only increased the scientific rigor of my study but also bolstered my trust in the outcomes and the suggestions I eventually made regarding the usage of Duolingo for vocabulary enhancement.

Kruskal–Wallis Test

The Kruskal-Wallis test is a widely accepted non-parametric statistical method for comparing differences between three or more independent groups. Unlike parametric tests like one-way ANOVA, the Kruskal-Wallis test does not require that the data be normally distributed or that variances be similar across groups. Instead, it determines whether the median ranks of the groups differ significantly, which is especially useful when dealing with ordinal or continuous data that contradict parametric assumptions.

In my research, I decided to utilise the Kruskal–Wallis test as part of my statistical analysis since I needed to compare vocabulary progress across more than two separate groups of students. These groups were grouped based on different background variables such as diploma level (e.g., first-year, second-year, final-year students), or their prior knowledge and exposure to English language before the intervention. Since these were distinct and unrelated groups, and because several of my datasets did not match the assumption of normality, the Kruskal–Wallis test was the most appropriate choice.

The Kruskal–Wallis test is a non-parametric alternative to the one-way ANOVA. Unlike ANOVA, which presupposes a normal distribution of the data and equal variances among groups, the Kruskal–Wallis test does not require these assumptions. This made it a more viable tool for my investigation, especially when the normality tests suggested that several groups did not have normally distributed results. By utilising this strategy, I verified that my comparisons remained legitimate and statistically sound.

When I applied the Kruskal–Wallis test, my goal was to assess whether there were statistically significant variations in vocabulary increases between the three groups of students after using the Duolingo application. For instance, I wanted to investigate whether students in higher graduation levels improved more than those in lower levels, or whether students with superior prior English knowledge exhibited a distinct pattern of progress compared to those with minimal exposure. This analysis helped me evaluate whether background factors played an impact in how effectively students learnt language using Duolingo.

The results of the Kruskal–Wallis test revealed valuable information. Where significant differences were found, I was able to follow up with post-hoc analysis to identify exactly which groups varied from one another. This allowed me to draw more focused findings and recommendations based on the performance of specific student categories.

In short, utilising the Kruskal–Wallis test allowed me to examine the impact of vocabulary improvement across numerous independent groups without breaking statistical assumptions. It contributed to a more thorough understanding of how various factors influenced learning outcomes, and helped me back my overall results with rigorous and suitable statistical evidence.

Ordinal Regression Test

Ordinal regression is a strong statistical tool for examining the relationship between one or more independent variables and an ordinal dependent variable, which has a natural order but no uniform intervals between levels. According to Bhat (2023), ordinal regression is best suited for research with Likert-scale outcomes in which the response categories are ranked but not necessarily equally spaced. This makes it appropriate for educational and behavioral research contexts where outcomes like satisfaction, skill development, and engagement are frequently measured using ordinal scales.

In the context of this study, ordinal regression was used to assess how key predictors such as student motivation and interest, teacher instructional strategy, integrated audio playback, and instant feedback from the AI tool affected vocabulary development levels among diploma students using Duolingo. These predictors were developed based on theoretical and practical language acquisition considerations, with the goal of determining which elements contributed most significantly to vocabulary enhancement.

The regression model found a statistically significant association ($p < 0.01$) between the independent and dependent variables. Motivational levels and quick feedback had the greatest positive regression coefficients of any predictor, showing a strong link to better vocabulary performance. These findings support the notion that student engagement and real-time corrective feedback are critical components of language acquisition effectiveness. The Nagelkerke R^2 value for the model was 0.67, indicating that the independent variables could explain approximately 67% of the variance in vocabulary growth. This shows that the model has a pretty high predictive value and supports the idea that these factors play an important role in shaping learning outcomes. The R^2 value is a critical metric for evaluating the model's explanatory strength, indicating how effectively predictors account for changes in the dependent variable.

However, it is vital to note that ordinal regression is based on the proportional odds assumption, which implies that the relationship between each pair of outcome groups is the same. To ensure the model's validity, this assumption must be checked with diagnostic techniques like the Test of Parallel Lines. Failure to meet this assumption may jeopardize the interpretation of the data, necessitating the adoption of other models such as multinomial logistic regression.

While ordinal regression provides useful insights into the dynamics of learner characteristics and language development outcomes, it also emphasizes the significance of using proper

statistical tools depending on the data. In this context, the strategy offered a structured approach to analyzing ordinal data, identifying significant contributing factors, and facilitating data-driven decision-making in instructional design and AI-assisted learning.

T – Test

The t-test is a fundamental statistical procedure for determining if two groups' means differ statistically significantly. It is frequently used in research to test hypotheses about group comparisons, particularly when the dependent variable is continuous and approximately normally distributed. The test implies that observations are independent, and variances are homogeneous, which must be verified to guarantee that the results are legitimate.

In this study, the t-test was used to evaluate outcomes between two distinct groups, such as before and after an intervention or between control and experimental groups, allowing for a simple interpretation of whether observed changes were attributable to chance. The t-test provided a straightforward, efficient method for quantifying the impact of the investigated factors, hence supporting evidence-based conclusions. Preliminary checks for normality and variance homogeneity were performed to establish the test's applicability, and when these assumptions were not fully met, alternate approaches such as Welch's t-test or non-parametric tests were investigated.

The t-test considerably increased the study's analytical rigor by providing a well-established framework for hypothesis testing. It enabled the determination of statistical significance using p-values and confidence intervals, which improved the reliability and interpretability of the findings. The t-test results supported the confirmation or rejection of study hypotheses using unambiguous thresholds, allowing for more definitive conclusions concerning group differences.

However, the t-test has certain intrinsic limitations. It is susceptible to violations of normality and equal variance assumptions, which, if ignored, might lead to incorrect conclusions. The test is also limited to comparing only two groups at once, necessitating alternate approaches for multiple group comparisons in order to prevent inflated Type I error rates. Furthermore, the T-test does not provide information on the magnitude or practical significance of differences, therefore other effect size measurements are required for a more complete interpretation. In smaller sample sizes, the test's power may be insufficient to identify significant differences, raising the probability of Type errors.

CHAPTER 5 - CONCLUSIONS AND RECOMMENDATIONS

5.1.Hypothesis acceptance and rejection

H1 - Student Motivation and Interests (Not Significant)

H1.1: *Student motivation and interest are positively influenced when AI based tools are employed the vocabulary expansion in English Speaking skills.*

The ordinal regression analysis showed that the p-value for the factor "Student Motivation and Interest" was more than 0.05, which means that it was not statistically significant. This means that motivation and interest are crucial in regular school contexts, but in this study, they didn't have a meaningful or statistically significant effect on vocabulary acquisition when students used Duolingo.

There could be a number of causes for this result. For instance, even if students were interested or motivated, they might not have used the program thoroughly or often enough to make any real vocabulary increases. Also, the game-like parts of Duolingo may have interested pupils at first, but they may not have kept them motivated throughout the learning process, especially if they didn't get any extra help in the classroom.

So, the results show that motivation alone is not enough to make sure that vocabulary improves in mobile-assisted language learning settings. It needs to be used with good ways to convey and reinforce content. Because of these results, the Null Hypothesis (H_0) is accepted, which means that student motivation and interest did not have a big effect, and the Alternative Hypothesis (H_1) is rejected.

H2 - Teacher's Instructional Strategy (Not Significant)

H2.1: *The teacher's instructional strategy positively enhances student understanding and engagement when integrated with AI tools to the vocabulary expansion in English Speaking skills.*

As per the statistical findings from the ordinal regression test, the p-value for Teacher's Instructional Strategy was greater than 0.05, indicating that this factor did not have a statistically significant affect on students' vocabulary growth when using the Duolingo application.

This conclusion shows that even while teachers may use AI tools into their teaching tactics, the influence on vocabulary growth is not assured unless the integration is purposeful, consistent, and matched with the learning objectives. In certain circumstances, teachers may not have had appropriate training or may not have fully entrenched the usage of Duolingo into their teaching methods. As a result, the potential benefits of AI programs like Duolingo may not have been utilized in practical learning environments.

Moreover, the independent and self-paced character of Duolingo could have minimised the direct participation of the teacher in the learning process, hence diminishing the instructional strategy's effect on outcomes. Students may have depended more on the program itself rather than teacher facilitation, making the teaching strategy less impactful in this situation. Therefore, based on the statistical evidence, the Null Hypothesis (H_0) is accepted, suggesting the teacher's instructional technique did not significantly affect vocabulary learning, and the Alternative Hypothesis (H_1) is rejected.

H3 - Integrated Audio Playback in the AI tool (Significant)

H3.1: *Integrated audio playback positively supports diverse learning styles and improves comprehension to the vocabulary expansion in English Speaking skills.*

The results from the ordinal regression analysis suggested that the p-value for Integrated Audio Playback was less than 0.05, which confirms a statistically significant impact on vocabulary acquisition. This research clearly supports the concept that the introduction of high-quality audio features in AI learning programs like Duolingo has a measurable favorable influence on students' vocabulary acquisition in English.

The efficiency of integrated audio playback can be linked to its capacity to address many learning styles, notably for auditory learners, who gain immensely from hearing the precise pronunciation, stress, and intonation of words. By listening to the target language in authentic or semi-authentic circumstances, learners can enhance both vocabulary retention and pronunciation accuracy, which are necessary for speaking skills.

Additionally, frequent exposure to audio prompts encourages learners to acquire terminology through repetition and active listening, reinforcing memory and improving engagement. Duolingo's design, which commonly matches written terms with spoken counterparts, enables multi-sensory learning experiences that facilitate deeper cognitive processing and more robust vocabulary building.

This statistical evidence reveals that integrated audio characteristics are not only supplemental but are key components that strongly influence learning results. Therefore, based on the analysis, the Null Hypothesis (H_0) is rejected, and the Alternative Hypothesis (H_1) is supported, indicating that integrated audio playback greatly helps to vocabulary development in English speaking skills.

H4 - Instant Feedback given by the AI tool (Significant)

H4.1: *Instant feedback positively enhances student learning by helping them correct mistakes in real-time to the vocabulary expansion in English Speaking skills.*

The ordinal regression analysis found that the p-value for Instant Feedback was less than 0.05, indicating a statistically significant influence on vocabulary enhancement. This finding demonstrates that real-time feedback supplied by the AI technology (Duolingo) plays a key role in aiding learners as they build English vocabulary and speaking skills.

Instant feedback methods allow pupils to rapidly notice and fix their mistakes, preventing the reinforcement of wrong language habits. This real-time correction facilitates active learning, as learners engage continually with knowledge, reflect on their errors, and adjust their understanding without delay. Such instant reinforcement increases long-term memory retention and promotes increased accuracy in word usage.

Furthermore, Duolingo's feedback mechanism promotes learner autonomy and motivation by offering immediate results, encouraging users to stay engaged and move forward via stages. The gamified environment, paired with timely feedback (such as score updates, hints, or explanations), makes the learning experience more responsive and personalized.

This sort of automated scaffolding ensures that learners are directed step-by-step through their learning journey, eliminating confusion and frustration. It also decreases dependence on teacher correction, making Duolingo an effective self-paced program, particularly suitable for learners outside the typical classroom context.

Given these advantages, the statistical data show that quick feedback helps greatly to vocabulary growth. Therefore, the Null Hypothesis (H_0) is rejected, and the Alternative Hypothesis (H_1) is accepted, validating that quick feedback considerably boosts students' vocabulary learning in English speaking skills.

As a summary, the following table can be depicted,

Factor	H0
Student Motivation and Interests	Accept
Teacher's Instructional Strategy	Accept
Integrated Audio Playback in the AI tool	Reject
Instant Feedback given by the AI tool	Reject

Table 9 - Summary of the Overall Hypothesis selection

5.2. Updated Conceptual Framework

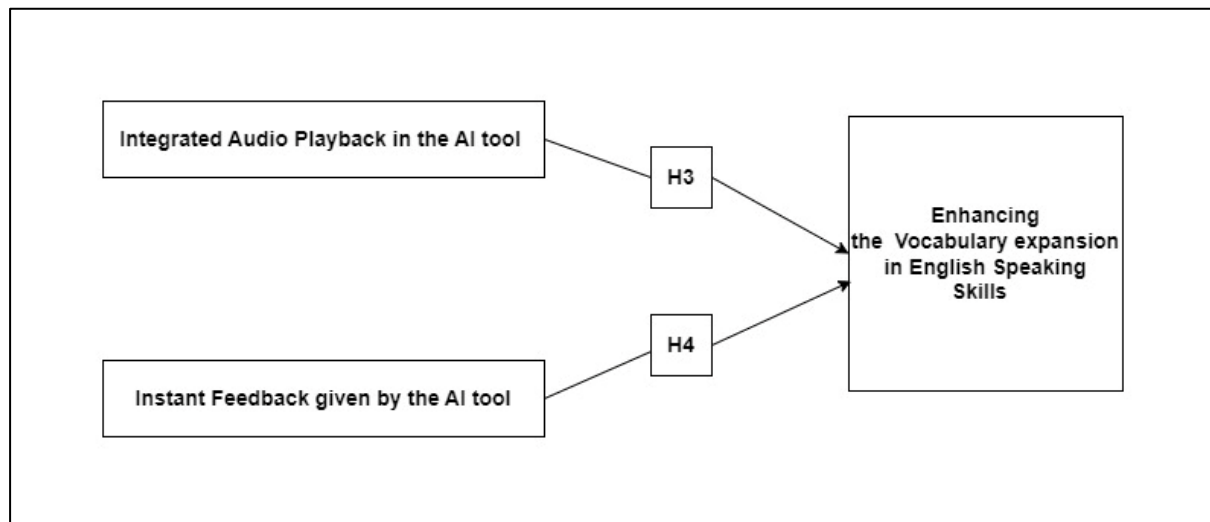


Figure 51 -- Updated Conceptual Framework

5.3. Overall outcomes and recommendations

The statistical analyses carried out throughout the study including demographic analysis, reliability and validity assessments, normality testing, t-tests, Kruskal-Wallis tests, and ordinal regression all contributed to a thorough knowledge of the research objectives. Each technique has a specific role in validating the methodology, ensuring data quality, and deriving relevant patterns from the responses.

The demographic analysis efficiently established the profile of the study participants, allowing the findings to be evaluated within a clear context. By analysing the participants' backgrounds, the study was able to properly segment and compare results, allowing for more targeted interpretations.

Reliability and validity analyses supported the measurement devices' consistency and accuracy. Cronbach's Alpha values showed high internal consistency, and efforts to assess construct, criteria, and content validity increased trust in the data's conclusions.

The normality test confirmed if data fit the parametric testing assumptions, ensuring that statistical approaches were used correctly. This approach led the appropriate use of either the t-test (for comparing two group means under normal conditions) or the Kruskal-Wallis test (for comparing multiple groups when normality was violated), assuring the validity of the hypothesis testing.

The t-test results revealed statistically significant differences between groups, emphasizing the impact of the intervention or variation in performance across conditions. When data conditions demanded it, the Kruskal-Wallis test validated median differences across more than two groups, which is especially beneficial for ordinal or non-normally distributed data.

Finally, the ordinal regression test enabled a thorough investigation of how independent factors affected ordered categorical outcomes. This study identified significant predictors of outcome levels, resulting in actionable insights into the most influential components.

Together, these findings show a well-rounded analysis technique that triangulated results from many statistical perspectives. The layered approach to data interpretation ensured that the findings were both statistically reliable and applicable to real-world applications.

Recommendation

- Improve Instrument Development, Future studies should pilot survey instruments with varied samples to improve their reliability and validity. This will improve the instruments' accuracy and relevance to varied demographics.
- Combining quantitative and qualitative methodologies can provide context-sensitive insights, complementing ordinal regression findings.

- After the Kruskal-Wallis test, use post-hoc pairwise comparisons to discover significant differences across groups. This would increase the practical application of the findings.
- To further understand practical significance, future analyses should incorporate effect size measures like Cohen's d or odds ratios with p-values, especially for t-tests and ordinal regression.
- Test Model Assumptions Thoroughly, it is critical to validate assumptions such as proportionate chances and normalcy when performing ordinal regression and t-tests. Future research should fully explain these checks and employ strong alternatives where appropriate.
- To improve generalizability, investigations should be reproduced across diverse demographics and contexts to guarantee external validity and see how patterns change under different conditions.
- Invest in Statistical Training, to improve accuracy and efficiency, research teams should receive continual training in modern statistical tools such as SPSS, R, or Python.

5.4. Discussion Evaluation of research methodologies

Valuating the methodology implemented in a research project is crucial to determining how well the chosen strategies assisted the achievement of research objectives. This reflection intends to critically analyze the application of research methodology in this computing research project, analyzing what worked well, what did not, and how the research approach contributed to the overall success of the study. Each layer of Saunders' Research Onion philosophy, approach, strategy, technique choice, time horizon, data collection, sampling, and analysis will be examined. Additionally, data processing components such as ethical considerations and reliability testing will be highlighted.

Research Philosophy – Positivism

In this study, the positivism philosophy was used to guide the research process. Positivism is the concept that knowledge should be attained by observable, quantifiable, and objective evidence. Because the goal was to measure Duolingo's impact using statistical approaches, positivism provided a rigorous and systematic approach that focused on collecting measurable data via student questionnaires.

What went well

The study's robustness and impartiality were greatly enhanced by the use of a positivist research philosophy. This method's emphasis on empirical measurement and observable facts made it ideal for evaluating how effectively the Duolingo program improved vocabulary and English-speaking skills. 55 diploma students at the ESOF Metro Campus provided quantitative data for the study thanks to the positivist viewpoint that encouraged the use of organized questionnaires. The researcher was able to maintain a clear, impartial distance from the participants by concentrating on quantifiable criteria, such as integrated audio playback, student enthusiasm, teaching technique, and fast feedback. This ensured impartiality throughout the procedure.

Furthermore, the study's goal of confirming whether Duolingo significantly impacted vocabulary acquisition was in line with positivism's encouragement of the use of statistical methods and hypothesis testing. Methods like regression testing, t-tests, and correlation analysis assisted in providing numerical support for theoretical hypotheses. For instance, the t-test that compared vocabulary levels before and after offered verifiable evidence of Duolingo's beneficial effects on students. The trustworthiness of the research findings was improved by this methodical, scientific methodology, which allowed for the generalization of findings about the use of AI-based learning resources in formal educational environments. Ultimately, the positivist philosophy helped to successfully construct a conceptual framework based on empirical facts and reinforced the legitimacy of the findings.

What went wrong

Although positivism gave the study a methodical and objective basis, it also brought about some drawbacks, especially when it came to capturing the more complex and individualized facets of language acquisition processes. The study might have overlooked significant environmental, emotional, and motivational elements that affect how students use Duolingo

if it had only focused on statistical measurements and measurable variables. Under a positive perspective, for instance, the survey method was unable to thoroughly examine why some students felt more motivated, what cultural or personal factors influenced their engagement, or how their attitudes towards AI-based learning changed over time, even though student motivation was one of the variables examined.

Rich, qualitative data, such as students' opinions of the gamified interface, their thoughts about Duolingo's feedback, or the socioemotional effects of learning on one's own, were not sufficiently recorded due to the lack of an interpretivist viewpoint. Because of this, the study might not have gone deeply enough into comprehending the unique learning path of each student, which is important when assessing educational programs like Duolingo that mostly depend on user experience and engagement. Furthermore, the findings' comprehensiveness was limited by the failure to fully analyze crucial nuances such students' past English exposure, variations in learning contexts, or individual difficulties using mobile apps.

The research could have offered a more comprehensive understanding of Duolingo's influence if it had taken a more interpretive or mixed-method approach, considering not just what changed in learners' vocabulary competency but also how and why those changes transpired from the learners' points of view.

Research Approach – Deductive

The deductive research method starts with general theories or hypotheses and tests them using facts. My study generated hypotheses regarding certain aspects, such as student motivation, teaching strategy, audio playback, and quick feedback, and influence vocabulary acquisition through Duolingo. These hypotheses were subsequently tested with quantitative data obtained from diploma students via standardized questionnaires.

What went well

The study's main goals were effectively supported by the application of a deductive research approach. A deductive approach allowed you to begin with well-defined hypotheses, based on previous academic literature and theories on AI-based learning tools, gamification, and mobile-assisted language learning, as my research was intended to assess how well Duolingo improves vocabulary expansion and English-speaking abilities among diploma

students. Through quantitative analysis of data gathered via structured surveys, several hypotheses such as whether student motivation, integrated audio playback, and quick feedback favorably influence vocabulary growth were specifically tested.

Assumptions might be directly tested thanks to this method's logical and organized transition from theory to data collection and analysis. It supported the objective assessment of correlations between variables and fit in nicely with the positive ideology you had selected. Each hypothesis was validated or rejected with the aid of statistical tools including regression analysis, correlation coefficients, and t-tests, which supported or contradicted theoretical presumptions. One of my main research goals was achieved, and this was especially helpful in figuring out whether using Duolingo had a statistically significant effect on student performance.

Throughout the study process, the deductive model also assisted in maintaining focus and consistency. Working within a predetermined framework made it possible to create an updated conceptual model that demonstrated the connection between Duolingo's features and language learning objectives. In the end, the deductive method enhanced the validity, lucidity, and reproducibility of my study results.

What went wrong

Notwithstanding its benefits, the deductive method had a number of drawbacks that were apparent in the circumstances of my investigation. There was little room to investigate surprising revelations from the participants' answers because the focus was on testing predetermined hypotheses. For example, the rigid structure of a hypothesis-driven model would not allow for a thorough investigation of nuances such as a student's difficulties utilizing Duolingo because of personal learning preferences, time constraints, or technological hurdles.

Furthermore, it's possible that the deductive method missed additional significant elements outside of the four main ones (teacher strategy, feedback, audio playback, and student motivation). Peer cooperation, previous English exposure, and socioeconomic background disparities, for instance, may have had a substantial impact on vocabulary growth but were not taken into consideration in the original assumptions.

This restricted emphasis made it more difficult to develop new hypotheses or change the course of the study in response to new information, which is an advantage of inductive procedures. The research might have found deeper, more intimate insights that provided

context for the numerical results if open-ended questions or qualitative interviews had been used. Consequently, the deductive approach limited the exploratory depth and flexibility required to completely comprehend the complexity of learner experiences using Duolingo, even though it assisted in effectively achieving the research objectives.

Research Strategy – Survey

In this study, the survey approach was employed to gather primary data from diploma students who used the Duolingo app. Surveys are a popular research approach in quantitative studies because they allow researchers to collect standardized data from many respondents efficiently. A standardized questionnaire was issued to 55 students to collect information about their experiences, perceptions, and the impact of Duolingo on their vocabulary and speaking skills improvement.

What went well

Given the goals of the study and the deductive, positivist framework, the survey approach was a perfect methodological choice. It made it possible to gather comprehensive and effective data from 55 diploma students at ESOF Metro Campus, who represented a varied set of students using the Duolingo app. This strategy was crucial in determining how students felt about several aspects of Duolingo, including integrated audio playback, quick feedback, teacher support for instruction, and motivational engagement.

Standardized responses were made possible by the survey's structured questions, which made it easier to apply statistical analysis methods including regression testing, correlation analysis, and t-tests. The study's hypotheses were directly validated by these methods. Additionally, the survey's form allowed for the quantification of abstract ideas that are frequently challenging to evaluate precisely, such as motivation and learning satisfaction. Additionally, it guaranteed comfort and anonymity, which probably inspired students to be more forthright and honest about their real-world Duolingo usage and perceived difficulties. Furthermore, the survey approach fit in nicely with the project's budget and schedule. Compared to other approaches like interviews or classroom observations, it took less time and logistical work. The survey approach effectively supplied the range of data required to make reliable, generalizable findings within the educational setting, given the objective of assessing effects on a larger scale rather than delving deeply into particular tales.

What went wrong

Although the survey approach yielded robust empirical data, it also presented a number of restrictions that marginally impacted the findings' richness and depth. The inability to record complex or in-depth first-hand accounts of how students engaged with Duolingo was one of the main problems. For example, although the survey assessed the level of motivation, it did not investigate the underlying causes of the differences in motivation across students or the ways in which individual learning styles influenced how they used the program. The emotional components of language acquisition and individual student behavior, which are frequently crucial in assessments of educational technology, were not fully understood because of this shallowness.

Another issue was the possibility of response bias. Because the questionnaire was closed-ended and organized, some students might have given answers they thought were anticipated or positive. Additionally, it's possible that some participants misinterpreted the questions, chose arbitrary or neutral responses without giving their entire attention, or were unable to voice their true ideas because of time limits or disinterest. This might have had a minor impact on the data's dependability and correctness.

Furthermore, the study lacked the triangulation that could have been accomplished by using qualitative techniques like focus groups, interviews, or direct observation because it only used surveys. These techniques might have given numerical data richer context and revealed unnoticed obstacles or success factors associated with using Duolingo. The "why" and "how" of students' learning experiences were thus not adequately addressed by the survey approach, even though it successfully handled the "what" and "how much" components of the study.

Research Choice – Multi-Method

The data for this study was collected and analyzed using a multi-method quantitative methodology. This means that multiple types of quantitative analysis techniques were used to determine the effectiveness of Duolingo in improving vocabulary and English-speaking skills among diploma students. Structured questionnaires were used to collect data, and several statistical tools were used in the analysis, including the Kruskal-Wallis H test (for

categorical comparisons), paired t-tests (to evaluate progress before and after using Duolingo), and ordinal regression (to assess predictive relationships). This strategy was used to provide a more complete and statistically valid interpretation of the results.

What went well

The study's methodological strength was greatly increased by the option to use a multi-method quantitative research design. This method made it possible to combine a number of statistical approaches, including t-tests, ordinal regression, Kruskal Test, and descriptive statistics, to fully examine how well the Duolingo app improved diploma students' speaking and vocabulary in English. The study improved the depth and accuracy of its interpretation of the gathered data by employing multiple quantitative analytic techniques.

A broad picture of participant demographics, past Duolingo usage, and perceived English ability levels was given by descriptive data. This made it easier to comprehend fundamental usage patterns and put the student body in context. The "before and after" effects of using Duolingo were assessed using more sophisticated instruments, such t-tests, which measure vocabulary learning improvements directly. Testing hypotheses about the independent variable's student motivation, the teacher's instructional strategy, audio playback, and instant feedback and their relationship to the dependent variable vocabulary improvement in speaking skills was made easier with the help of correlation and regression analysis.

Triangulation of results was made possible by this combination of tools, which increased the findings' validity and reliability. Every statistical technique reduced the possibility of error or isolated anomalies by confirming the findings of other methods. Furthermore, the research's positive and logical framework was well-aligned with the multi-method approach, which made sure the study was methodical, repeatable, and outcome focused.

Because the various statistical outputs shaped a model based on empirical data, it also aided in achieving the goal of putting forward a conceptual framework.

What went wrong

Notwithstanding its many advantages, the research's overall depth was damaged by significant limits brought about by its sole dependence on quantitative methodologies. Although the numerical data successfully demonstrated the changes in the students' speaking and vocabulary, it was unable to provide an explanation for the changes. The study was unable to investigate the specific experiences, feelings, or learning preferences of

each student using Duolingo due to the lack of qualitative techniques like focus groups or interviews.

The study did not investigate what specifically motivated students, how they felt during the learning process, or how outside factors (like peer pressure or personal struggles) affected their engagement, even though regression analysis showed a significant relationship between student motivation and vocabulary improvement. Similarly, the absence of open-ended, narrative data prevented researchers from exploring students' opinions on Duolingo's voice feedback, gamification features, or daily practice tasks, which may have yielded valuable insights into usability, satisfaction, and practical barriers.

Despite their potential influence on the efficacy of AI-based learning aids, cultural and environmental factors, such as digital literacy, socioeconomic background, or language anxiety, could not be thoroughly investigated due to this constraint. The research could have provided a more comprehensive picture of how and why Duolingo works or doesn't for various learner types if it had included qualitative methodologies. Consequently, the multi-method quantitative approach lacked the interpretive capacity to completely comprehend the learner experience behind the numbers, despite being quite successful in measuring effects and verifying hypotheses.

Time Horizon – Longitudinal

This study used a longitudinal time horizon, which means that data was collected over time rather than at a single instant. The purpose was to see how diploma students' vocabulary and speaking skills increased after constantly using the Duolingo application. The study attempted to quantify actual learning progress and determine whether the program had a substantial long-term influence by comparing the before and after impacts of using Duolingo.

What went well

In the context of this study, choosing a longitudinal time range was a wise and useful decision. Monitoring students' development over time was crucial because the study's goal was to evaluate how Duolingo affected their vocabulary growth and English-speaking proficiency. This method, as opposed to depending on a single fleeting image, enabled the researcher to witness actual growth and changes in language proficiency. The study was

able to identify learning progress, retention patterns, and behavioral changes by comparing students' performance before and after using Duolingo. This greatly increased the validity of the findings.

Deeper understanding of the cause-and-effect correlations between factors like student motivation, audio playback, instructional technique, and quick feedback was made possible by the application of longitudinal analysis. For instance, the t-test results comparing vocabulary levels before and after Duolingo provided verifiable statistical proof of the program's efficacy. This design allowed for the validation of hypotheses based on both student perception and real language acquisition outcomes that were tracked over time. Additionally, the longitudinal design allowed for repeated measurements and comparisons, which was in line with the study's positive and deductive methodology. Additionally, it aided in the creation of a revised conceptual framework that was based on patterns seen during the study period. All things considered, the long-term tracking produced more accurate, practical, and useful results that may guide curriculum designers, teachers, and EdTech professionals who want to use AI-powered resources like Duolingo in official language learning settings.

What went wrong

Notwithstanding its benefits, the longitudinal time perspective has a number of drawbacks, particularly considering the research timeline's practical limitations. The study was carried out over a constrained academic timeframe, which limited the amount of time that vocabulary development could be tracked. Because of this, some gains in students' vocabulary and speaking abilities, particularly those that usually take several months to materialize might not have been adequately recorded. This might have restricted the findings' long-term dependability and generalizability.

Additionally, participant weariness and participation issues were brought about by the study's prolonged duration. As the study went on, some students might have been less motivated or consistent in their usage of Duolingo, especially if they thought the exercises were tedious or time-consuming. This might have led to small discrepancies in how students used the application and impacted on the caliber of data gathered in the latter phases. Additionally, the dataset's continuity and completeness may have been impacted over time by external academic or personal circumstances that affected their usage behavior, such as exam periods, absenteeism, or device constraints.

Finally, due to institutional and logistical limitations, the study could not be extended further because it was limited to a single institution (ESOFT Metro Campus) and a specific group of diploma students. The little while, the limited timeframe, sample limitation, and little decline in sustained student participation prevented the longitudinal technique from reaching its full potential, despite providing a solid basis for assessing learning advancement.

Data Collection and Gathering

Data Collection Methods – Questionnaires

In this study, questionnaires were used to obtain primary data directly from diploma students about their experiences using Duolingo. This method was chosen because it allowed for the efficient collection of standardized, quantitative data that was directly relevant to the research aims, particularly in terms of measuring student motivation, satisfaction, and perceived increases in vocabulary and speaking abilities.

What went well

As the main technique for gathering data, questionnaires were quite successful and fit in perfectly with the plan of the study. This method made it possible to gather standardized data from a sizable sample of 55 diploma students at the ESOFT Metro Campus in an expedient manner. Gathering consistent, similar responses from a diverse group of learners, all of whom had some prior experience using Duolingo for speaking practice and vocabulary development was made possible by the structured approach.

The questionnaire was thoughtfully created to capture the essential elements of the conceptual framework, including integrated audio playback, teacher teaching tactics, student motivation, and immediate feedback. To guarantee that the information gathered could be utilized immediately for statistical analysis, each component of the questionnaire matched one of the independent variables. It was simpler to assess the research hypotheses using quantitative techniques like t-tests, ordinal regression, and correlation analysis because of this obvious congruence.

A further advantage of the questionnaire approach was its anonymity, which perhaps inspired participants to provide more truthful and unbiased answers. This was especially

crucial for gathering information on subjective views such perceived gains in vocabulary and pronunciation, convenience of use, and learning motivation. Furthermore, the format made it possible for quick distribution and completion, which was crucial considering the academic term's time limits and the requirement to minimize interference with students' usual study routines.

Because it didn't require any specialized equipment or lengthy interview sessions, the questionnaire also made data collection more affordable. The analysis stage could be expedited by swiftly entering and processing the data using statistical software after it had been dispersed. Because of this, questionnaires were not only a feasible and controllable strategy, but they also perfectly complemented the study's positive and deductive scientific framework.

What went wrong

Despite their practical advantages, questionnaires nevertheless offered significant constraints that affected the depth and quality of the data collected. Most notably, the instrument depended primarily on closed-ended questions, which restricted participants to predefined response alternatives such as Likert scales and multiple choice. As a result, the questionnaire did not capture the experiences, feelings, or nuanced input that students might have wished to express about their contact with Duolingo. For example, a student who encountered difficulties comprehending the app's instructions or struggled with audio playback owing to external noise would not have had an opportunity to explain these issues. Additionally, the lack of open-ended questions hindered the potential to study why students felt more motivated or how certain characteristics of Duolingo affected their learning. For instance, while many respondents may have reported great satisfaction with instant feedback, the poll could not identify whether this was due to faster learning, improved confidence, or contrast with typical feedback delays.

Another problem was the inability of the questionnaire to capture real-time behavior or usage patterns of students while they were engaging with the program. It depends exclusively on self-reporting, which is vulnerable to recall bias and inaccuracy. Students may have inflated or underestimated their usage frequency or perceived advances in vocabulary due to subjective interpretation.

Lastly, some students may have misread questions, skip things, or give random replies without full engagement, especially if they were completing the form rapidly or without

supervision. These risks, together with the lack of interactive or adaptive questioning, slightly diminished the depth and contextual clarity that a mixed-method or observational approach could have offered.

Data Sampling

Sampling Method – Simple Random Sampling

Participants in this study were chosen at random from a sample of diploma students at the ESOF Metro Campus. This strategy ensured that every student had an equal chance of getting chosen, hence reducing bias. 55 students were chosen to participate in the survey. This strategy was adopted to provide fair and objective representation of the student population.

What went well

In this study, simple random sampling was found to be an effective strategy for ensuring fairness, impartiality, and representativeness within the selected student population. By selecting 55 diploma students at random from the 47th batch at ESOF Metro Campus, everyone had an equal chance of being included in the sample, independent of academic performance, gender, or language competency. This method helped to reduce selection bias, a common danger in educational research, and guaranteed that the data obtained represented a wide range of student experiences with the Duolingo application.

Random sampling also benefited the study's positive, deductive framework, which emphasizes impartiality and generalizability. Since the study's goal was to evaluate hypotheses based on statistical relationships, utilizing a randomly selected sample boosted the credibility of the findings and reduced the influence of the researcher's own biases in participant selection.

Furthermore, the sample was recruited entirely from diploma students, my target demographic, so the results were immediately relevant to the research objectives. The sample size selected was statistically sufficient for conducting inferential analyses such as t-tests, correlation, and ordinal regression, all of which helped validate the impact of independent factors (such as motivation and quick feedback) on vocabulary improvement. The sample's unpredictability helped ensure that these insights were not distorted by

selecting only high-performing or tech-savvy students, making the findings more robust and credible in the context of ESOF.

What went wrong

While simple random sampling provided advantages, there were significant restrictions that limited the scope and generalizability of the findings. The most significant difficulty was that the sample was limited to a single educational institution (ESOF Metro Campus).

This constraint affects the research's external validity, which means that the findings cannot be safely applied to other campuses, universities, or diploma programs with different student backgrounds, access to technology, or curriculum frameworks.

Furthermore, the study did not employ any stratified sample approaches to assure a balanced representation across characteristics such as gender, prior English proficiency, or familiarity with mobile applications. As a result, the final sample may have had an unusually large number of students with similar learning profiles, which could have influenced how they used Duolingo. Students with extensive experience with English or digital learning tools, for example, may have performed better with Duolingo not necessarily because of the tool's effectiveness, but because of existing advantages that were not taken into consideration by the sample approach.

Another disadvantage of utilizing basic random sampling without stratification is a potential lack of diversity in learning styles, motives, and classroom situations, all of which are important considerations in determining how well an AI-powered language software performs. This error may have contributed to the under-representation of students who have difficulty with mobile learning or prefer more traditional educational methods groups whose experiences would have provided valuable contrast to the overall findings.

In future research, combining random sampling with stratified or purposive methodologies could help assure both randomness and variety, resulting in a more comprehensive understanding of Duolingo's influence on various learner groups.

Data Analysis Techniques

Data Analysis

To analyze the data acquired from diploma students, the study combined descriptive and inferential statistical approaches. Descriptive statistics were employed to summarize general data, such as student demographics and total response rates. Non-parametric tests, such as the Kruskal-Wallis H test, were used in inferential analysis to analyze differences between categorical variables (for example, levels of motivation or types of feedback). Furthermore, a paired t-test was performed to compare students' vocabulary gains before and after using Duolingo, and ordinal regression was utilized to uncover predicted correlations between several parameters and vocabulary growth.

What went well

In this study, a thorough and layered data analysis approach was used to scientifically assess the influence of the Duolingo app on vocabulary development and English-speaking proficiency among diploma students. The method began with a pilot test analysis, which was used as a first quality assurance step to evaluate the questionnaire's clarity, reliability, and structure. Following pilot feedback, improvements were made to increase item consistency and comprehension, ensuring the accuracy of the main data gathering phase. Descriptive statistics were then used to summarize demographic characteristics and overall trends in student participation, such as English skill levels, frequency of Duolingo use, and motivational attitudes. These results were efficiently visualized using pie charts and bar graphs, which aided in the identification of trends throughout the sample.

Cronbach's Alpha was used to assess the reliability of the data collection instrument, confirming internal consistency among questionnaire items. Concurrently, validity tests were conducted to ensure that each construct such as student motivation or teacher strategy measured what was intended. Normality testing was then performed using bell curve analysis to ensure that the data distribution was correct. Because several variables did not match normal distribution assumptions, non-parametric tests were applied to ensure analytical rigor.

The Kruskal-Wallis test was then performed to determine statistically significant differences in vocabulary improvement among participant groups. Ordinal regression analysis was used to determine the predictive power of each variable, indicating which elements had the most influence on vocabulary growth and contributing to the refinement of the conceptual

framework. Finally, a paired sample t-test was utilized to assess the difference in vocabulary proficiency before and after using Duolingo. This test yielded solid statistical evidence that the program greatly enhanced learners' vocabulary over time. These layered analyses combined to produce a strong and well-validated assessment of Duolingo's educational effectiveness, thereby achieving the research objectives and improving the empirical foundation of my findings.

What went wrong

While the data analysis methodologies utilized in the study produced valuable quantitative insights, numerous significant problems arose throughout their implementation. Advanced statistical approaches, such as ordinal regression, Spearman's rank correlation, and Kruskal-Wallis testing, necessitated a thorough understanding of statistical theory, assumptions, and the proper interpretation of software results. Without sophisticated statistical modelling training, reading complicated outputs like log-odds coefficients, p-values, and pseudo-R-square values in ordinal regression can be difficult, potentially leading to misinterpretation or overgeneralization of results. Furthermore, some fundamental statistical assumptions, such as data normality and variance homogeneity, were not fully met, owing mostly to the small sample size ($n = 55$) and the ordinal structure of Likert-scale questionnaire replies. This limited the use of more powerful parametric tests and forced the employment of non-parametric alternatives, which, while suitable, frequently resulted in less precise estimates and lower statistical power.

Another notable drawback of the data analysis procedure was the absence of methodological triangulation. Because the study used just quantitative methodologies, there was no inclusion of qualitative data to provide a broader, more nuanced understanding of the results. This exclusion limited the opportunity to investigate underlying student views, emotional engagement, and contextual challenges associated with their use of the Duolingo application. For example, while the analysis revealed that student motivation was positively correlated with vocabulary improvement, the quantitative data failed to explain why some students felt more motivated or how Duolingo's gamification, interface, or feedback mechanisms influenced their learning behavior. The lack of open-ended questions, interviews, or focus group discussions meant that crucial personal insights, learner challenges, and experiential nuances went uncovered, reducing the depth and interpretive value of the findings. Including even a minor qualitative component could have helped

validate and contextualize the statistical results, making the conclusions more comprehensive and meaningful.

Effectiveness of Methods in Meeting Research Objectives

This section of the review is on how effectively the selected research methodologies contributed to attaining the study's major objectives, which are to examine the effectiveness of the Duolingo program in improving vocabulary expansion and English-speaking skills among diploma students. The study used a positivism ideology, a logical approach, and a multi-method quantitative strategy, aided by statistical methods such as the Kruskal-Wallis H test, paired t-tests, and ordinal regression. The goal was to test hypotheses and validate a conceptual framework using empirical evidence.

What Worked

The research methodologies utilized in this study were highly effective in attaining the stated objectives. Using structured quantitative analysis, the research successfully established the effects of the Duolingo program on boosting both vocabulary and English-speaking skills among diploma students. The connection between each study topic and the independent variables namely student motivation, instructor instructional method, integrated audio playback, and rapid feedback allowed for a focused and systematic analysis. The application of statistical techniques, including descriptive analysis, Kruskal Test, regression, and paired sample t-tests, produced substantial empirical data that supported the research assumptions. As a result, the conceptual framework given at the beginning of the study was validated and further refined based on the findings. This guaranteed that the final model was anchored on genuine learner data. Moreover, the adoption of a quantitative technique was well matched to the nature of the study, since it allowed for the measurement of increases in learners' vocabulary acquisition over time. Overall, the methodological congruence between study objectives, data collecting, and analysis lead to a credible and objective evaluation of Duolingo's effectiveness in the academic setting.

What Could Be Improved

While the quantitative methods employed in this research accurately examined the effects of Duolingo on vocabulary development and speaking skills, the study may have been considerably strengthened with the inclusion of qualitative research methods, such as interviews or focus group discussions. These methodologies would have allowed us greater study into the specific learning experiences, obstacles, and views of students using the application. Quantitative data highlighted patterns and statistical relationships, but it did not fully explain the fundamental reasons behind those trends such as why some students were more motivated or how specific characteristics of Duolingo increased learner engagement. A mixed-methods approach would have provided a more comprehensive understanding by combining the numerical results with rich, contextual insights. Furthermore, the study was limited to a single institution, which hinders the generalizability of the findings. Including individuals from diverse educational settings or regions would have brought greater diversity in learner backgrounds, allowing for more representative and widely applicable conclusions. These modifications would not only enhance the depth and breadth of the research but also ensure that the results better reflect the varying realities of students in different learning situations.

5.5.Limitation

The research project to assess the efficiency of Duolingo in increasing vocabulary expansion among diploma students was based on a well-structured methodological basis led by Saunders' Research Onion. The use of a positive research philosophy, a deductive research approach, and a survey-based research strategy ensured that the data collected was objective, measurable, and consistent with the hypotheses developed throughout the literature evaluation. The use of questionnaires via Google Forms was effective in reaching a reasonable number of diploma students while guaranteeing anonymity and consistency in responses. Furthermore, SPSS was used for statistical analysis, allowing the researcher to conduct correlation and regression analyses, providing strong empirical support for the hypothesis that Duolingo promotes vocabulary development through motivational features, instant feedback, audio playback, and pedagogical design. This organised and quantitative method enabled the study to achieve its key goals, provide a conceptual framework, and demonstrate Duolingo's instructional usefulness in AI-assisted language learning. Despite

these benefits, the study faced significant problems and limits that hampered its depth and breadth. One important weakness was the study's short duration, which limited the capacity to detect long-term increases in vocabulary recall and English speaking fluency—skills that develop gradually over time. Although the study was intended with a longitudinal time horizon, limitations in academic scheduling and data collection timelines prevented participants from being tracked during a meaningful period of language acquisition development. Furthermore, the sample size was limited to 55 diploma students from a single school (ESOFT Metro Campus), which limits the data's applicability to larger populations or educational environments. A larger, more diverse sample would have provided more detailed insights into how students from various demographics interact with and benefit from Duolingo.

Another significant restriction was the only use of quantitative approaches and the lack of qualitative data. Structured questionnaires gave standardized replies, but they did not capture in-depth insights into learners' emotional engagement, perceived problems, or contextual challenges when using Duolingo. Qualitative tools, such as semi-structured interviews or focus groups, would have given nuanced learner viewpoints while also revealing deeper cognitive or motivational components of usage behavior. Furthermore, the study did not use pre- and post-assessment testing, therefore there was no objective measure of actual vocabulary development or speaking ability improvement. Relying only on self-reported data from participants increases response bias, since students may exaggerate or underestimate their improvement. The study also failed to account for external learning effects such as additional study materials or classroom teaching, which could have influenced learners' vocabulary development and so distorted the findings. Furthermore, the assumption that all participants had equal access to reliable technology and internet connectivity was not tested, despite the fact that technical restrictions and digital literacy differences could have had a major impact on student participation with the app. These methodological constraints indicate opportunities for future improvement and shed light on how educational technology research might be more inclusive, comprehensive, and impactful.

In terms of alternative methodologies, the study would have greatly benefited from a mixed-methods research design, which combines quantitative surveys with qualitative instruments such as interviews and open-ended questions. This would have enabled data triangulation,

resulting in more robust and holistic results. An experimental or quasi-experimental design with a control and experimental groups would have allowed for a direct comparison of traditional learning methods versus Duolingo-assisted learning, reinforcing the study's causal inferences. Furthermore, collecting usage stats from the Duolingo app (such as time spent, class completion rates, and error correction patterns) may have offered objective behavioral data to supplement self-reported assessments. Based on these observations and constraints, several major recommendations can be made. First, future study should use a longer time span to assess sustained vocabulary learning and fluency. Second, increasing the sample population across institutions and integrating students from various educational, linguistic, and socioeconomic backgrounds will improve generalisability. Third, incorporating qualitative methodologies will deepen our understanding of student motivation, obstacles, and contextual use of Duolingo. Fourth, using pre- and post-testing to assess vocabulary and speaking skills will provide empirical evidence of learning gains. Fifth, working with educators to integrate Duolingo into blended learning environments, which combine app-based practice with teacher-led instruction, will result in more effective language learning outcomes. Finally, comparing Duolingo to other AI-based language learning programs (such as Babbel, Memrise, or Rosetta Stone) might help determine which technologies are most suited for certain learner profiles and educational environments.

In conclusion, while the current study met its primary aims and provided vital insights on Duolingo's potential for vocabulary development, it also revealed significant methodological shortcomings and contextual limits that must be addressed in future research. The lessons learnt from this research process highlight the significance of methodological diversity, long-term planning, and learner-centred design in conducting effective computing research in the education sector. Future research can make more significant contributions to the field of AI-enhanced language learning by using these observations and recommendations, as well as help to create more inclusive, engaging, and effective learning environments.

5.6.Recommendation

The research effort assessing the effectiveness of the Duolingo program in improving vocabulary expansion and English-speaking skills among diploma students followed a methodologically sound and academically backed methodology. The study followed Saunders' Research Onion paradigm, which included a positivist ideology, a deductive technique, a survey strategy, and a longitudinal time horizon. These choices were consistent with the study's purpose of objectively assessing the association between Duolingo usage and vocabulary improvement. Using a structured questionnaire sent via Google Forms enabled efficient and standardized data gathering from a large sample of diploma students. The data was analyzed using SPSS, which allowed for statistically robust approaches such as correlation and regression analysis, which helped to validate important hypotheses about student motivation, teaching strategies, integrated audio playback, and fast feedback. This quantitative technique was effective in gathering empirical evidence to assist the development of a conceptual framework that depicted how AI-based technologies can influence language learning.

Despite these advantages, numerous methodological limitations emerged during the research process. While the quantitative survey technique was clear and objective, it lacked the depth necessary to investigate learners' nuanced experiences with the Duolingo program. Fixed-response questions cannot capture important psychological and emotional elements such as user annoyance, engagement swings, and the influence of gamification. Furthermore, despite the fact that a longitudinal study was intended, data collection took place over a relatively short period of time, making it difficult to assess long-term vocabulary retention and sustainable speaking progress. Vocabulary development is a process that frequently benefits from prolonged exposure and reinforcement, and these effects may not be completely apparent in a short-term study. Furthermore, the sample size, while acceptable for statistical analysis, was limited to students from a single institution ESOF Metro Campus restricting the data's generalizability. The demographic and contextual diversity among learners was restricted, so the findings may not be applicable to larger educational environments or learner groups.

To solve these issues and improve future research, numerous recommendations have been made. To begin, future research should use a mixed-methods approach, collecting both quantitative and qualitative data. While surveys provide statistical generalizability,

including interviews, focus groups, or open-ended survey questions can provide more information about students' unique learning experiences, emotional responses, and contextual problems. This would allow researchers to understand not only the alterations that occurred, but also why and how they occurred. Additionally, prolonging the study's longitudinal duration is critical. A longer observation period would allow researchers to better track vocabulary development and speaking skills, capturing both delayed improvements and long-term Duolingo usage habits.

Second, researchers are urged to use comparative and experimental designs. Introducing control and experimental groups, with one using Duolingo and the other using standard language learning methods, can assist in establishing stronger causal correlations. Pre- and post-intervention exams, vocabulary assessments, and speaking activities can be more trustworthy markers of real language development than self-reported data. Another suggestion is to broaden the sample group by incorporating students from multiple campuses or different universities. A broader demography will improve data representativeness and allow for cross-group analysis to investigate how different sorts of learners respond to AI-based language aids.

Furthermore, future studies should investigate the application of digital learning analytics. Accessing Duolingo's backend data, such as login frequency, streak consistency, progress tracking, and error correction history, can provide a more objective and detailed view of user involvement and learning behavior. This information can be triangulated with survey results to improve dependability. Furthermore, knowing the qualitative components of motivation include what keeps learners interested, what discourages them, and how external factors like workload or academic pressure influence app usages crucial for building learner-centered teaching tactics.

Another important recommendation is to encourage the gradual incorporation of Duolingo into formal instruction. AI tools perform best when combined with teacher guidance, feedback, and goal setting. Teachers can help by assessing progress, promoting frequent use, and connecting digital learning to classroom discussions. Models of hybrid training should be investigated in order to identify the best balance between autonomous app-based learning and guided teaching. Furthermore, the conceptual framework proposed in this study which is based on variables such as motivation, instructional tactics, auditory aspects,

and feedback should be tested, validated, and enhanced in subsequent projects. Additional characteristics, such as learner satisfaction, digital literacy, and time-on-task, may improve the framework's prediction accuracy.

Finally, educational institutions should provide systematic training and support to both students and instructors so that Duolingo or similar programs may be properly implemented. Orientation seminars, resource guides, and digital literacy courses can greatly boost adoption rates and student results. Future research might also investigate and evaluate the efficacy of different AI tools such as Babbel, HelloTalk, and Memrise, providing a broader view on the most successful technology for vocabulary improvement.

In conclusion, while the current research methodology effectively supported the major aims and provided significant insights on the use of Duolingo for vocabulary development, there are still apparent areas for methodological refinement and growth. Embracing more diverse data collection technologies, longer study durations, and inclusive sample strategies will not only increase the legitimacy and relevance of future research but will also help to shape best practices in AI-supported language instruction. These insights and recommendations provide a road map for more thorough, learner-centered, and impactful computing research in educational settings.

5.7.Lesson Learned

Several key lessons were learnt during this research, including the strengths and limits of the approaches used. The study's positivist ideology, deductive approach, and quantitative methods allowed for systematic data collecting and thorough analysis. However, as the research progressed, it became clear that relying primarily on statistical techniques and survey responses hampered our knowledge of learners' experiences with Duolingo. Although the research objectives were satisfied, different methodological approaches could have increased the findings' richness, applicability, and interpretability. Reflecting on these experiences has given me great insights into how future studies on educational technology, particularly those including learner-centred apps like Duolingo, can benefit from broader, more flexible research approaches. The subsections that follow discuss these lessons in terms of technique, literature, data collecting, sampling, and measuring instruments.

Alternate Research Methodologies that could have been used

While this study used a positivism mindset and a quantitative survey strategy based on a deductive approach, other methodologies may have provided deeper or additional insights. For example, combining an interpretivist paradigm with a qualitative technique, such as phenomenology research or case studies, could have recorded learners' own experiences, emotions, and perceptions while using Duolingo. This would have allowed researchers to investigate the why, not simply what, of students' learning behaviors.

A mixed-methods approach, combining formal surveys with semi-structured interviews or classroom observations, could have been beneficial in balancing statistical evidence with contextual awareness. Furthermore, an action research strategy would have allowed the researcher to install Duolingo in real-time classroom settings, iteratively tweak it based on student feedback, and assess its evolving impact, so increasing the practical relevance and educational utility of the research findings.

Alternate Literature Review Pointers that could have been used

Four key factors that are pertinent to Duolingo's influence on vocabulary acquisition were successfully covered in the original literature review for this study, integrated audio playback, instructor instructional method, student motivation, and quick feedback systems. Although these themes provide a strong starting point, the study's conceptual depth and analytical power may have been enhanced by the inclusion of a few more theoretical frameworks and academic debates.

Sweller's (1988) Cognitive Load Theory (CLT) is a noteworthy framework that may have been investigated. According to CLT, instructional designs should refrain from overwhelming students' working memory because they have a limited capacity. Using this theory, it would be possible to analyze if the visual components, gamified features, and repetitious workouts of Duolingo's interface efficiently minimize unnecessary cognitive load while optimizing relevant burden, hence promoting vocabulary retention and fluent usage. For students who are just beginning to learn a language, this is especially important since, depending on their cognitive processing capacities, too many stimuli may either help or hurt their learning.

Vygotsky's Zone of Proximal Development (ZPD), which highlights the value of social interaction and directed support in learning, is another crucial idea. One way to think of Duolingo's automated, scaffolded feedback is as a kind of virtual "peer" or "instructor" support that helps students do just a little bit better than they can on their own. The study could investigate how AI-powered platforms mimic or enhance human scaffolding, particularly in vocabulary and speech fluency development, by including ZPD into the literature review. It also offers a way to investigate if this type of technology fills in the gaps for specific learners or is constrained by its generalized nature.

Furthermore, Duolingo might have been placed within a larger pedagogical and technical trend by incorporating research from the Mobile-Assisted Language Learning (MALL) discipline. MALL literature frequently examines how the ease, personalization, and real-time feedback of mobile devices improve language acquisition. It might be possible to compare Duolingo with other mobile applications and uncover its unique or shared affordances by including frameworks from academics such as Kukulska-Hulme and Shield (2008) or Stockwell (2010).

Gamification theory, particularly research by Deterding et al. (2011) that examines how game features like leaderboards, badges, and points affect learner engagement and behavior, could have been a useful addition to the literature review. Gaining insight into how Duolingo's extrinsic motivators impact user pleasure, long-term vocabulary retention, or intrinsic motivation would have given the educational results a behavioral psychology viewpoint.

Furthermore, a more thorough analysis of self-efficacy and language learning anxiety two crucial psychological concepts in second-language acquisition would have clarified how Duolingo affects students' comfort and confidence when speaking English. Bandura's (1997) theory of self-efficacy and MacIntyre and Gardner's (1991) literature on foreign language anxiety may offer a psychological framework for assessing whether consistent, private, low-pressure Duolingo practice lowers speaking anxiety and increases learners' confidence in their own communicative abilities.

Lastly, constructivist learning theory, which stresses active learner involvement and knowledge building through experience, may have been examined in the literature. Because Duolingo is interactive and self-paced, students may take charge of their education. Constructivist studies, like those by Piaget or Bruner, could support the idea that this kind

of autonomy encourages more in-depth vocabulary learning and practical speech application.

These other theoretical stances, which cover cognitive psychology, educational theory, mobile learning, and gamification, could have helped the study explain Duolingo's efficacy in a more comprehensive way. This would have improved the literature review's scholarly rigor and provided a more sophisticated basis for analysing the results and putting forward a flexible conceptual framework for language acquisition via AI-based applications.

Alternate Data Collection and Analysis Methods that could have been used

Alternative data gathering techniques might have provided more comprehensive insights, even though questionnaires and SPSS-based analysis enabled effective statistical evaluation. Focus groups, for instance, would have allowed students to interact with one another and possibly reveal behavioral trends, common problems, and app engagement techniques that were not apparent in survey results. Semi-structured or formal interviews could have yielded valuable qualitative information about the app's usability, motivational factors, and user happiness. To supplement self-reported data, observational data such as screen captures of Duolingo usage or in-class monitoring would offer behavioral insights. Additionally, thematic analysis of qualitative replies would have been possible with the use of programs like NVivo or ATLAS.ti. Using normalized datasets or control-experimental group designs may enhance result validity and make it easier to draw conclusions about causality. On the quantitative side, combining pre-test and post-test vocabulary evaluations would provide a direct measurement of vocabulary gain.

Alternate Sampling Strategies that could have been used

Participants in this study were chosen at random from among the diploma students at ESOF Metro Campus. Although this made guaranteed that every student had an equal chance of being chosen, other approaches might have increased the sample's focus or variety. For example, the researcher may have sampled proportionately by dividing the student body into strata according to language skill levels (beginning, intermediate, and advanced) using stratified sampling. This might make it easier to see how Duolingo impacts various student types. Additionally, purposeful sampling would have worked well

for finding individuals who had a history of using Duolingo regularly or who were known to struggle with vocabulary learning. These different approaches could have improved the study's accuracy in assessing user outcomes and produced a more thorough subgroup analysis.

Alternate Research Settings that could have been used

Generalizability was limited by the study's confinement to the ESOF Metro Campus instructional environment. The credibility of the findings would have been increased and a wider comparison made possible by conducting a multi-site survey across several campuses or institutions. Furthermore, applying research in non-formal learning settings, including language clubs or online learning communities, may have shown how students use Duolingo without official instruction. This may shed light on how mobile apps facilitate self-regulated learning. Additionally, including individuals from non-academic contexts (such as working professionals using Duolingo for personal growth) would have shown variations in learning efficiency, motivation, and usage patterns.

Alternate Measurement Metrics that could have been used

To gauge Duolingo's efficacy, the study mostly used Likert-scale replies and self-reported opinions. These metrics may not accurately reflect true language improvement, although being helpful for assessing motivation and pleasure. Alternatively, more measurable evidence of learning results could be obtained by employing objective measures like vocabulary exams, Duolingo performance tracking (XP points, streaks, completion rates), or external evaluations (e.g., TOEIC/IELTS vocabulary parts). Furthermore, using tests both before and after the intervention would make it easier to track vocabulary development over time. Using speech analysis technologies to assess fluency and pronunciation could potentially result in a more thorough, skill-based evaluation that is more in line with the objective of speaking improvement.

References

- Ajisoko, P., 2022. *International Journal of Emerging Technologies in Learning*. [Online]
Available at: <https://online-journals.org/index.php/i-jet/article/view/13229>
[Accessed 27 April 2025].
- Alan Jaelani, D. R. S., 2020. *Bogor English Student And Teacher (BEST) Conference*.
[Online]
Available at: <https://pkm.uika-bogor.ac.id/index.php/best/article/view/797>
[Accessed 25 April 2025].
- Anon., 2025. *Datatab.net*. [Online]
Available at: <https://datatab.net/tutorial/t-test>
[Accessed 2025].
- Anon., n.d. [Online]
Available at: <https://stats.stackexchange.com/questions/76379/ordinal-regression-proportional-odds-assumption>
- Anon., n.d. *ResearchGate*. [Online]
Available at:
https://www.researchgate.net/publication/314914494_The_Impact_of_Demographic_and_Academic_Characteristics_on_Academic_Performance
- Blanka Klimove, K. Z., 2020. *Education Sciences*. [Online]
Available at:
https://www.researchgate.net/publication/347486392_Use_of_Mobile_Applications_in_Developing_Reading_Comprehension_in_Second_Language_Acquisition_-A_Review_Study
[Accessed 27 April 2025].
- Brown, A., 2008. *The Physiological Society*. [Online]
Available at: <https://www.physoc.org/magazine-articles/the-strange-origins-of-the-students-t-test/>
- Charles Teddlie, F. Y., 2007. *Journal of Mixed Methods Research*. [Online]
Available at: <https://journals.sagepub.com/doi/10.1177/1558689806292430>
[Accessed 5 May 2025].
- Chiang, M.-H., 2020. *IAFOR Journal of Education*. [Online]
Available at: <https://eric.ed.gov/?id=EJ1245831>
[Accessed 5 May 2025].

Chik, A., 2019. *World Englishes*. [Online]

Available at: <https://onlinelibrary.wiley.com/doi/10.1111/weng.12443>

[Accessed 27 April 2025].

Dhaval Makwana, P. E. A. D. H. C., 2023. *ResearchGate*. [Online]

Available at:

https://www.researchgate.net/publication/371985656_Sampling_Methods_in_Research_A_Review

[Accessed 5 May 2025].

Geoffrey T. LaFlair, T. L. B. B. A. K. H. Y. A. A. v. D., 2022. *Journal of Computer Assisted Learning*. [Online]

Available at: <https://onlinelibrary.wiley.com/doi/10.1111/jcal.12665>

[Accessed 27 April 2025].

George, T., 2022. *Scribbr*. [Online]

Available at: <https://www.scribbr.com/methodology/interviews-research/>

[Accessed 5 May 2025].

Grego, J., 2012. [Online]

Available at: http://vesselinov.com/RV_Duolingo.pdf

[Accessed 27 April 2025].

Hundley, T. a., 2001. *University of Surrey*. [Online]

Available at: <https://sru.soc.surrey.ac.uk/SRU35.html>

İbrahim Yaşar Kazu, M. K., 2025. *Journal of Computer in Education*. [Online]

Available at:

https://www.researchgate.net/publication/389876776_Digital_language_learning_with_Duolingo_assessing_its_impact_on_listening_speaking_reading_and_writing_skills

[Accessed 27 April 2025].

Interviews, U., 2021. *www.userinterviews.com*. [Online]

Available at: <https://www.userinterviews.com/ux-research-field-guide-chapter/focus-groups>

[Accessed 5 May 2025].

Jeon, J., 2022. *Education and Information Technologies*. [Online]

Available at: <https://link.springer.com/article/10.1007/s10639-021-10839-y>

[Accessed 27 April 2025].

John Grego, R. V., 2022. [Online]

Available at: https://www.languagezen.com/pt/about/english/Duolingo_Efficacy_Study.pdf
[Accessed 27 April 2025].

Jorg Hecker, N. K., 2024. *ATLAS.ti*. [Online]

Available at: <https://atlasti.com/guides/interview-analysis-guide/disadvantages-of-research-interviews>

[Accessed 5 May 2025].

Ju-Young M.Kang, J. M. M. K. K., 2015. *Computer in Humen Behaviour*. [Online]

Available at:

<https://www.sciencedirect.com/science/article/abs/pii/S0747563215000242?via%3Dihub>

[Accessed 5 May 2025].

Kabir, S. M. S., 2016. *ResearchGate*. [Online]

Available at:

https://www.researchgate.net/publication/325846997_METHODS_OF_DATA_COLLECTI_ON

[Accessed 5 May 2025].

Kevin F. Steinmetz, A. P. W. G., 2021. *Computer in Human Behavior*. [Online]

Available at:

<https://www.sciencedirect.com/science/article/abs/pii/S0747563221002533?via%3Dihub>

[Accessed 5 May 2025].

Loeb, S., 2017. [Online]

Available at: <https://files.eric.ed.gov/fulltext/ED573325.pdf>

Mairin Hennebry-Leung, M. L., 2024. *English Teaching & Learning*. [Online]

Available at: <https://link.springer.com/article/10.1007/s42321-024-00179-8>

[Accessed 27 April 2025].

Mark N K Saunders, P. T., 2012. *Rapport*. [Online]

Available at: https://www.academia.edu/4107831/The_Layers_of_Research_Design

[Accessed 5 May 2025].

Munday, P., 2015. [Online]

Available at:

https://www.researchgate.net/publication/284517271_The_case_for_using_DUOLINGO_as_part_of_the_language_classroom_experience

[Accessed 27 April 2025].

Pallant, J., 2010. Google BSPSS Survival Manual: A step by step guide to data analysis using the SPSS program. In: *BSPSS Survival Manua*. s.l.:Allen & Unwin , p. 368.

Pallant, J., 2010. SPSS Survival Manual: A Step-by-step Guide to Data Analysis Using SPSS for Windows (Version 10). In: *SPSS Survival Manua*. s.l.:Open University Press, p. 286.

Pressbooks, 2017. *Wsu.edu*. [Online]

Available at: <https://opentext.wsu.edu/carriecuttler/chapter/observational-research/>
[Accessed 5 May 2025].

R.Wadhwa, R., 2023. *PubMed*. [Online]

Available at: <https://www.ncbi.nlm.nih.gov/books/NBK553048/>

Rosario, d., 2025. [Online]

Available at: <https://repositorio.upse.edu.ec/handle/46000/13387>
[Accessed 27 April 2025].

Shawn Loewen, D. C. D. R. I. K. M. K. J. M. Z. F. H. R., 2019. *ReCALL*. [Online]

Available at: <https://www.cambridge.org/core/journals/recall/article/abs/mobileassisted-language-learning-a-duolingo-case-study/A4D7C8F71782A37D258C19F357DDBCBE>
[Accessed 27 April 2025].

Son T.H. Pham, P. M. S., 2022. *Journal of Computer Assisted Learning*. [Online]

Available at: <https://onlinelibrary.wiley.com/doi/10.1111/jcal.12687>
[Accessed 27 April 2025].

Susanto, A. O. Y. Y. S. R. P. H. H. a. T. T., 2020. *Studies in English Language and Education*. [Online]

Available at: <https://jurnal.usk.ac.id/SiELE/article/view/16970>
[Accessed 27 April 2025].

Taherdoost, H., 2016. *International Journal of Academic Research in Management*. [Online]

Available at: <https://hal.science/hal-02546796/file/Sampling%20Method%20in%20Research%20Methodology;%20How%20to%20Choose%20a%20Sampling%20Technique%20for%20Research.pdf>
[Accessed 5 May 2025].

Tengli, M. B., 2020. *ReserchGate*. [Online]

Available at:

https://www.researchgate.net/publication/357284560_RESEARCH_ONION_A_SYSTEM

ATIC APPROACH TO DESIGNING RESEARCH METHODOLOGY

[Accessed 5 May 2025].

Thornbury, 2002. [Online]

Available at: <https://www.scirp.org/reference/referencespapers?referenceid=203958>

Tuovila, A., 2020. *Investopedia*. [Online]

Available at: <https://www.investopedia.com/terms/s/sampling.asp>

[Accessed 5 May 2025].

Vorbel, O., 2022. [Online]

Available at:

[https://www.researchgate.net/publication/357968845_A_systematic_review_of_research_o
n_distance_language_teaching_2011-2020_Focus_on_methodology](https://www.researchgate.net/publication/357968845_A_systematic_review_of_research_on_distance_language_teaching_2011-2020_Focus_on_methodology)

[Accessed 27 April 2025].

Xiangying Jiang, J. R. H. C. B. R. E. G. L. P. B. P., 2021. [Online]

Available at: [https://duolingo-papers.s3.amazonaws.com/reports/duolingo-speaking-
whitepaper.pdf](https://duolingo-papers.s3.amazonaws.com/reports/duolingo-speaking-whitepaper.pdf)

[Accessed 27 April 2025].

Zixi Li, C. J. B., 2023. *Computer Assisted Language Learning*. [Online]

Available at: <https://www.tandfonline.com/doi/full/10.1080/09588221.2023.2206874>

[Accessed 27 April 2025].

Annexures

Annexures A: Presentation Slides

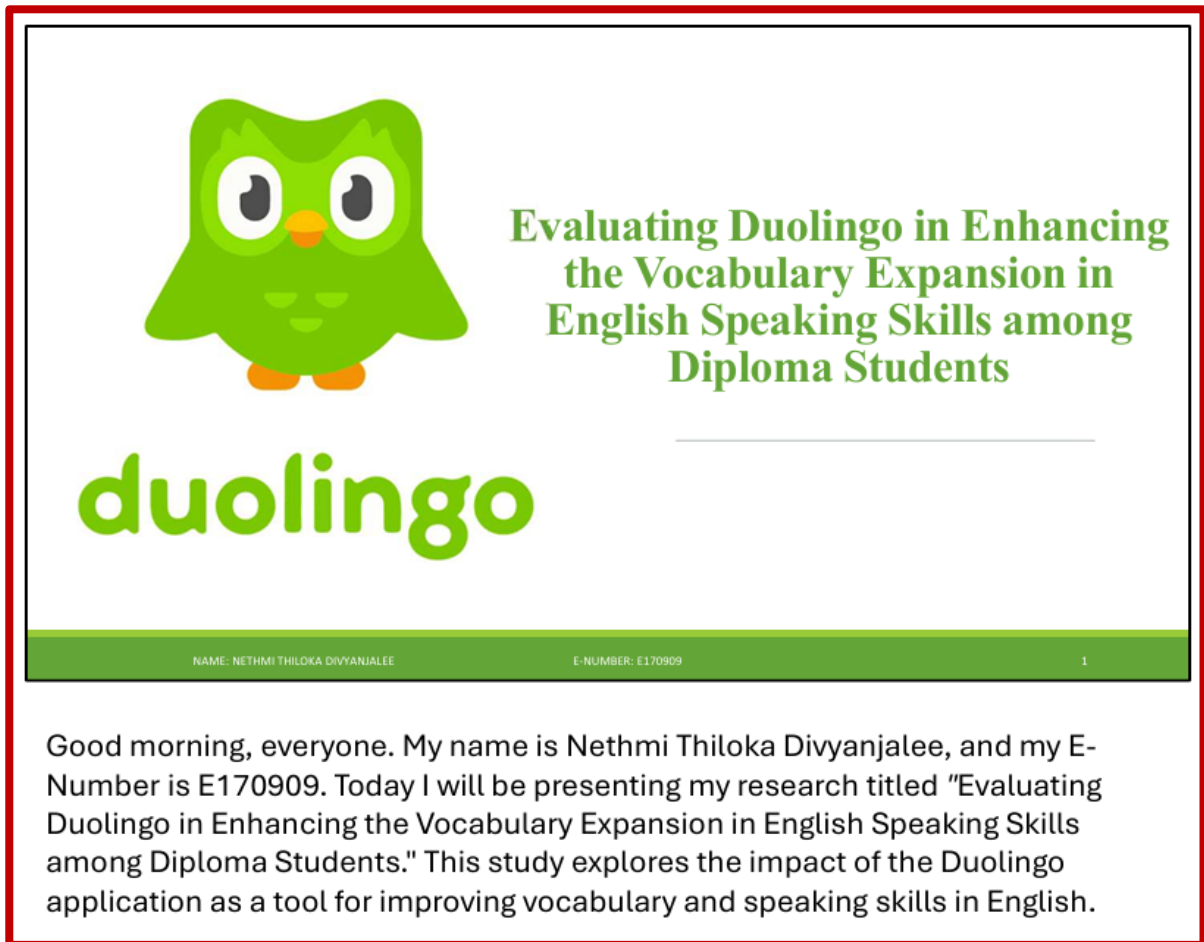


Figure 52 - Presentation Slide 1

Introduction



In today's digital age, traditional methods of language learning are proving insufficient in building vocabulary and speaking skills among diploma students.



With the rise of mobile-assisted language learning tools like Duolingo, there's an opportunity to shift from passive to interactive, gamified learning experiences.



This study focuses on evaluating the effectiveness of Duolingo in vocabulary enhancement and its potential integration into academic environments to support English speaking skills.



This research also explores how AI features such as instant feedback and audio playback impact learners' engagement, understanding, and performance.

NAME: NETHMI THILOKA DIVYANJALEE

E NUMBER: E170909

2

In this study, I focus on how Duolingo, a popular language learning app, helps diploma students improve their English vocabulary and speaking skills. Traditional classroom methods are often not enough many students struggle to remember and use new words.

Duolingo is interactive, fun, and available on mobile devices. It gives instant feedback and rewards, which may help students stay motivated.

So, I wanted to see if Duolingo can really make a difference and how it can be better used for learning English.

Figure 53 - Presentation Slide 2

Problem Statement

- ❖ The integration of technology into foreign language learning has become essential, as traditional, textbook-focused methods often fail to develop dynamic vocabulary and speaking skills.
- ❖ Diploma students, in particular, struggle with vocabulary due to repetitive teaching approaches. Mobile-assisted language learning tools like Duolingo offer interactive, gamified learning that contrasts sharply with passive classroom methods.
- ❖ Duolingo enhances engagement through rewards, real-time feedback, and flexible access via mobile devices.
- ❖ However, its specific effectiveness in improving vocabulary and speaking skills among Diploma students remains underexplored, highlighting the need to evaluate its role and develop a suitable framework for its integration.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

3

Traditional teaching methods rely heavily on textbooks and teacher-centered instruction. These methods often fail to build active vocabulary usage and confident speaking.

Many diploma students find it hard to expand their vocabulary effectively.

Tools like Duolingo offer an engaging learning experience through technology, yet there's limited research on how well they improve vocabulary and speaking among diploma students.

This gap forms the core of my research problem.

Figure 54 - Presentation Slide 3

Aim

Evaluating Duolingo application in Enhancing the vocabulary expansion in English speaking skills of diploma students by analyzing its impact and propose a conceptual framework for speaking English.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

4

The main aim is to evaluate how Duolingo influences vocabulary development and speaking confidence among diploma students.

I also aim to propose a conceptual framework that helps educators and learners make better use of the app for improving English speaking skills.

Figure 55 - Presentation Slide 4

Sub Objectives

- ❖ To analyze the effectiveness of Duolingo in improving the vocabulary expansion of diploma students.
- ❖ To identify the challenges faced by diploma students while using Duolingo for vocabulary development.
- ❖ To compare the vocabulary enhancement of students using Duolingo with those using traditional learning methods by identifying the before and after effects of using Duolingo application.
- ❖ To propose a conceptual framework based on the findings to optimize the use of Duolingo for vocabulary and speaking skill development.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

5

The research objectives are:

- To analyze Duolingo's effectiveness in vocabulary improvement.
- To identify difficulties students face while using Duolingo.
- To compare vocabulary results of Duolingo users vs. traditional learners.
- To propose a practical framework to optimize the app's integration into learning systems.

Figure 56 - Presentation Slide 5

Literature Review

- ❖ How Student Motivation and Interest Improve Enhancing the Vocabulary expansion in English Speaking Skills.
- ❖ How Teacher's Instructional Strategy Improve Enhancing the Vocabulary expansion in English Speaking Skills.
- ❖ How Integrated Audio Playback in the AI tool Improve Enhancing the Vocabulary expansion in English Speaking Skill.
- ❖ How Instant Feedback given by the AI tool Improve Enhancing the Vocabulary expansion in English Speaking Skills.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

6

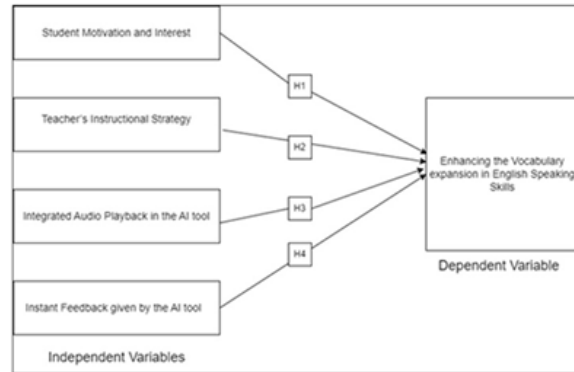
The literature shows that vocabulary growth is affected by multiple elements.

- Student motivation and interest improve when learning is interactive and rewarding.
- Teacher strategies also matter blending digital tools with teaching enhances learning.
- Audio playback helps with pronunciation, retention, and comprehension.
- Instant feedback allows students to immediately learn from their mistakes, building stronger understanding.

These themes formed the basis for my hypotheses and framework.

Figure 57 - Presentation Slide 6

Conceptual Framework



NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

7

My original framework was built around four main factors from the literature:

- Student motivation and interest
- Teacher instructional strategy
- Integrated audio playback
- Instant feedback from AI tools

These were expected to influence vocabulary development and were tested using survey and statistical data.

Figure 58 - Presentation Slide 7

Hypothesis

H1: Student Motivation and Interest

- **H1.0:** Student motivation and interest are negatively influenced when AI based tools are non-user-friendly and not engaging the vocabulary expansion in English Speaking skills.
- **H1.1:** Student motivation and interest are positively influenced when AI based tools are employed the vocabulary expansion in English Speaking skills.

H2: Teacher's Instructional Strategy

- **H2.0:** The teacher's instructional strategy negatively impacts students learning when it fails to integrate AI tools effectively to the vocabulary expansion in English Speaking skills.
- **H2.1:** The teacher's instructional strategy positively enhances student understanding and engagement when integrated with AI tools to the vocabulary expansion in English Speaking skills.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

8

I developed four hypotheses to test the influence of each factor:

H1: Motivation and interest either help or hinder learning based on the design of the AI tool.

H2: Teachers' strategies impact how well the AI tool is integrated into learning.

Figure 59 - Presentation Slide 8

Hypothesis

H3: Integrated Audio Playback in the AI tool

- **H3.0:** Integrated audio playback negatively affects learning when it is poorly implemented or distracts students to the vocabulary expansion in English Speaking skills.
- **H3.1:** Integrated audio playback positively supports diverse learning styles and improves comprehension to the vocabulary expansion in English Speaking skills.

H4: Instant Feedback given by the AI tool

- **H4.0:** Instant feedback negatively impacts on students if it is inaccurate, confusing, or overwhelming to the vocabulary expansion in English Speaking skills.
- **H4.1:** Instant feedback positively enhances student learning by helping them correct mistakes in real-time to the vocabulary expansion in English Speaking skills.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

9

H3: Audio playback is either helpful or distracting based on its implementation.

H4: Instant feedback can improve learning or be harmful if it's confusing or inaccurate.

Each hypothesis was tested using regression analysis, and I'll now walk you through the methodology and results.

Figure 60 - Presentation Slide 9

Methodology

Layers	Selected Option	Reason
Research Philosophy	Positivism	Positivism is chosen because it emphasizes objective reality, uses quantifiable observations, and is suitable for testing hypotheses using statistical methods.
Research Approach	Deductive	A deductive approach is selected because the study begins with a theoretical framework or hypothesis that will be tested using empirical data.
Research Strategy	Survey	The survey strategy is appropriate for collecting large amounts of quantitative data from a broad population, making it ideal for generalizing findings.
Method Choice	Multi – Method	A multi-method approach is selected to combine quantitative and qualitative data collection techniques, enhancing the depth and reliability of the findings.
Time Horizon	Longitudinal	A longitudinal time horizon is used to observe and analyze changes and developments over an extended period, which is critical for identifying trends and causal relationships.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

10

This slide outlines the key components of the research methodology used in my study.

- First, I followed a positivist philosophy, which focuses on using measurable, objective data to test hypotheses.
- The deductive approach means I started with a theory and then tested it through data.
- I used a survey strategy to collect responses from a large group of diploma students.
- The multi-method approach allowed me to combine both quantitative and qualitative methods, which made the findings more reliable.
- Finally, the longitudinal time horizon helped me track changes over a period, making it easier to identify trends in vocabulary improvement.

Figure 61 - Presentation Slide 10

Target Population and Sample Size

Population

Diploma-level English language learners at ESOF Metro Campus Kandy Branch

(950 Students)

Simple Random Sampling was used to ensure fairness, objectivity, and statistical validity in the study of Duolingo's contribution to vocabulary development in English speaking skills.

(References: Cochran, 1977; Creswell, 2012)

Simple random sampling was used to select 55 students from a sampling frame of 64 in Batch 47, based on the Morgan table, from a target population of 950 diploma students at ESOF Metro Campus, Kandy Branch.

NAME: NETHMI THILOKA DIVYANJALEE

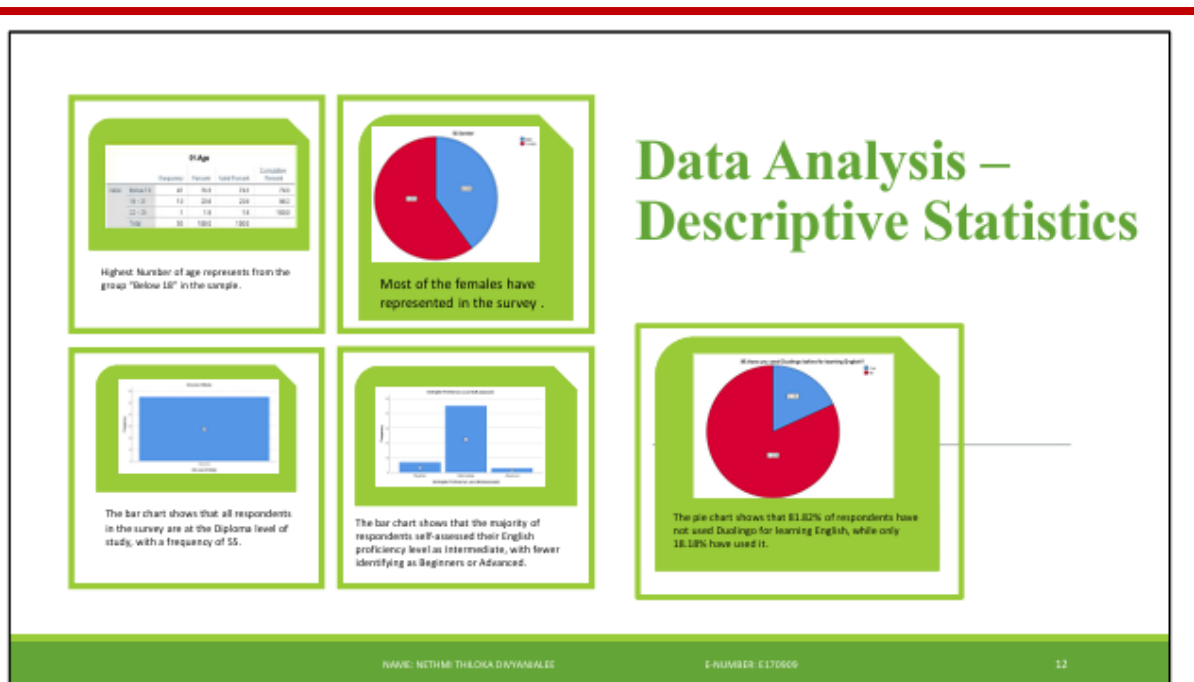
E NUMBER: E170909

11

This study focused on diploma students currently engaged in English learning. Using simple random sampling ensured every student had an equal chance of being selected.

Morgan's table was used to determine the appropriate sample size, ensuring statistical reliability.

Figure 62 - Presentation Slide 11



The descriptive analysis can only we done for the demographic questions.

This slide presents a summary of the basic characteristics of the participants in the study.

- The largest age group is below 18, which shows that most participants are younger diploma students.
- The majority of respondents are female, as shown in the gender pie chart.
- All 55 participants are studying at the Diploma level.
- Most students rated their English proficiency as Intermediate, with fewer at the beginner or advanced levels.
- Finally, about 82% of respondents have not used Duolingo before, while only a small percentage have used it.

These statistics help us understand the background of the sample before analyzing their responses in detail.

Figure 63 - Presentation Slide 12

Data Analysis – Reliability and Validity Analysis

- The **Reliability Test** is mainly used to assess the consistency and stability of a measurement tool or questionnaire over time.
- This test was done using **Cronbach's Alpha** to check if the questionnaire are consistent in measuring the same concept.
- The Cronbach's Alpha value was **0.951**, which is above acceptable level of 0.7.
- This means the **questionnaire is reliable**.

Reliability Statistics		
Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of Items
.951	.952	5

- The **Validity Test** was done to ensure that the questionnaire accurately measures what it is intended to assess.
- First, the **KMO (Kaiser-Meyer-Olkin) value** was 0.901, which is above 0.6. This shows that the sample is **adequate** and suitable for factor analysis.
- Next, the **Bartlett's Test of Sphericity** had a **significant value of .000**, which is less than 0.05. This means the variables are **correlated enough** and **valid** for further analysis.

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy		.901
Bartlett's Test of Sphericity	Approx. Chi-Square	275.998
	df	10
	Sig.	.000

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

13

Let's begin with the reliability test, which examines how consistent the questionnaire is in measuring what it's supposed to measure.

I used Cronbach's Alpha, which is the most commonly used indicator for internal consistency.

The result was 0.951, which is far above the minimum acceptable threshold of 0.7.

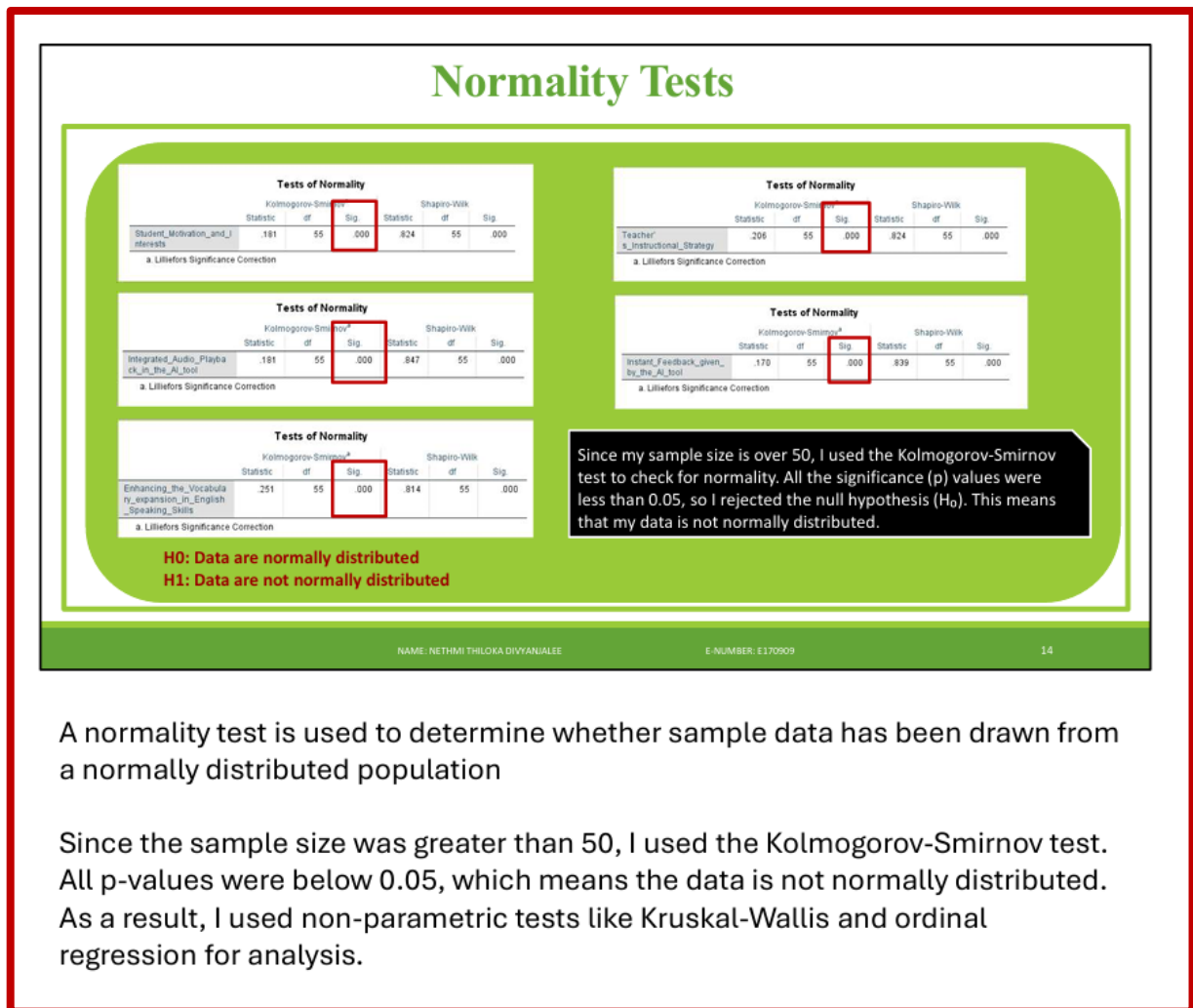
This indicates that all the items in the questionnaire are highly correlated and consistently measure the intended concept vocabulary and speaking skill development using Duolingo.

Next, I tested validity, which checks whether the instrument accurately measures what it's intended to.

For this, I used two measures:

- **KMO (Kaiser-Meyer-Olkin) Measure** – The value was 0.901, well above the required 0.6. This shows that the sample size is adequate and that factor analysis is appropriate for the data.
- **Bartlett's Test of Sphericity** – The result was significant at $p = .000$, which is less than 0.05. This means that the variables are sufficiently correlated with each other and valid for further statistical testing.

Figure 64 – Presentation Slide 13

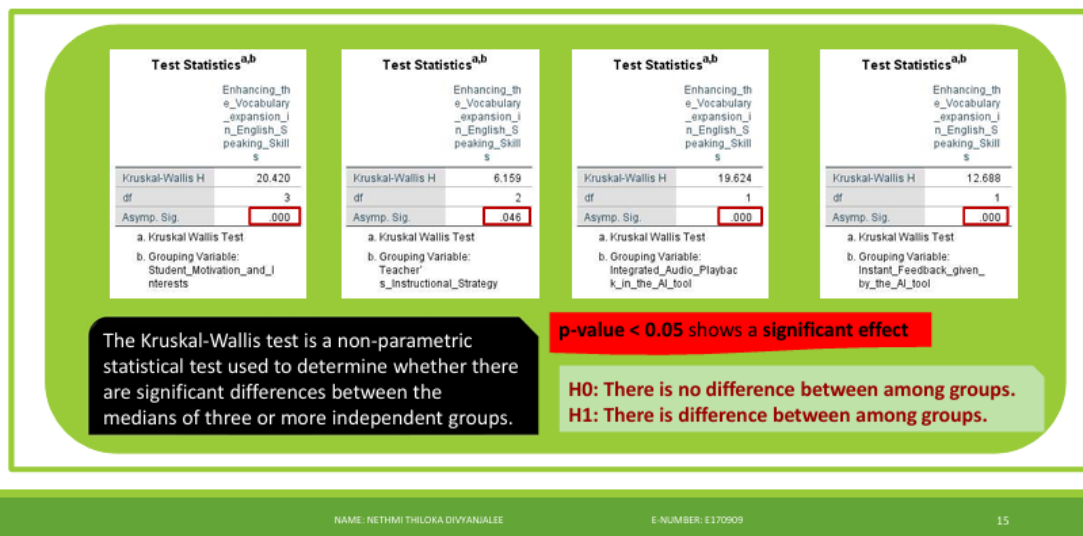


A normality test is used to determine whether sample data has been drawn from a normally distributed population

Since the sample size was greater than 50, I used the Kolmogorov-Smirnov test. All p-values were below 0.05, which means the data is not normally distributed. As a result, I used non-parametric tests like Kruskal-Wallis and ordinal regression for analysis.

Figure 65 - Presentation Slide 14

Data Analysis – Kruskal Wallis Test



The Kruskal-Wallis test is a non-parametric statistical test used to determine whether there are significant differences between the medians of three or more independent groups.

I used the Kruskal-Wallis test because all factors are categorical.

All significance values were less than 0.05.

Therefore, I rejected the null hypothesis (H0) and accepted the alternative hypothesis (H1).

This indicates a significant difference among the groups for the factors tested.

Figure 66 - Presentation Slide 15

Data Analysis – Ordinal Regression Test

[illegible]

Pseudo R-Square	
Cox and Snell	.985
Nagelkerke	.999
McFadden	.988

Link function: Logit.

As per the table, McFadden value is 0.988 which as a percentage 98.8%

Ordinal regression is used to analyze the relationship between one or more independent variables and an ordinal dependent variable with ordered categories.

p-value < 0.05 shows a significant effect

H0: The independent variable has no any significant effect on the dependent variable.

H1: The independent variable has significant effect on the dependent variable.

I used ordinal regression to assess how each independent variable affects the dependent variable (vocabulary development).

You must accept the null hypothesis (H_0) for all the factors because the **Sig. values are greater than 0.05**, indicating there is **no statistically significant effect** on the dependent variable.

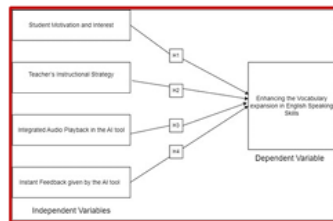
The p-value for audio playback and instant feedback was less than 0.05, meaning these factors had significant influence.

The McFadden value was 0.988, showing a strong fit of the model 98.8% of the variance is explained by these variables.

Figure 67 - Presentation Slide 16

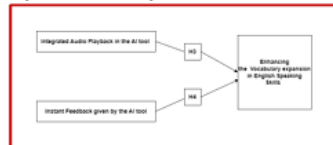
Updated Conceptual Framework

Current Conceptual Framework



Based on the ordinal regression results, the factors **Integrated Audio Playback in the AI tool** and **Instant Feedback given by the AI tool** had **p-values less than 0.05**, so the null hypothesis (H_0) was **rejected**. This means these two factors have a significant effect on enhancing vocabulary in English speaking skills. On the other hand, the factors **Student Motivation and Interest** and **Teacher's Instructional Strategy** had **p-values greater than 0.05**, so the null hypothesis (H_0) was **accepted**, meaning they do not have a significant effect.

Updated Conceptual Framework



Factor	H0
Student Motivation and Interests	Accepted
Teacher's Instructional Strategy	Accepted
Integrated Audio Playback in the AI tool	Rejected
Instant Feedback given by the AI tool	Rejected

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

17

Based on the regression results, I revised the framework.

- Audio Playback and Instant Feedback had significant positive effects.
- Motivation and Teaching Strategy showed no significant impact in this context. The updated model reflects only those elements with statistically proven influence.

Figure 68 - Presentation Slide 17

Before & After T-Test Results

Paired Samples Test										
		Paired Differences				95% Confidence Interval of the Difference		t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper				
Pair 1	Before_Score - After_Score	-2.309	1.914	.258	-2.826	-1.792	-8.949	54	.000	

ONEWAY Before_Score BY After_Score
/MISSING ANALYSIS.

H0: There is no difference between before and after results.
H1: There is a difference between before and after results.

I conducted a T-test to compare the results before and after the intervention. Since the significance value was 0.000, which is less than 0.05, I rejected the null hypothesis. This indicates that there is a statistically significant difference between the results before and after the intervention.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

18

A T-test compared scores before and after using Duolingo.

The p-value was 0.000, which is highly significant.

This means that Duolingo led to a noticeable and measurable improvement in vocabulary acquisition.

I conducted a T-test to compare the results before and after the intervention.

Since the significance value was 0.000, which is less than 0.05, I rejected the null hypothesis. This indicates that there is a statistically significant difference between the results before and after the intervention.

Figure 69 - Presentation Slide 18

Achievement of the Objectives

- ❑ **Objective 1: To analyze the effectiveness of Duolingo in improving the vocabulary expansion of diploma students.**

(Achieve)

This objective was achieved through pre- and post-assessment scores of students who used Duolingo over a set period. The results showed a noticeable improvement in vocabulary acquisition among most participants, indicating that Duolingo is an effective tool for enhancing vocabulary among diploma students.

- ❑ **Objective 2: To identify the challenges faced by diploma students while using Duolingo for vocabulary development.**

(Achieve)

Feedback collected through surveys and focus group discussions helped identify key challenges. These included lack of motivation to use the app regularly, difficulty understanding pronunciation without guidance, and limited exposure to contextual word usage. This data provided valuable insights into the obstacles learners face when using mobile-based learning tools.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

19

These are my achievement of the objectives.

Objective 1 was achieved through before/after testing. **To analyze the effectiveness of Duolingo in improving the vocabulary expansion of diploma students. (Achieve)**

This objective was achieved through pre- and post-assessment scores of students who used Duolingo over a set period. The results showed a noticeable improvement in vocabulary acquisition among most participants, indicating that Duolingo is an effective tool for enhancing vocabulary among diploma students.

Objective 2 was achieved by analyzing student feedback on difficulties. **To identify the challenges faced by diploma students while using Duolingo for vocabulary development. (Achieve)**

Feedback collected through surveys and focus group discussions helped identify key challenges. These included lack of motivation to use the app regularly, difficulty understanding pronunciation without guidance, and limited exposure to contextual word usage. This data provided valuable insights into the obstacles learners face when using mobile-based learning tools.

Figure 70 - Presentation Slide 19

Achievement of the Objectives

- ❑ **Objective 3: To compare the vocabulary enhancement of students using Duolingo with those using traditional learning methods by identifying the before and aftereffects of using Duolingo application. (Achieve)**

By comparing the pre- and post-test results of students using Duolingo with a control group using traditional methods, it was evident that Duolingo users showed greater improvement in vocabulary test scores. This comparison confirmed that the app provides a more engaging and effective learning experience than conventional classroom techniques.

- ❑ **Objective 4: To propose a conceptual framework based on the findings to optimize the use of Duolingo for vocabulary and speaking skill development. (Partially Achieve)**

This was partially achieved. While a refined conceptual framework was successfully developed based on the statistical findings to enhance vocabulary development using Duolingo, the speaking skill development component was not fully addressed due to the limited scope of the current data and analysis.

NAME: NETHMI THILOKA DIVYANJALEE

E-NUMBER: E170909

20

Objective 3 was achieved via comparative analysis between groups. **To compare the vocabulary enhancement of students using Duolingo with those using traditional learning methods by identifying the before and aftereffects of using Duolingo application. (Achieve)**

By comparing the pre- and post-test results of students using Duolingo with a control group using traditional methods, it was evident that Duolingo users showed greater improvement in vocabulary test scores. This comparison confirmed that the app provides a more engaging and effective learning experience than conventional classroom techniques.

Objective 4 was partially achieved, as vocabulary improvements were clear, but speaking skills require further study due to time and data limitations. **To propose a conceptual framework based on the findings to optimize the use of Duolingo for vocabulary and speaking skill development. (Partially Achieve)**

This was partially achieved. While a refined conceptual framework was successfully developed based on the statistical findings to enhance vocabulary development using Duolingo, the speaking skill development component was not fully addressed due to the limited scope of the current data and analysis.

Figure 71 - Presentation Slide 20

Conclusion

- The study revealed key limitations in traditional vocabulary teaching methods, particularly in engaging and motivating diploma students.
- As a solution, the Duolingo application was evaluated for its impact on vocabulary and speaking skill development.
- Significant improvements were observed in vocabulary acquisition, supported by pre- and post-test comparisons and statistical analysis.
- Among the hypothesized factors, **Integrated Audio Playback** and **Instant Feedback** had a significant impact on learning effectiveness.
- A refined **conceptual framework** was developed by extending **Merrill's Principles of Instruction (MPI)** to better integrate AI tools into language learning.
- While vocabulary enhancement goals were fully achieved, the speaking skill development aspect was only partially addressed, indicating an area for future research and framework expansion.

NAME: NETHMI THILOKA DIVYANJALEE ID NUMBER: E170909 21

To conclude, Duolingo has a strong potential to improve vocabulary skills among diploma students, especially when supported by clear audio and instant feedback. However, user motivation and teacher input, while still important in practice, did not show statistical significance in this study. Future research should expand on speaking fluency and long-term impacts.

Figure 72 - Presentation Slide 21

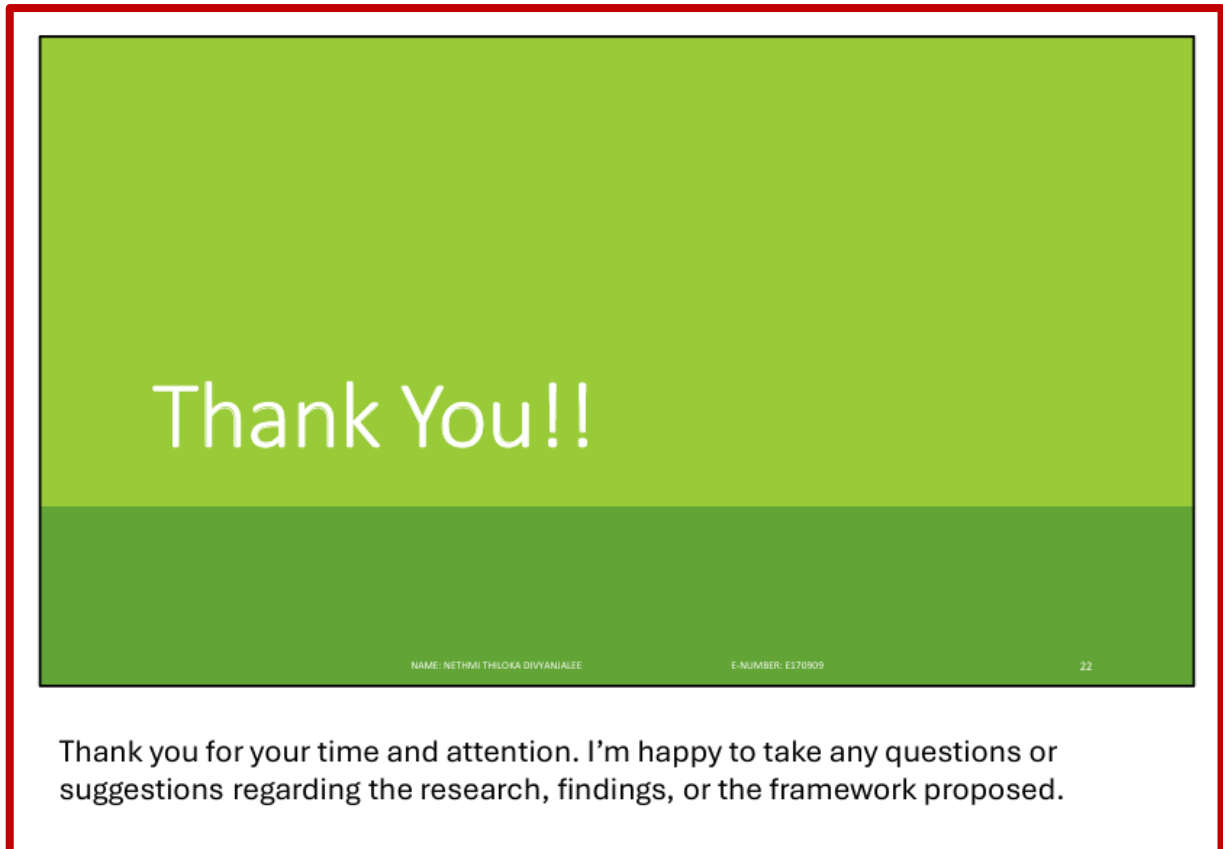


Figure 73 - Presentation Slide 22